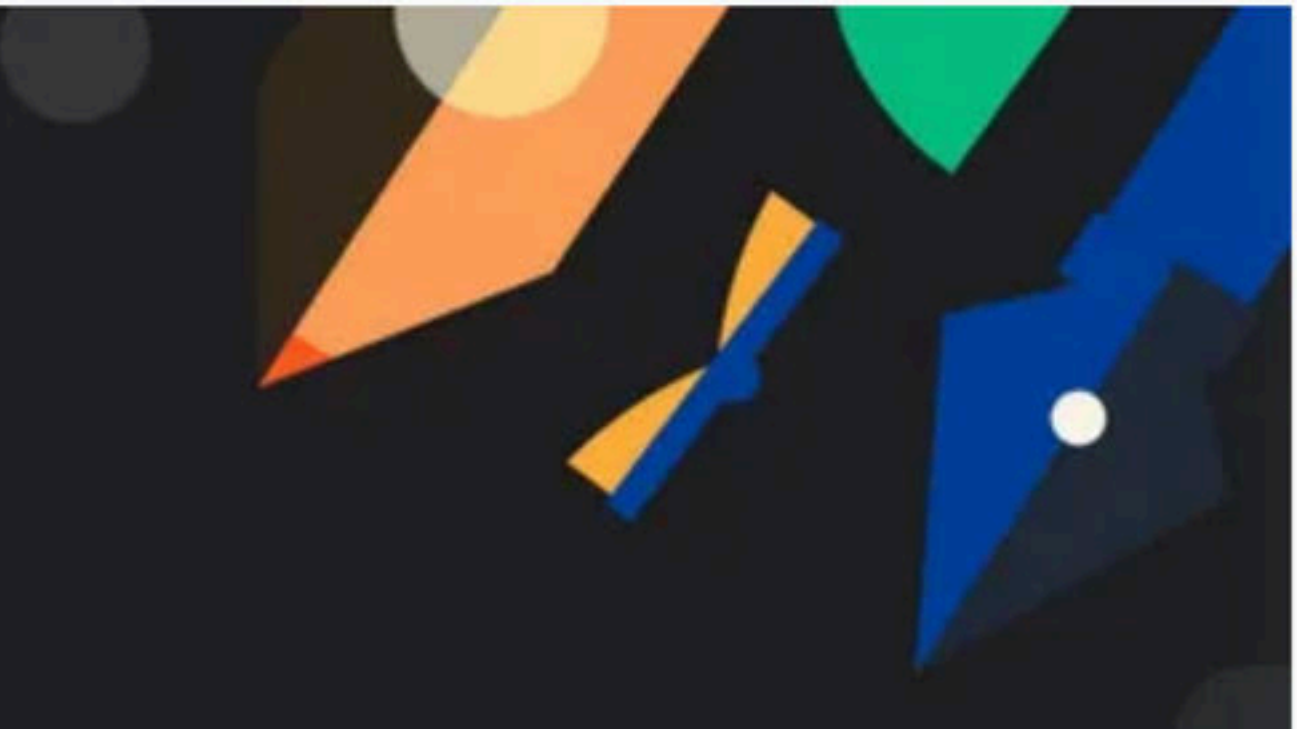




2nd Doubts Class with Lakshay

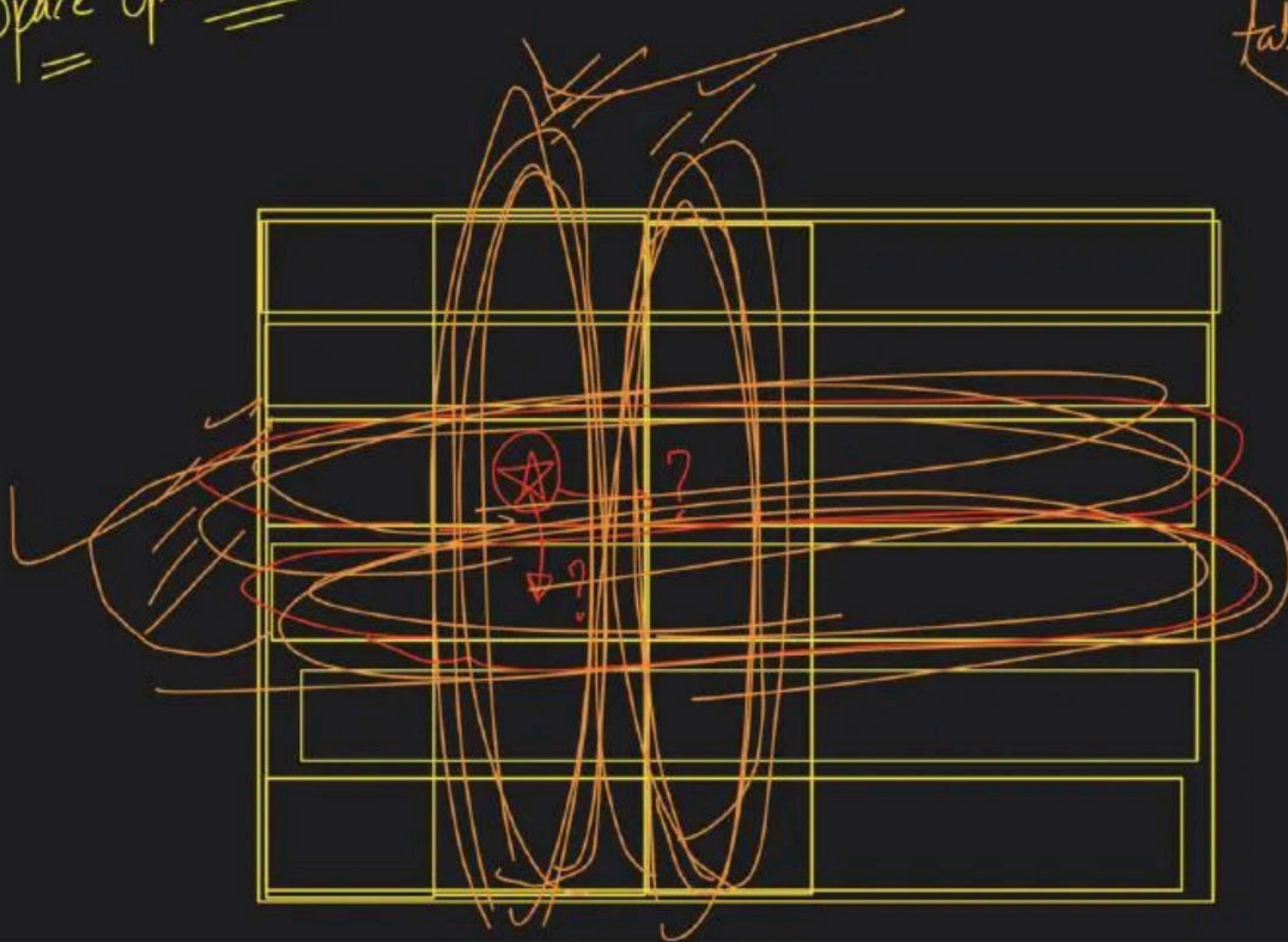
Special class

DP Class - 4 [Join here]

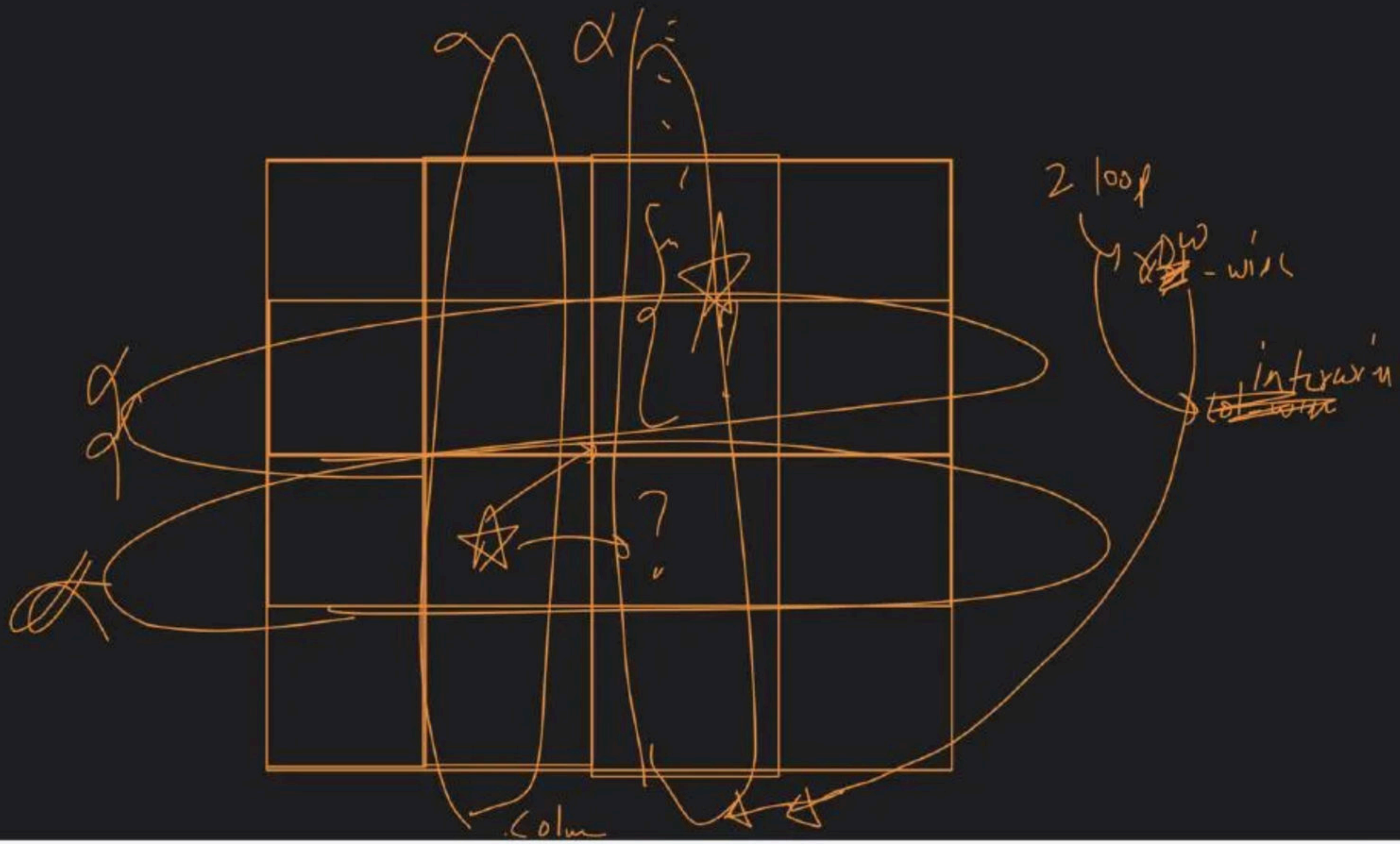
Special class

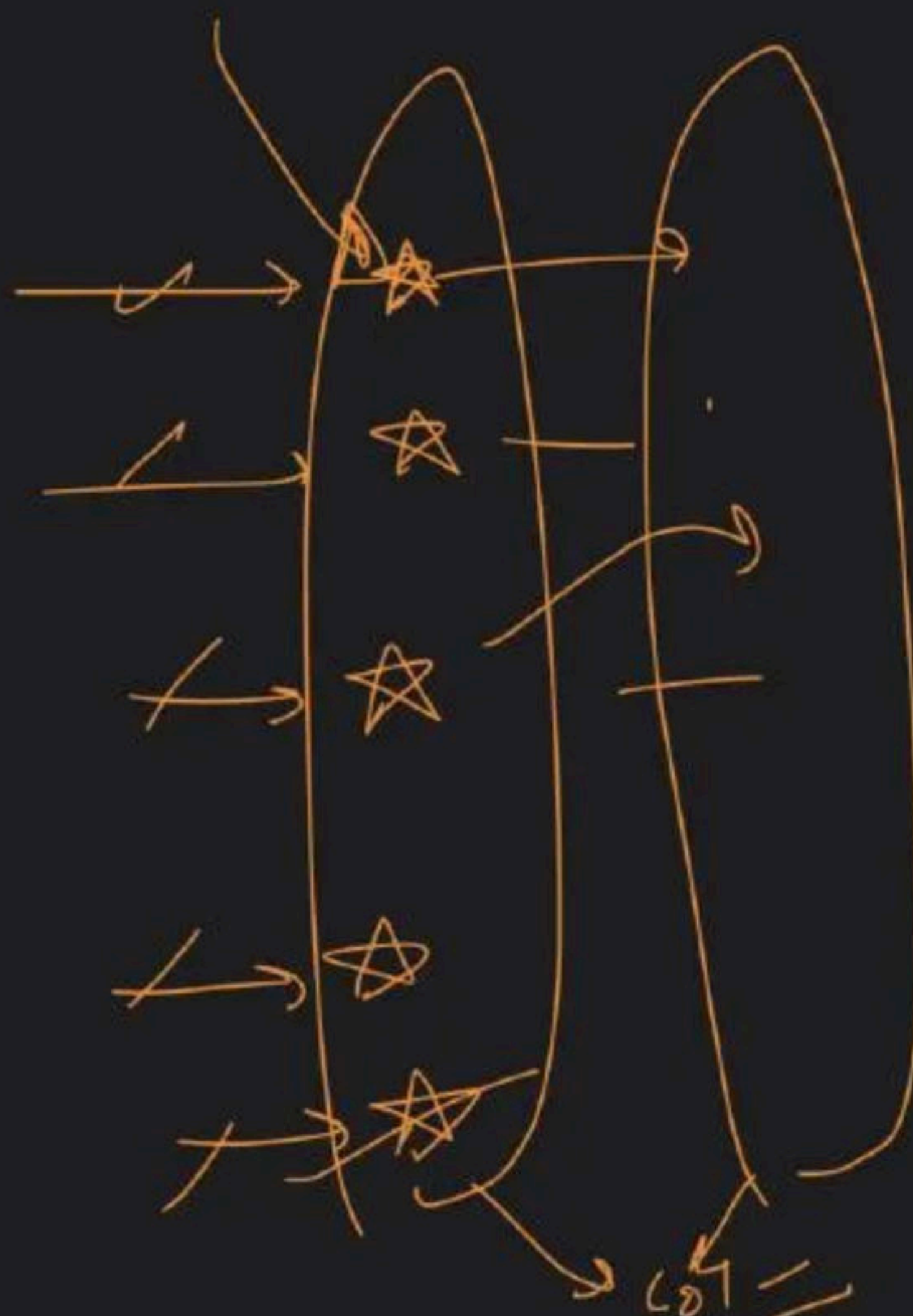
① Interchanged
=

Space Optimisation



tab → 2 loop
→ row-wise
→ col-wise



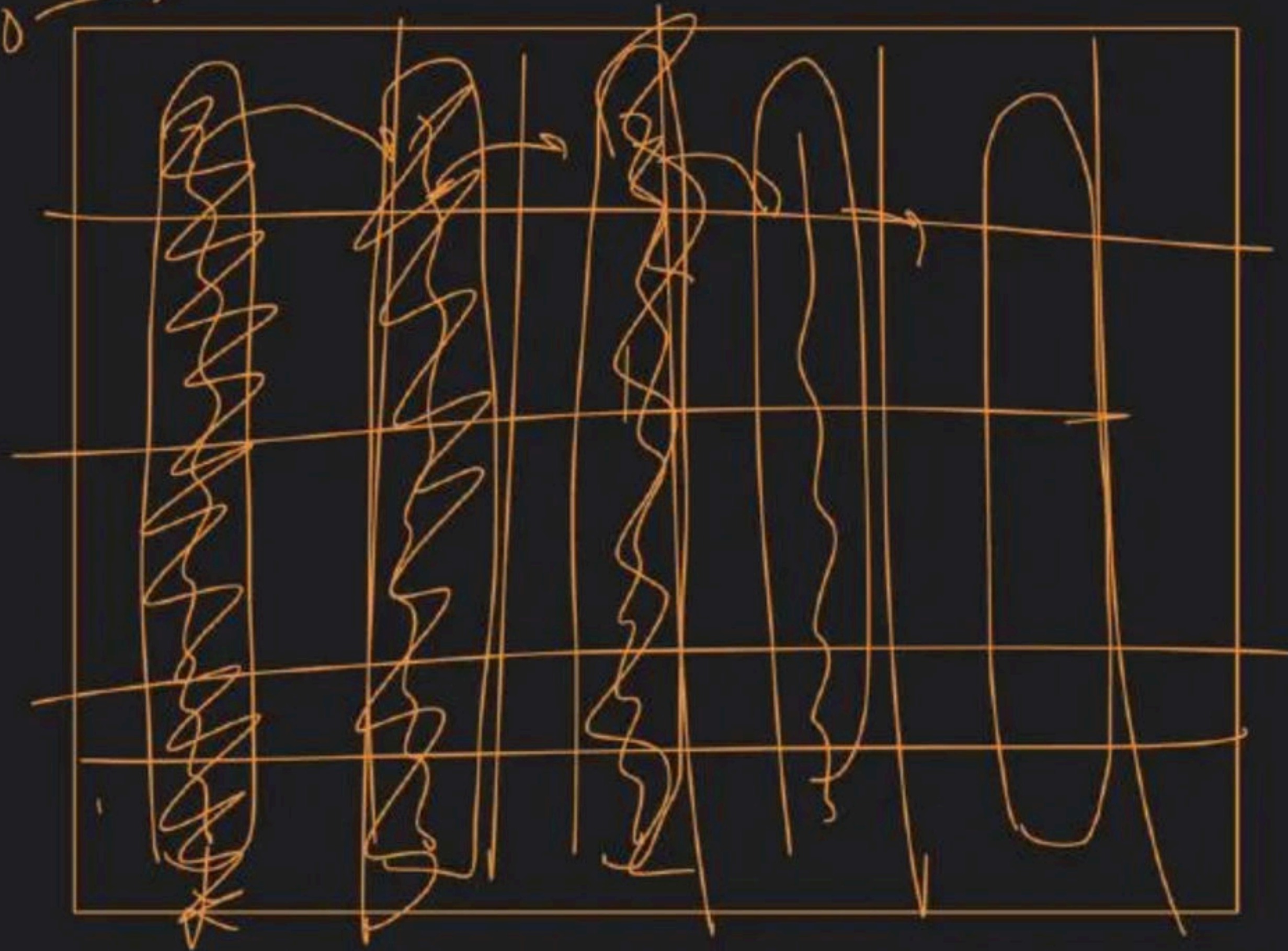


2 loop

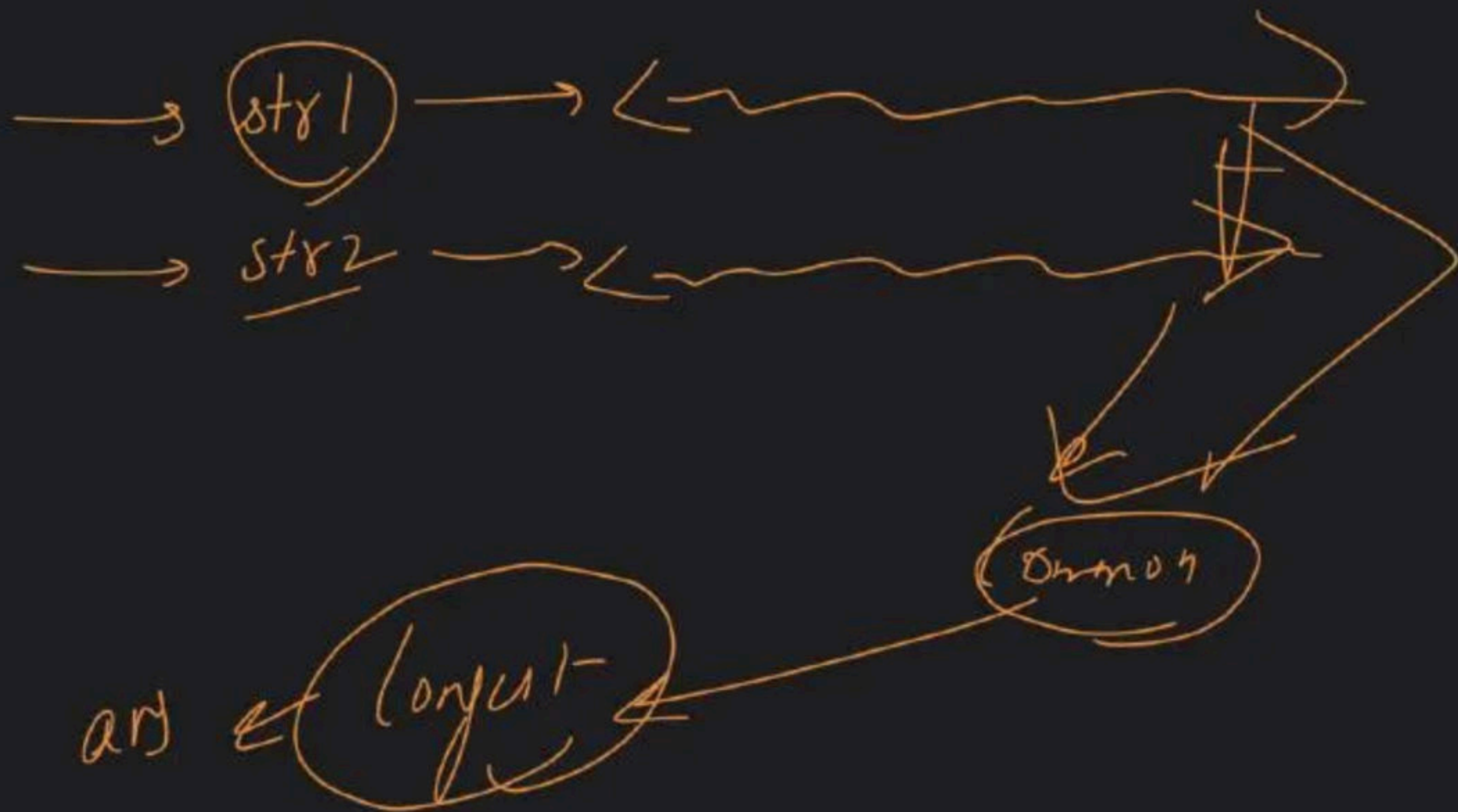
row win

interview

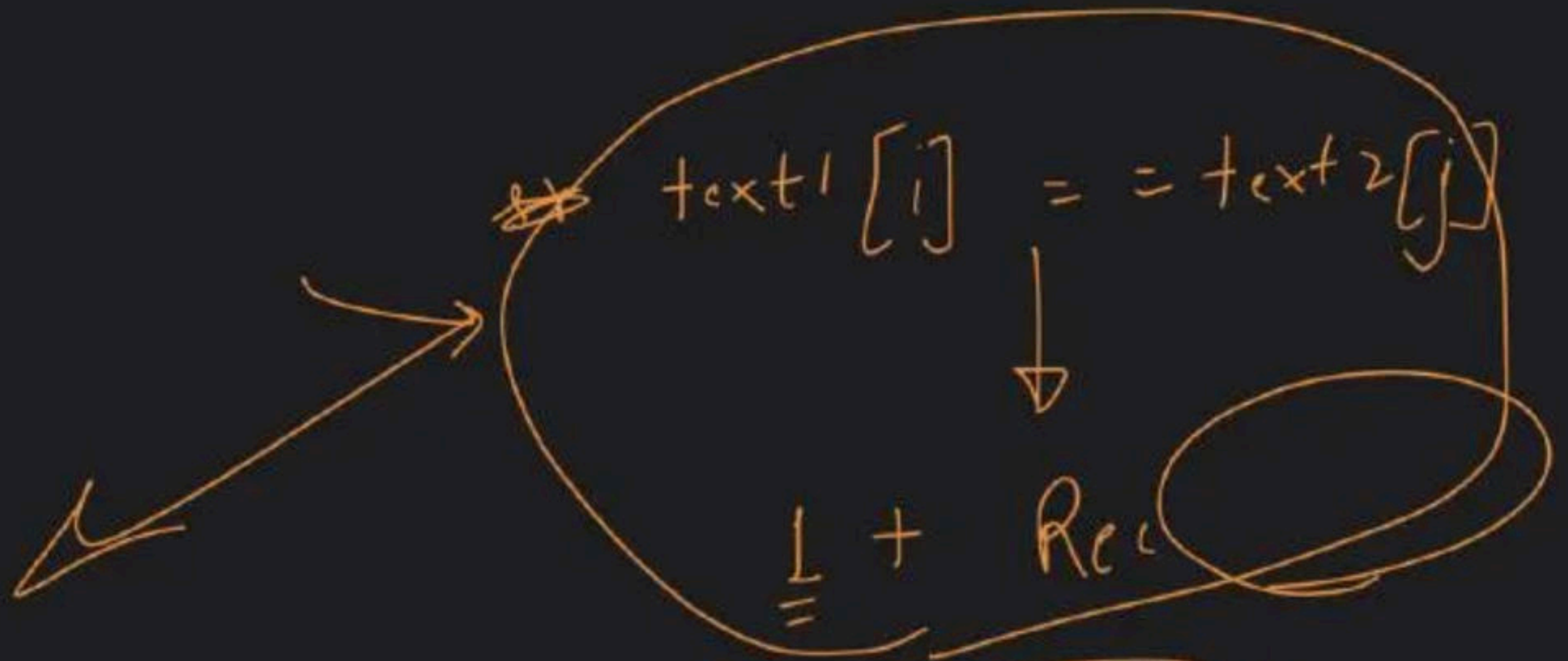
2d $\rightarrow$ $(n \times n)$ $\rightarrow$ tab (n^2)
S.O $\rightarrow n+n \rightarrow 2n \sim (n) //$



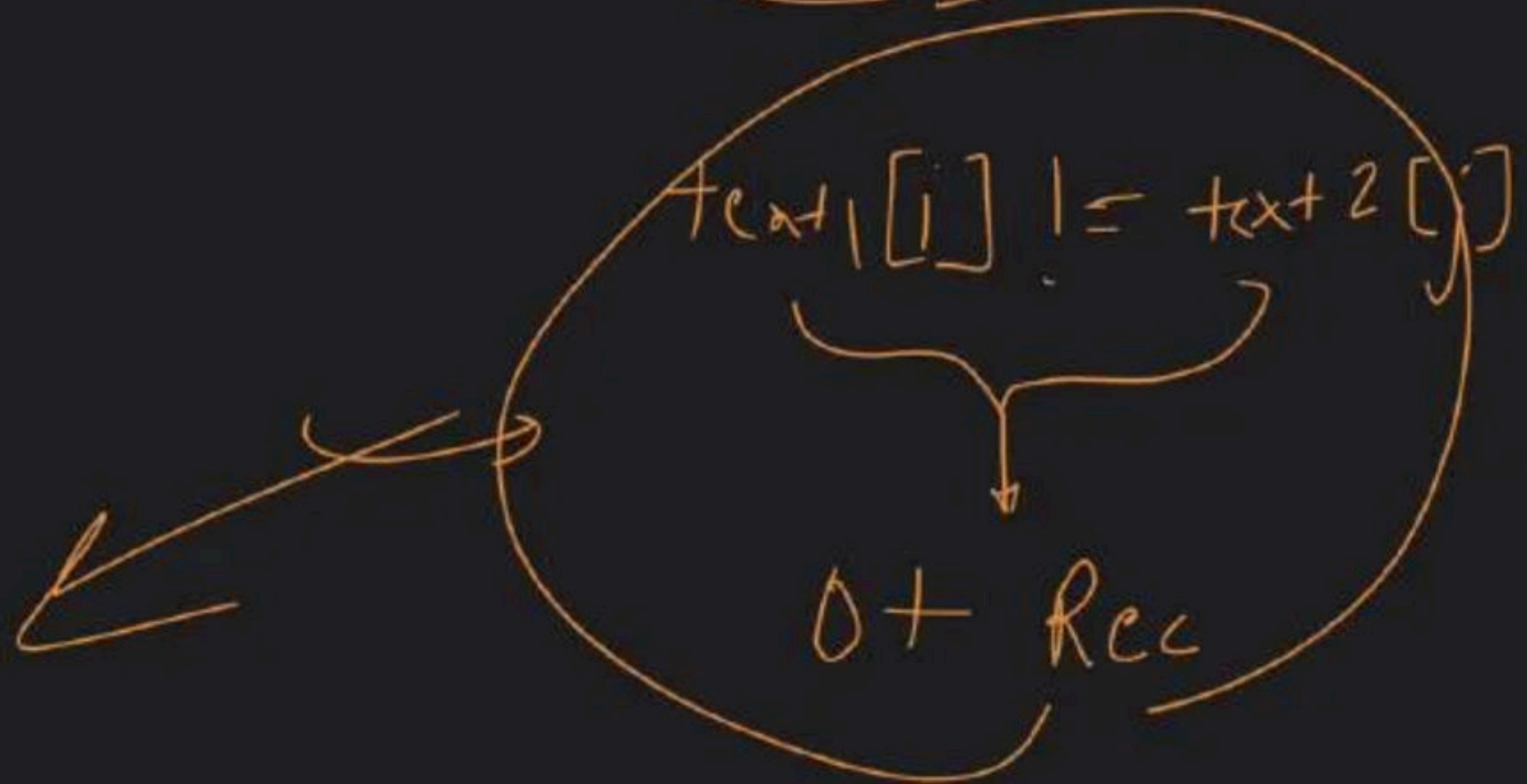
→ Longest Common Subsequence

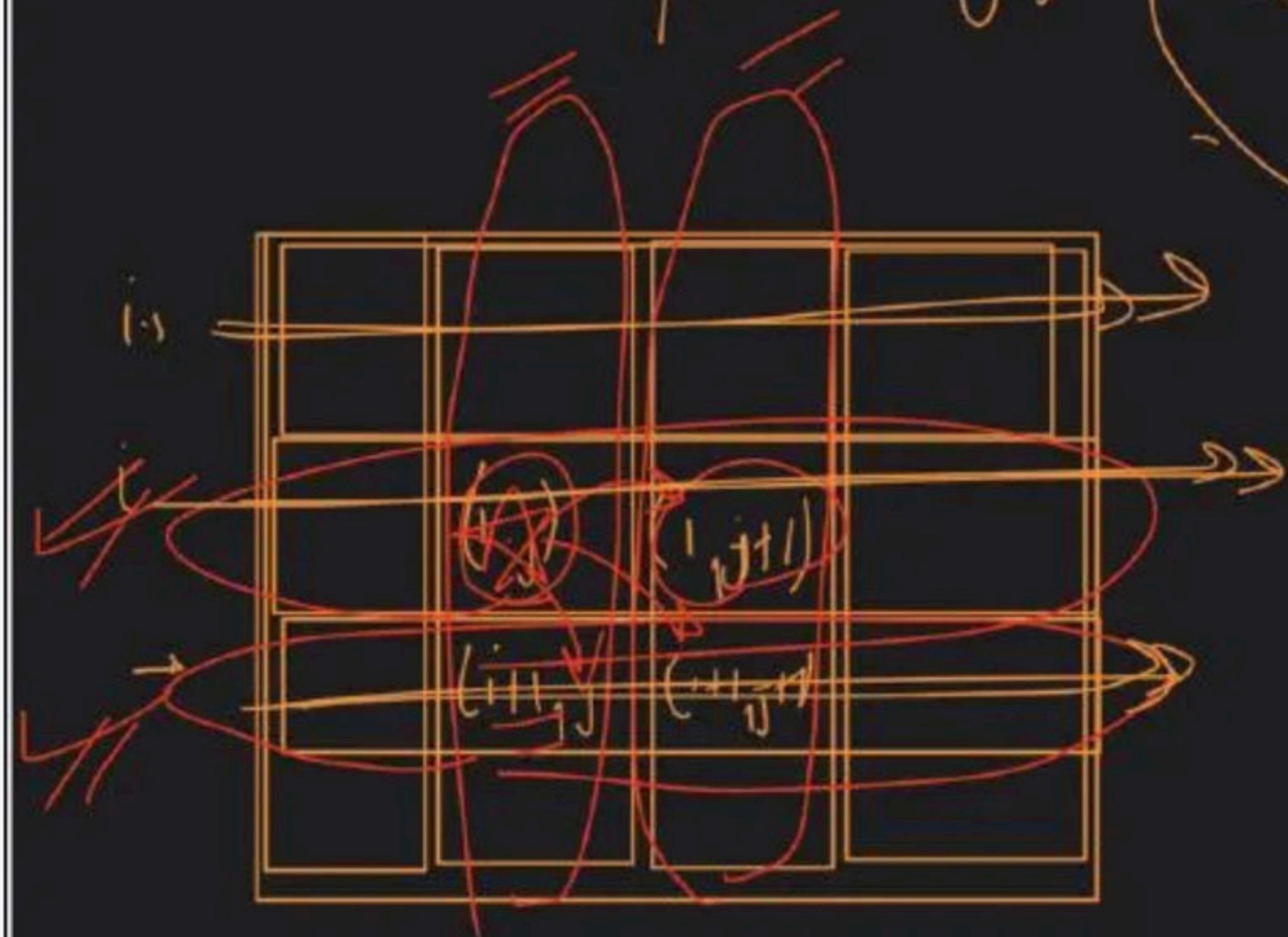


abcd e
↑
i



a c e
↑
j



$$dp[i][j]$$
$$dp[i+1][j+1]$$
$$dp[i+1][j]$$
$$d_p [i] [j+1]$$


PPO
 PPI $\rightarrow$ ① infer $k \times 2$ u

PPI

$$|x|$$

① infer $k \times 2$ in

10 days

due per

not per

~~over pool~~

[Handwritten signature]

1

12

 $\mu \leq \gamma$

人

40k pm

→ 2018

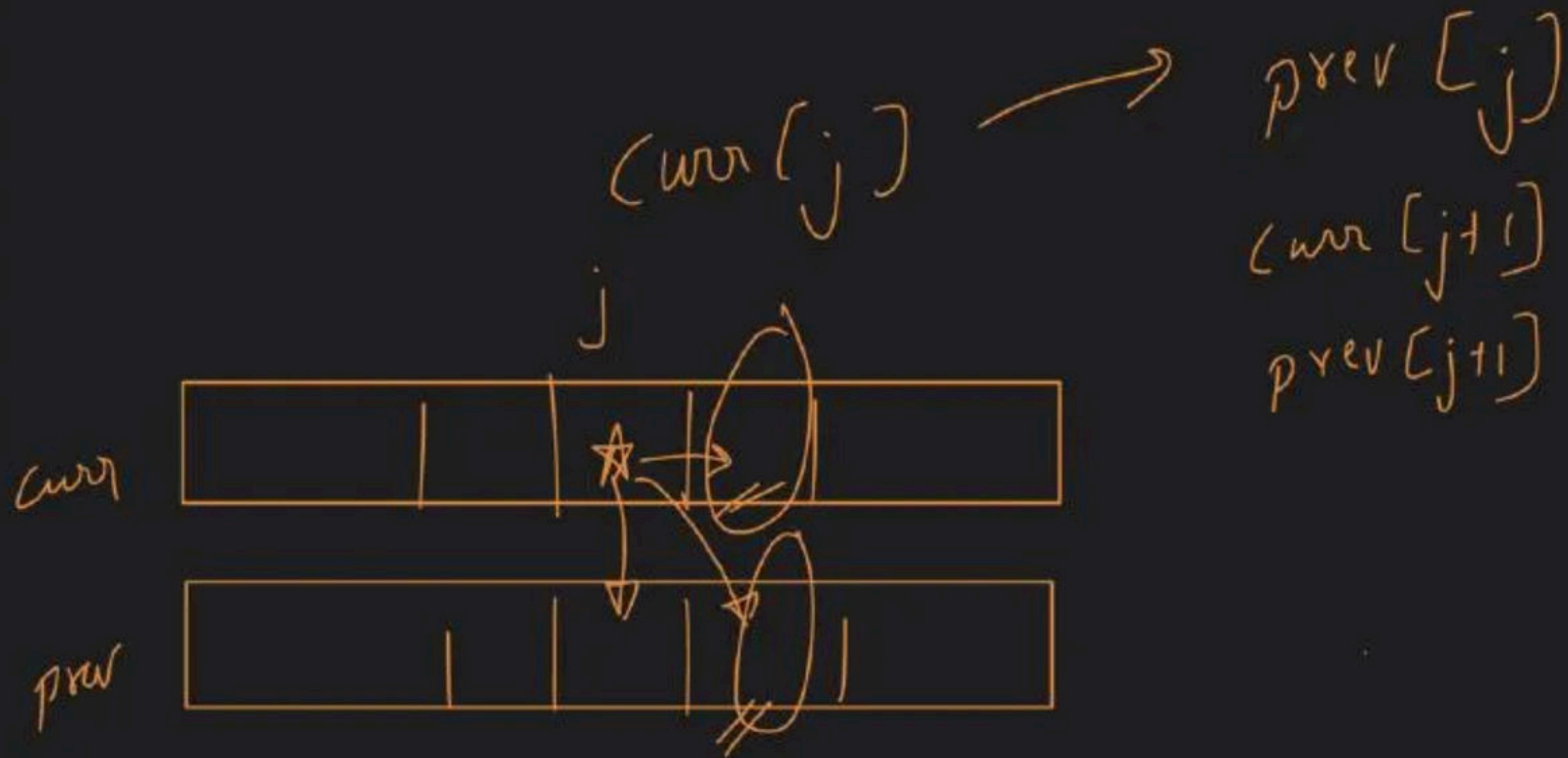
1L PM

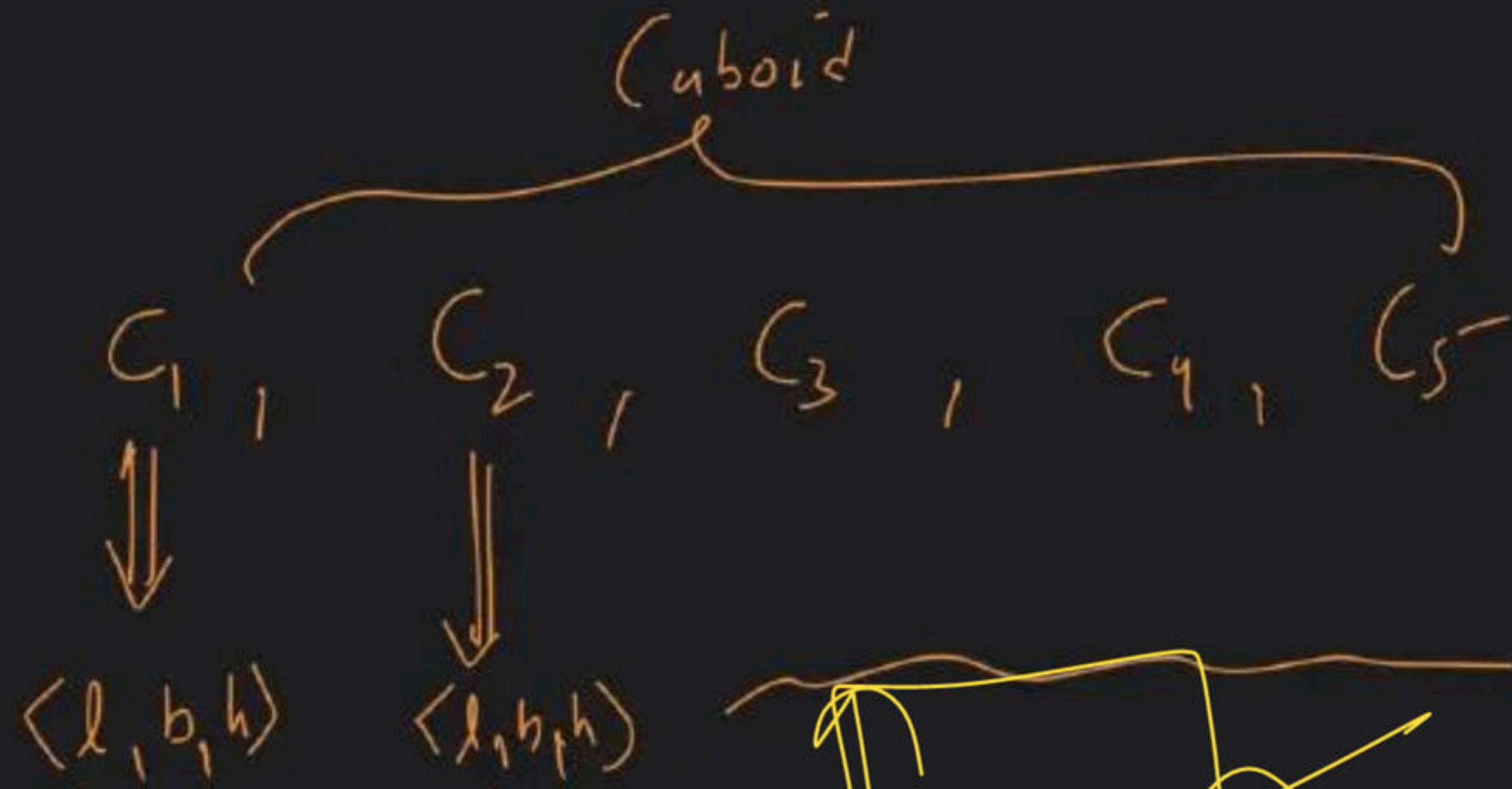
→ 2019 ≥

intern

1.5L → 2L



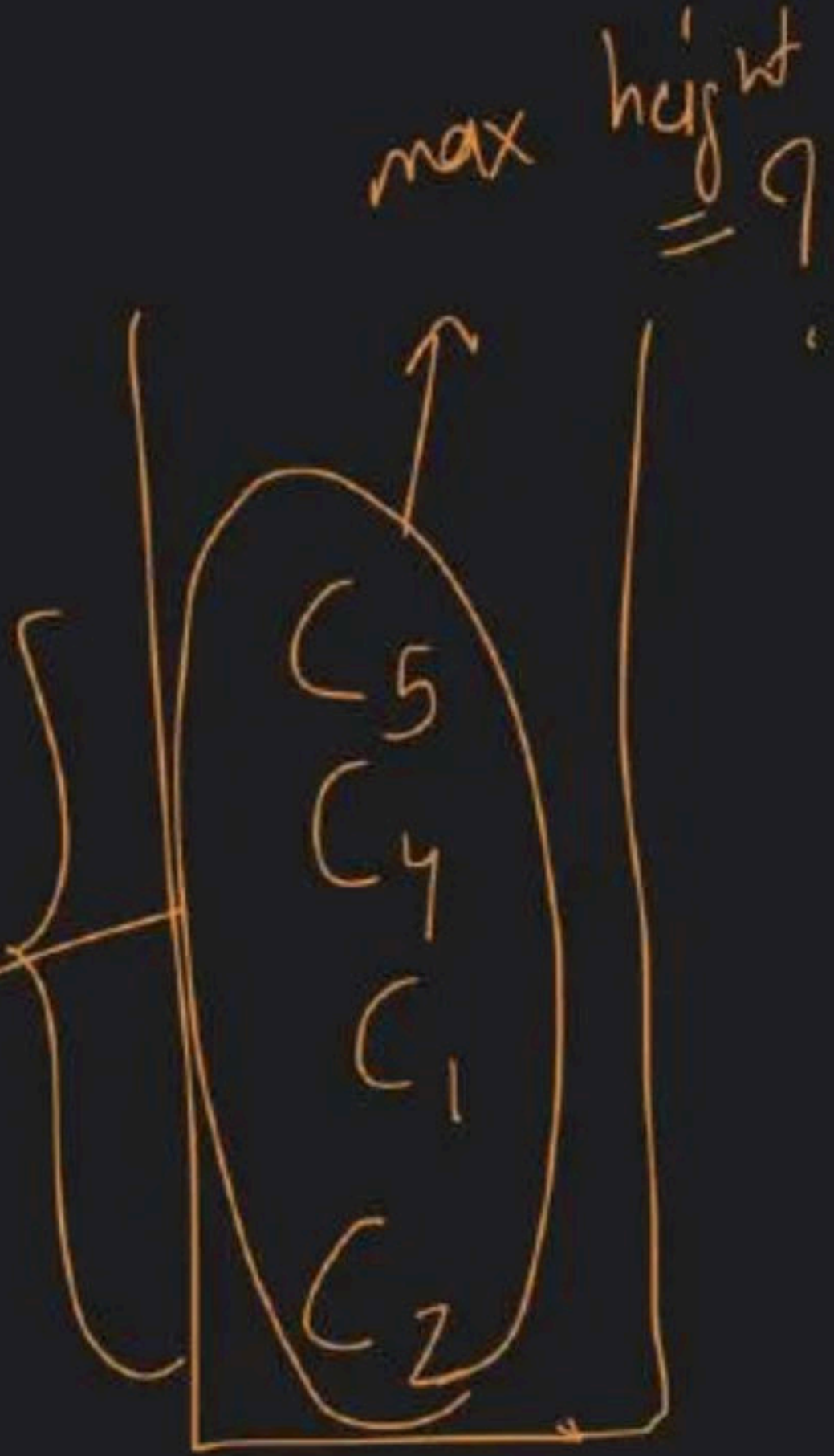


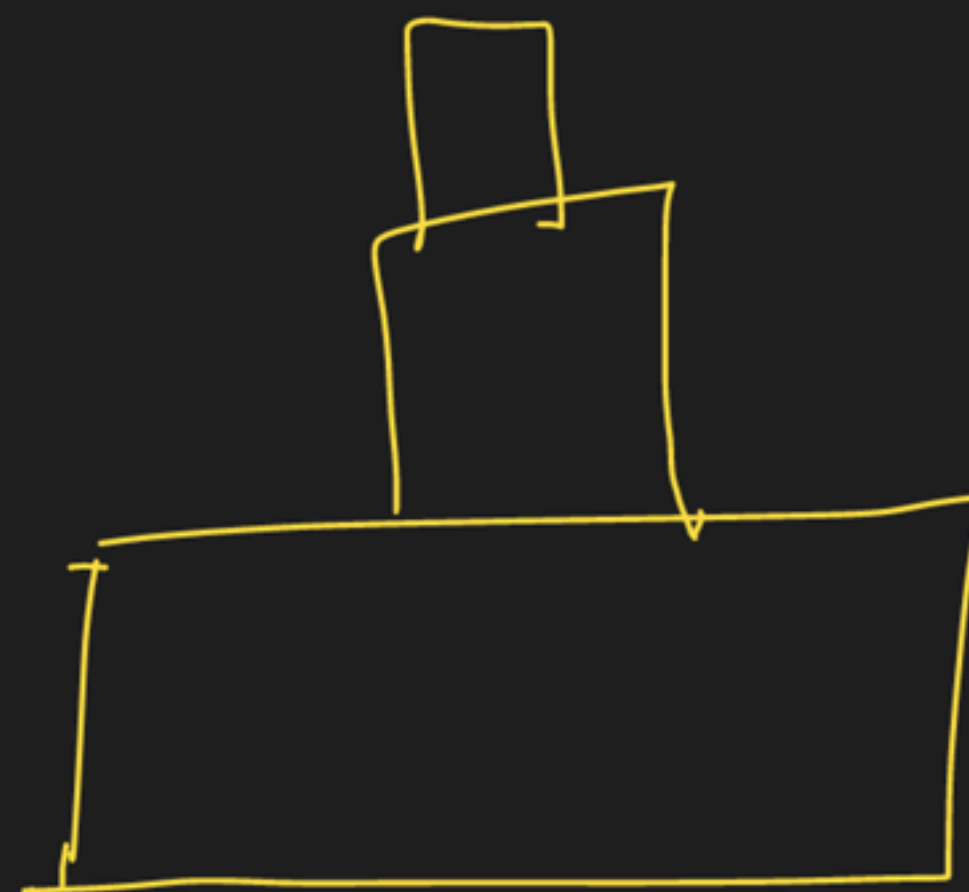


1 minutes

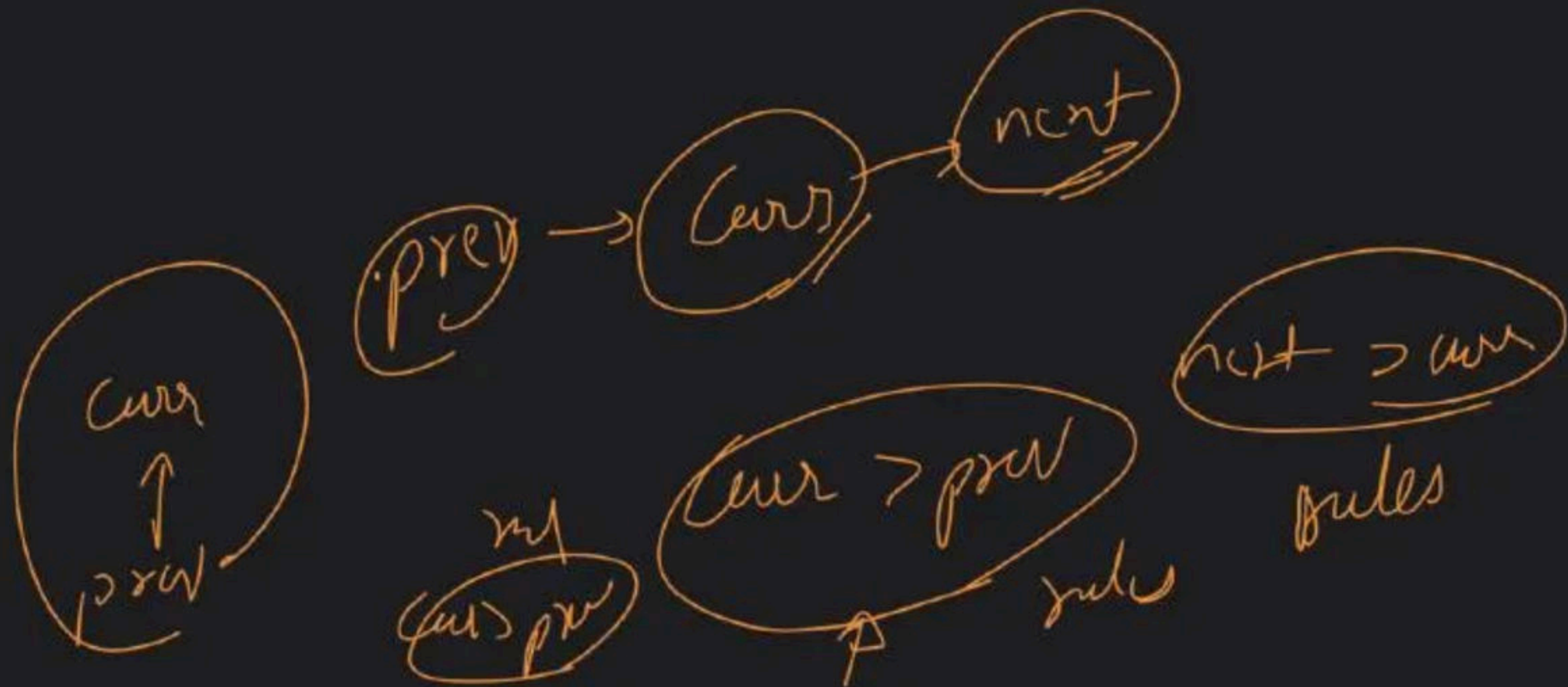


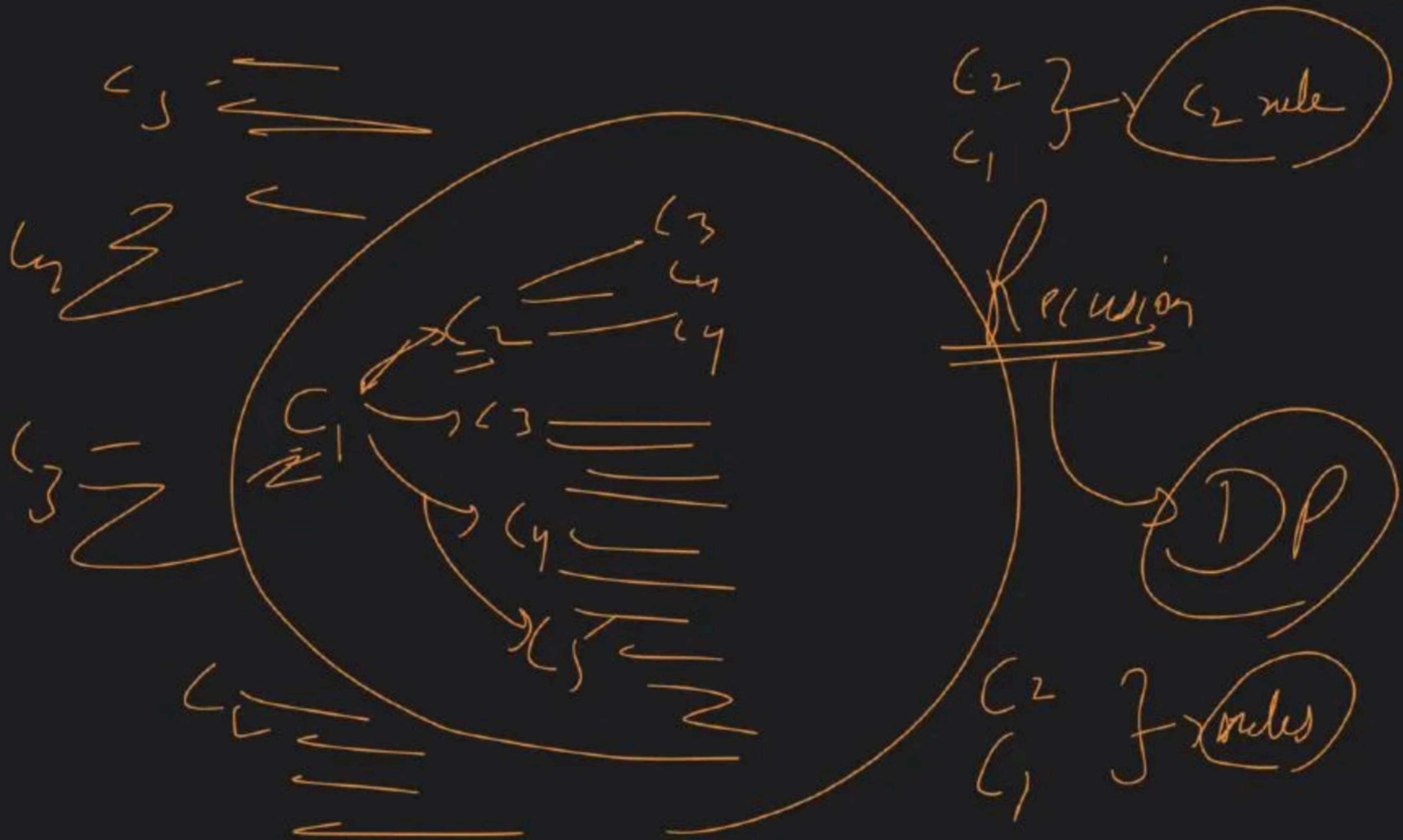
as per rules





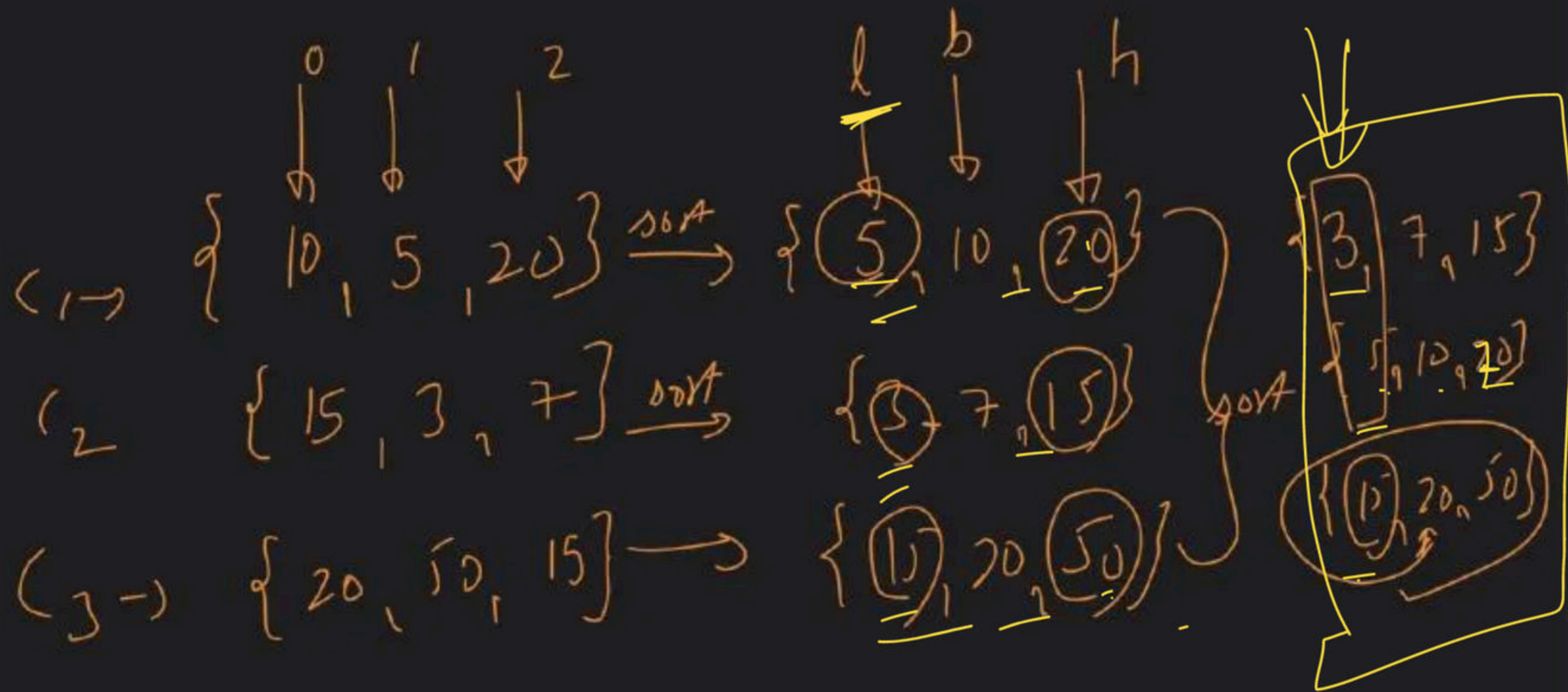
Longest Increasing Subsequence





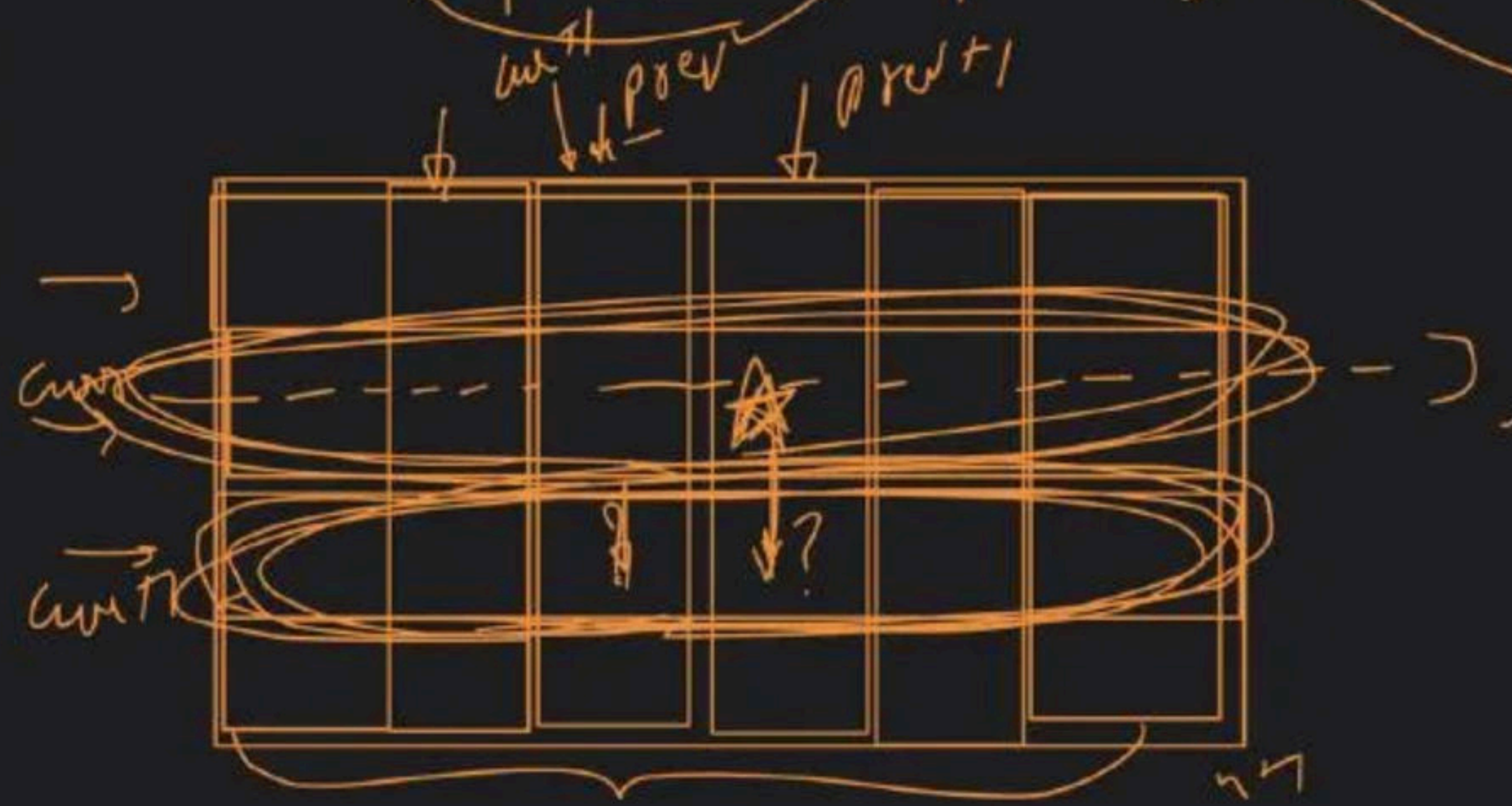
sort → $n \log n$

$\left\{ \begin{array}{ccc} l & b & h \\ - & - & \bigcirc \\ \uparrow & \uparrow & \uparrow \end{array} \right\}$
, $O(1)$



max $\rightarrow$ height $\rightarrow$

$dp[curr][prev+1]$



next
 $dp[curr+1][prev+1]$

$dp[curr+1][curr+1]$

50, 45, 20

95, 37, 53

45, 23, 12

- [20, 45, 50] → h

- [37, 53, 95] → h

- [12, 23, 45] → h

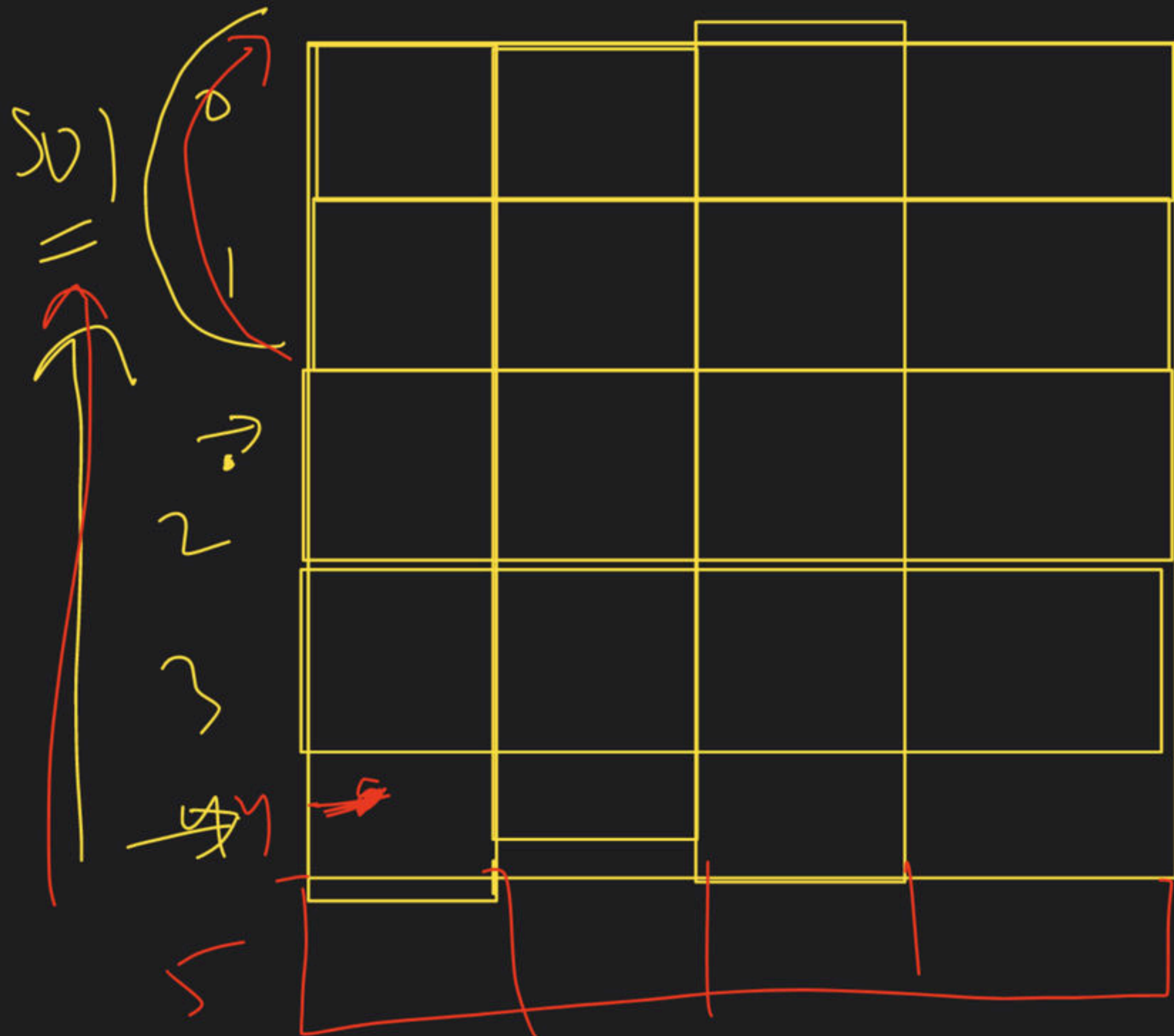
- 20, 45, 50

37, 53, 95

2, 23, 45
20, 45, 50
37, 53, 95

[12, 23, 45]
[20, 45, 50]
[37, 53, 95]

⇒ ..



$$5+1$$

Φ

$$cur = 2$$

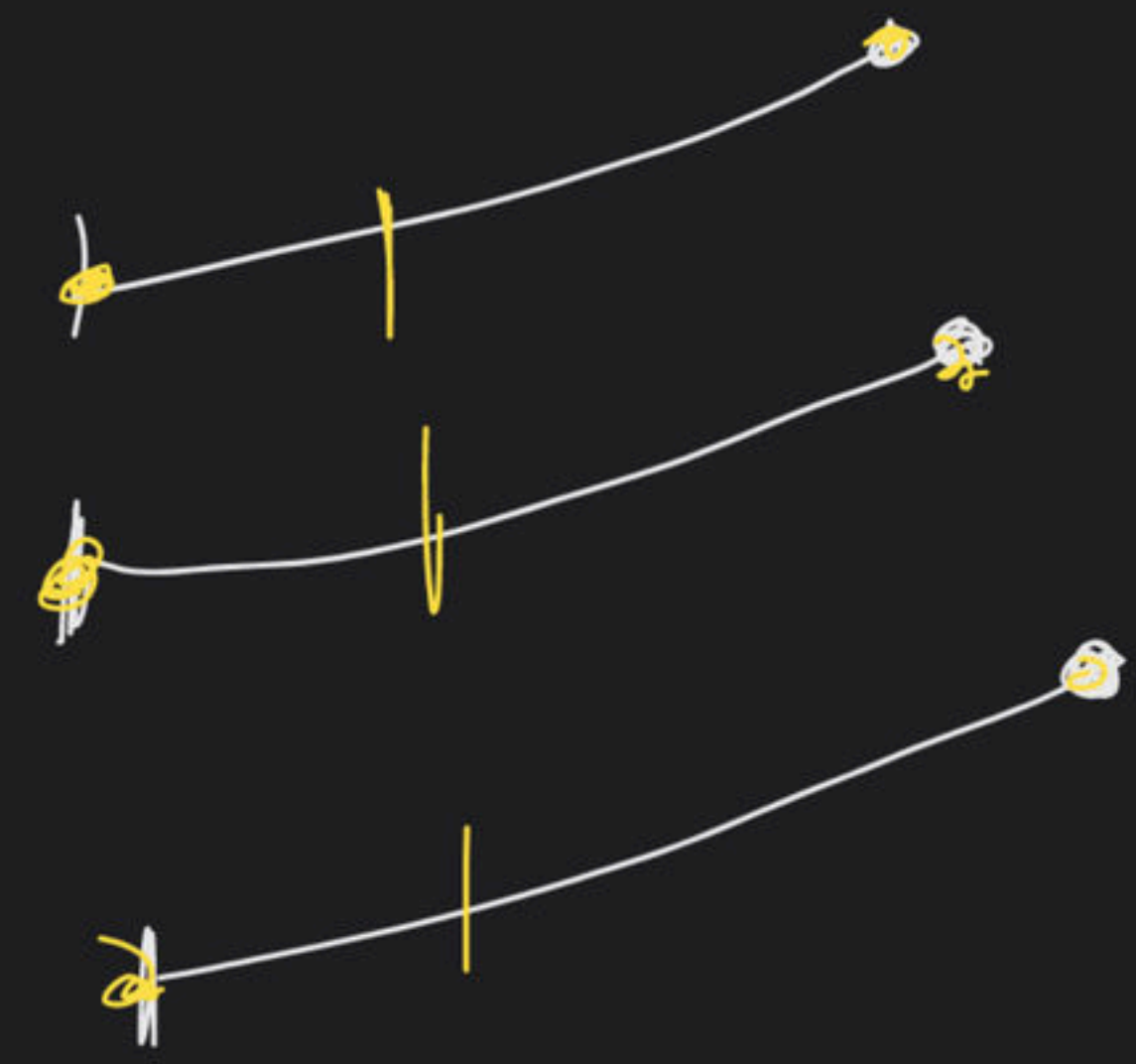
$$prev = 1 \rightarrow$$

$$prev = 3 \rightarrow -1$$

A

B

C



small
=
Big

$$0 \ [20, 48, 50]$$

$$1 \ [37, 53, 95]$$

$$2 \ [12, 23, 45]$$

$$\cancel{B \geq 12}$$

α

95+

$$1, 2$$

50+

$$0, 1$$

$$(-1, 0)$$

95

50

$$\underline{\underline{145}}$$

$(12, 23, 45)$
 $(20, 45, 50)$
 $(32, 53, 95)$

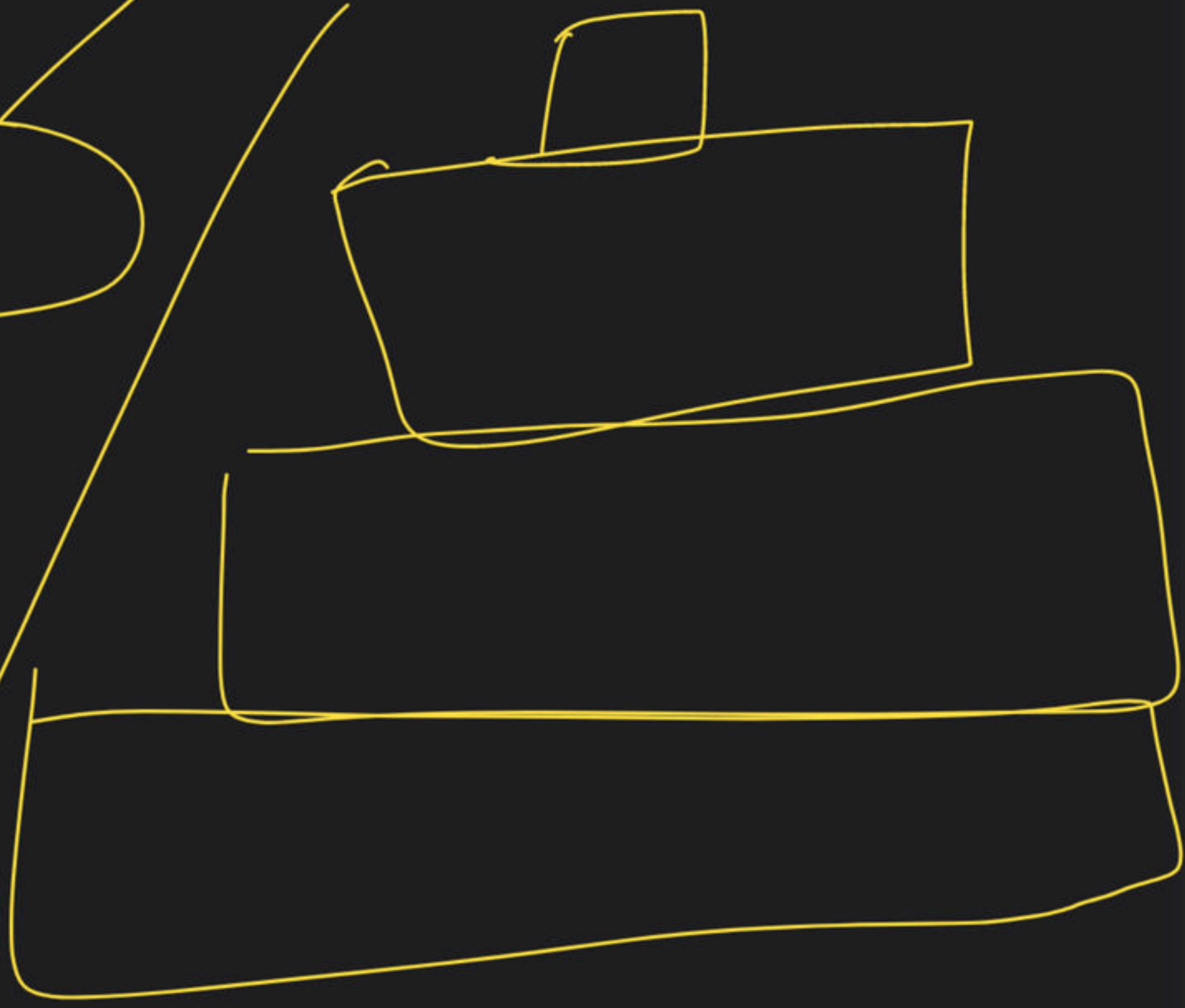
$-1, 0$

45+

$0, 11$

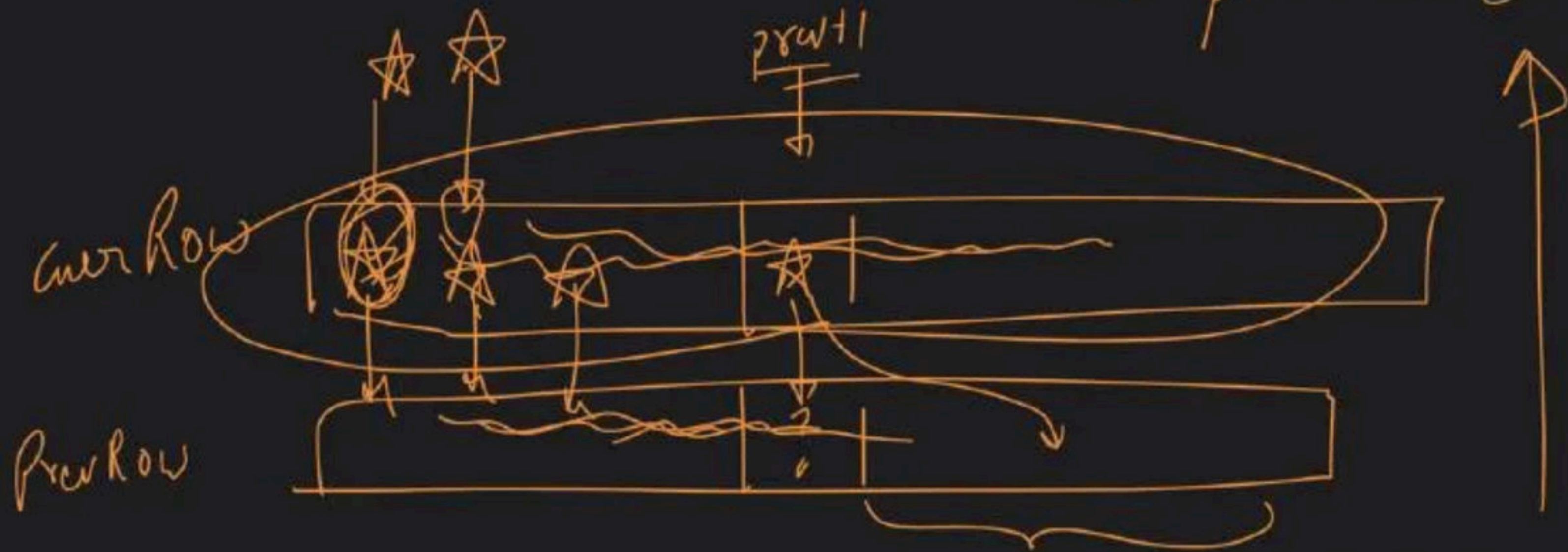
50+

$1, 2$



Ragini?

$curRow[prev+1]$ $\rightarrow$ $prevRow[prev+1]$ ✓
 $\rightarrow$ $prevRow[cur+1]$



?

e

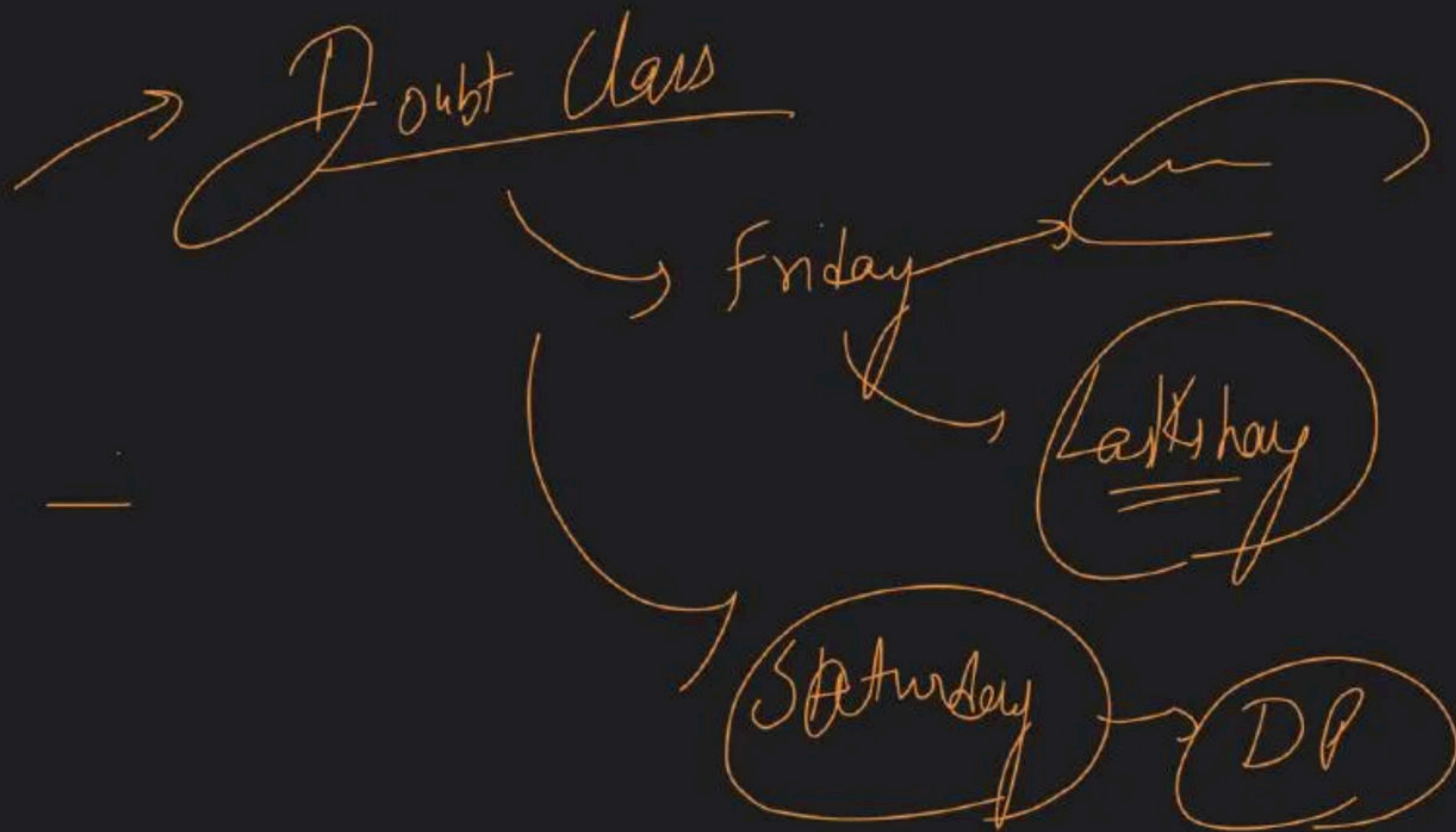


x	y	z	z	z	+	+	-	-
---	---	---	---	---	---	---	---	---

cur, row



2

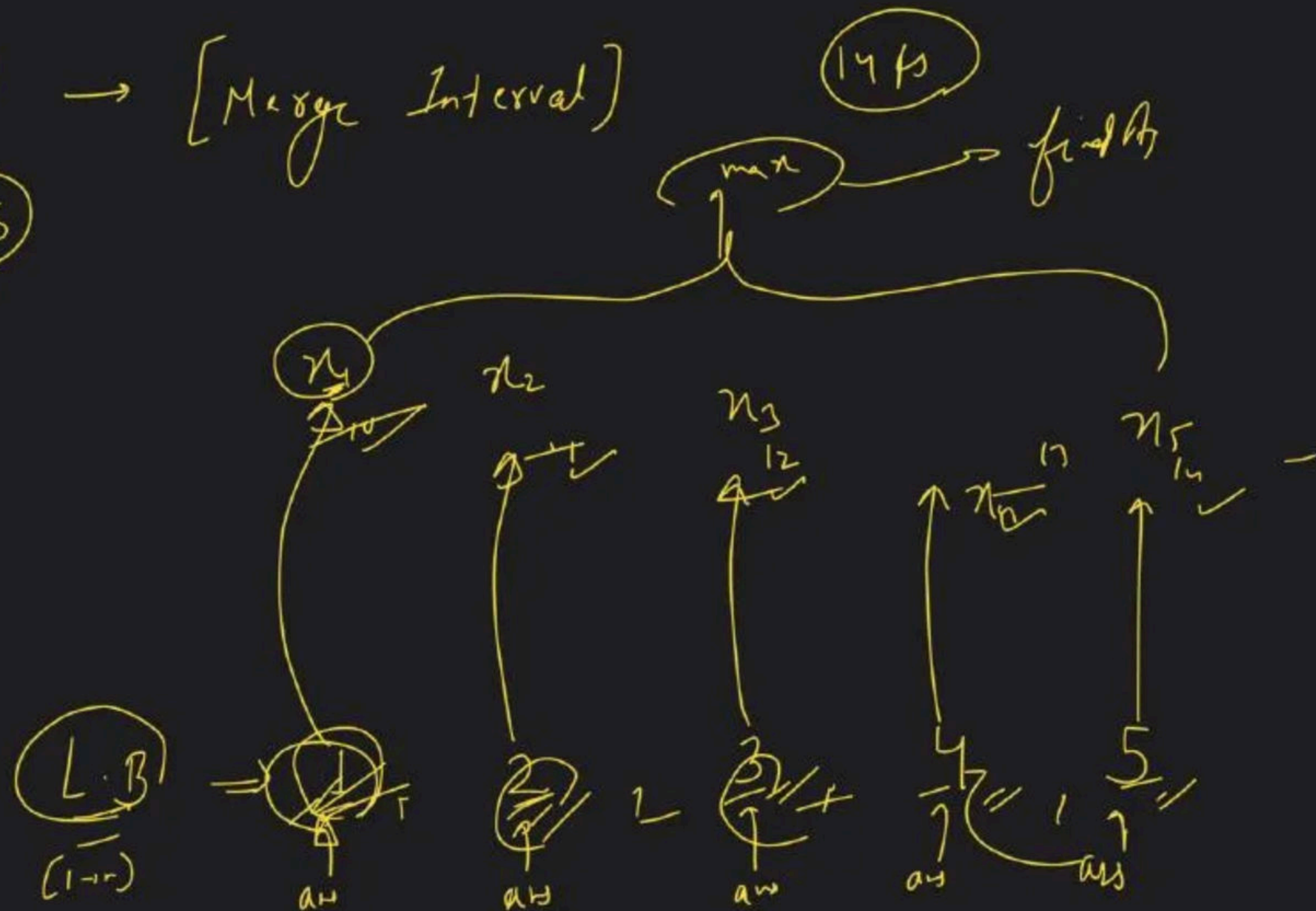


DP Class - 5

Special class

→ DP → [Merge Interval]

$n=5$



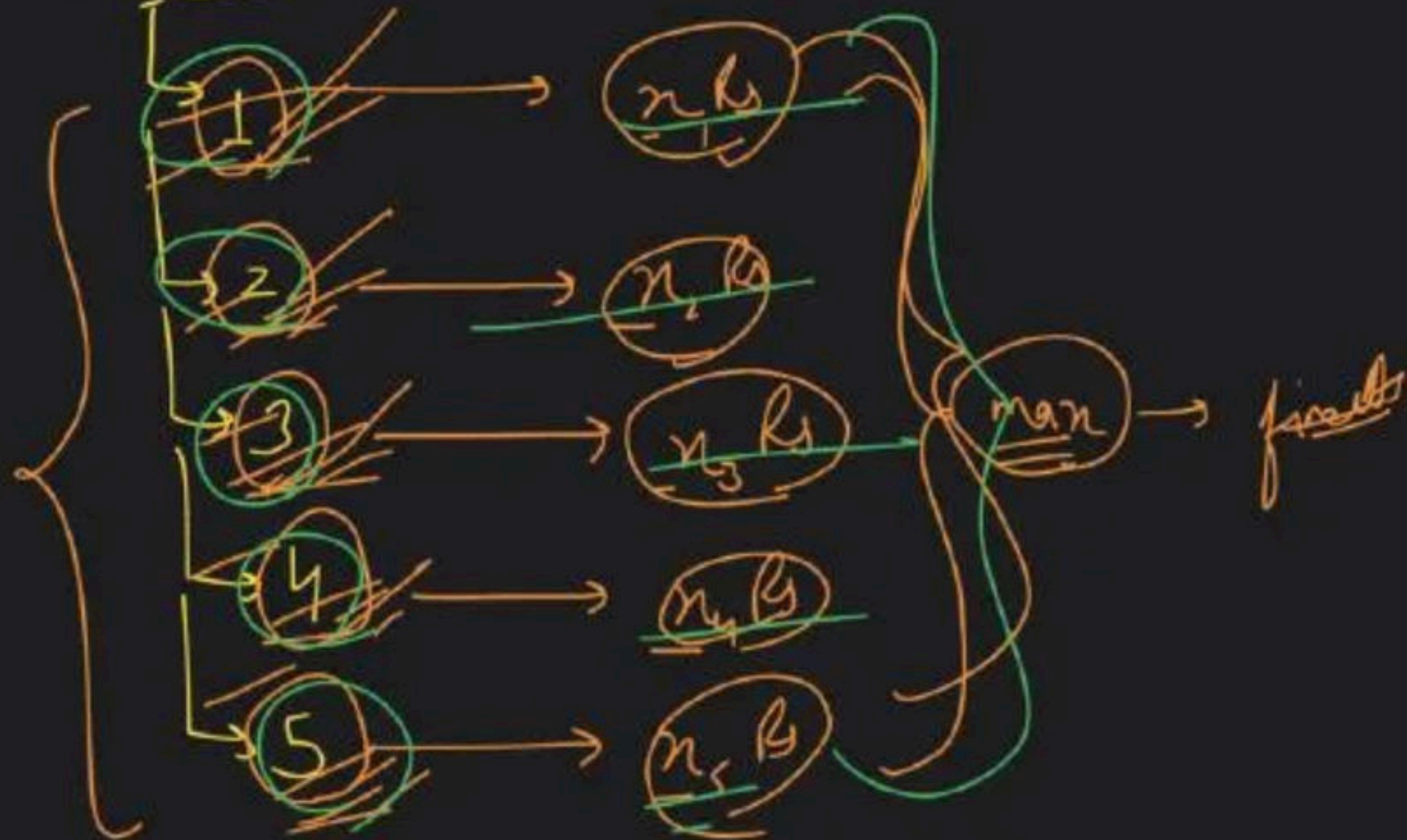
L.B

Student → guns

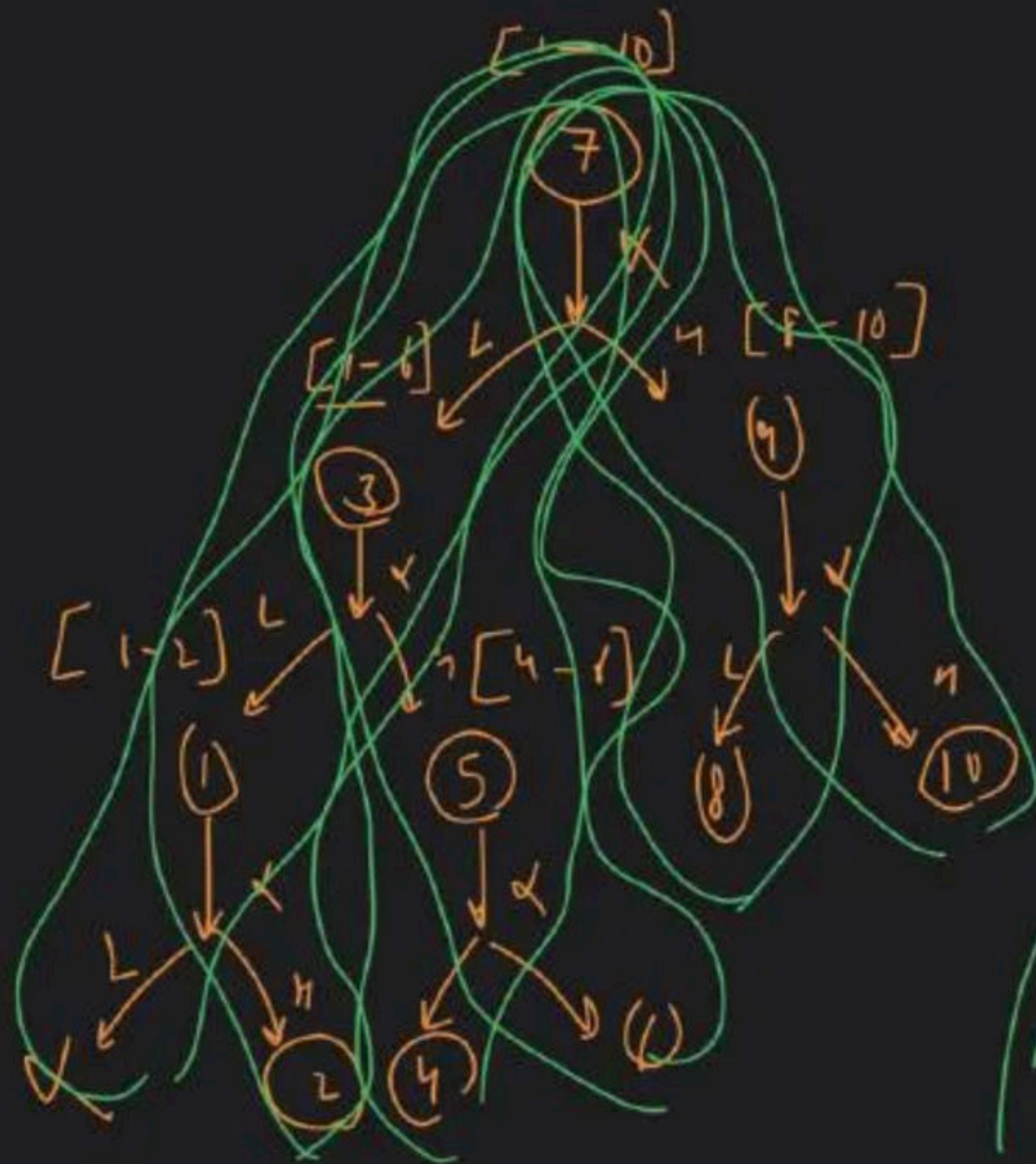
n=5

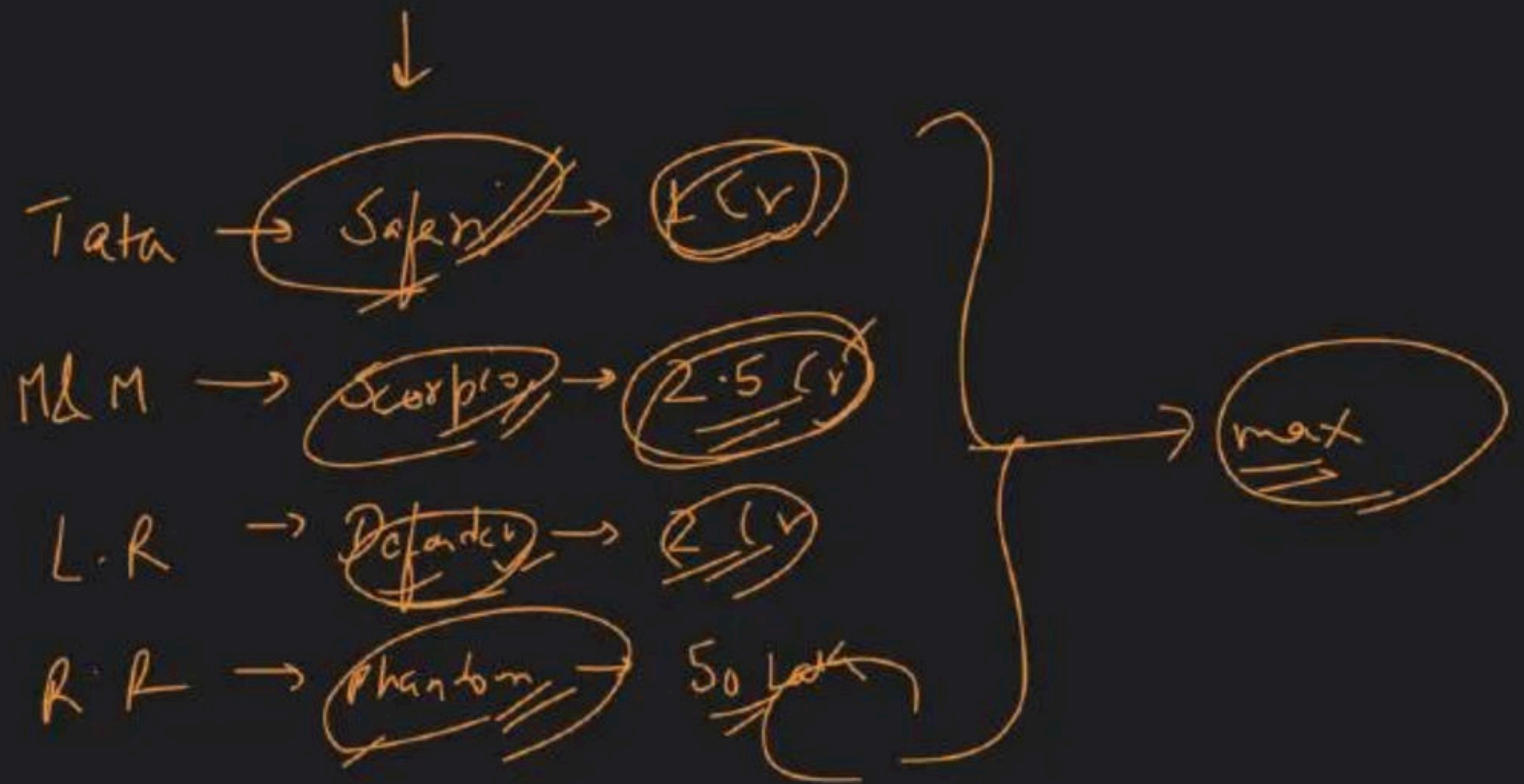
diag

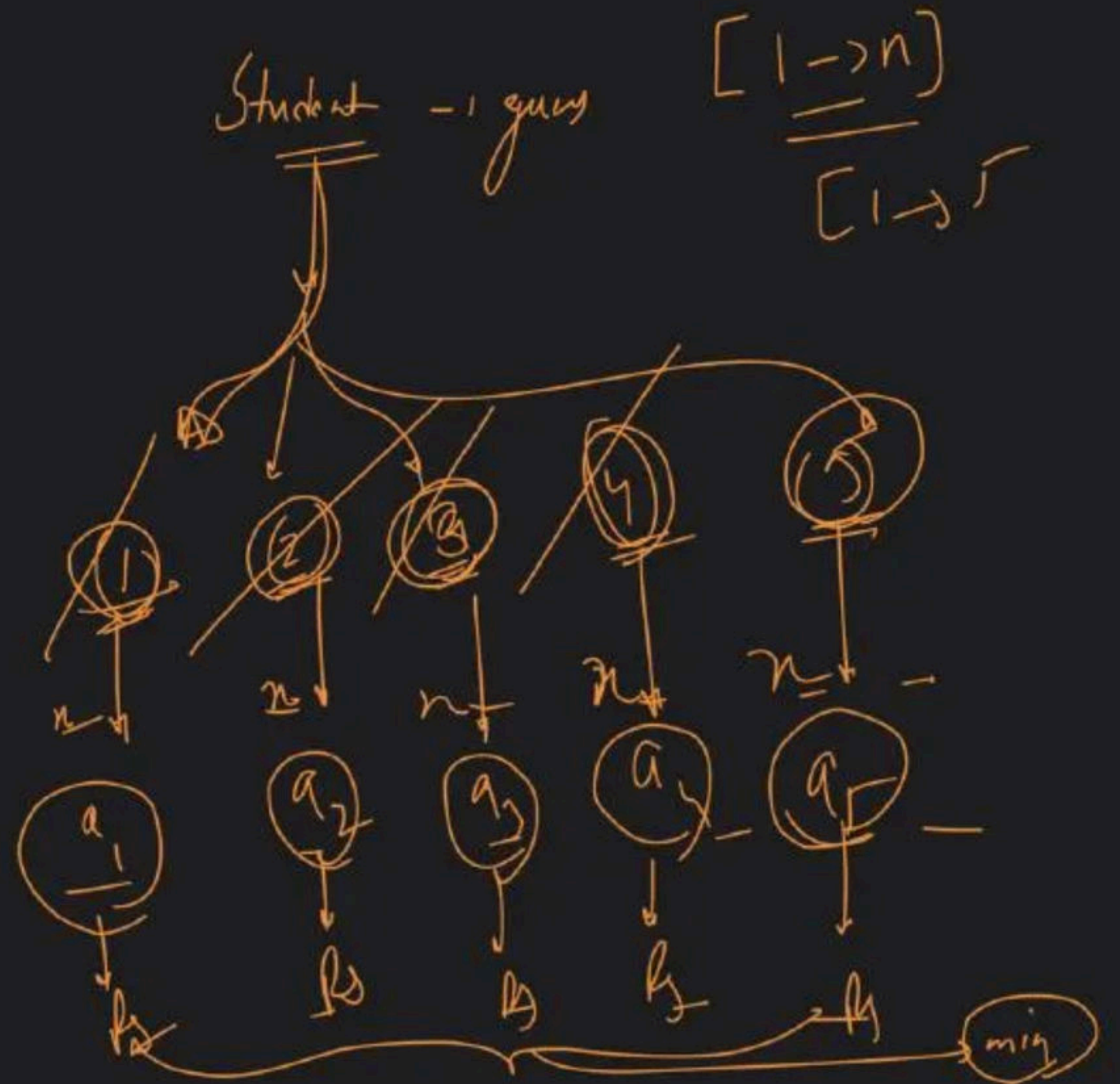
any → number

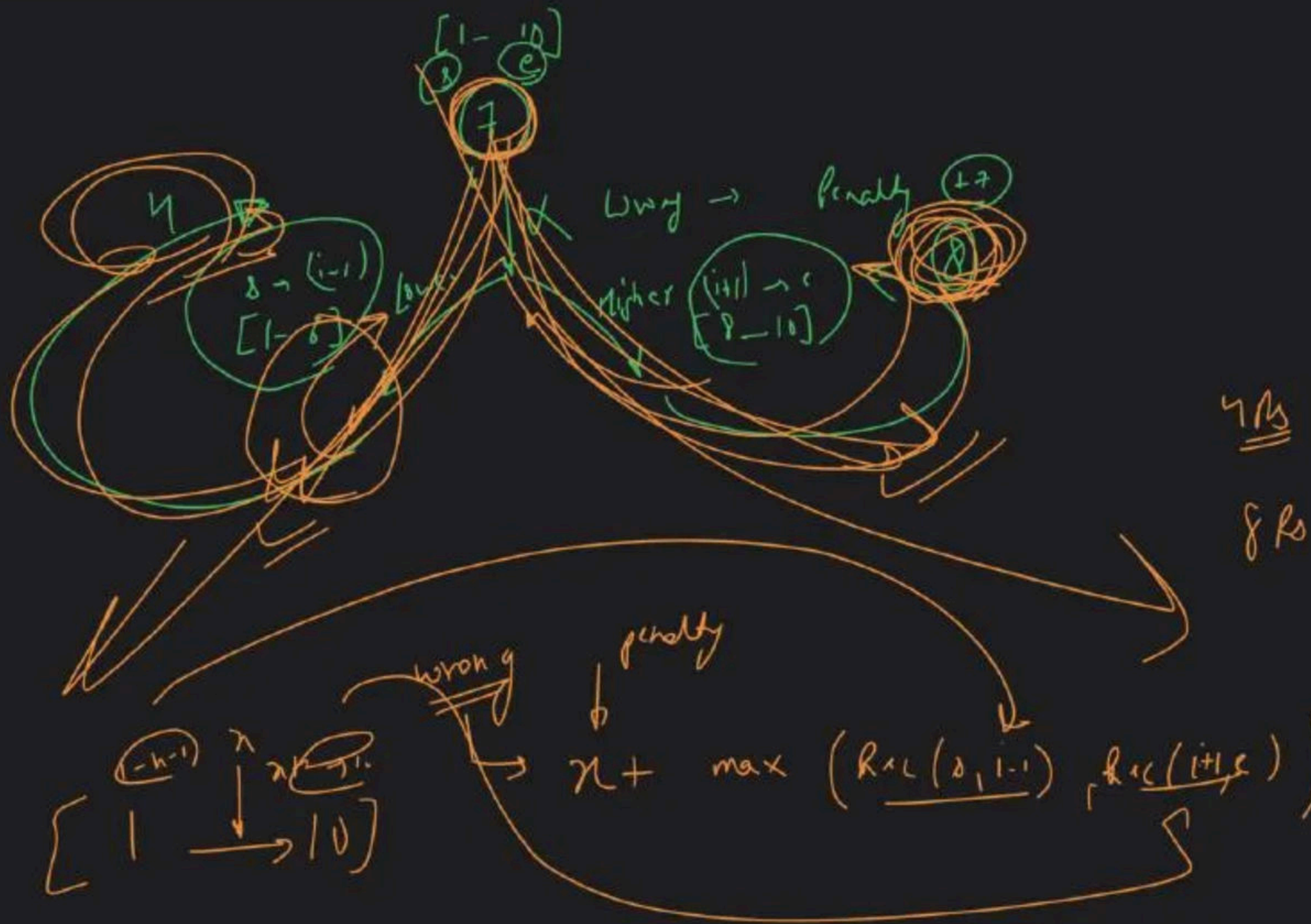


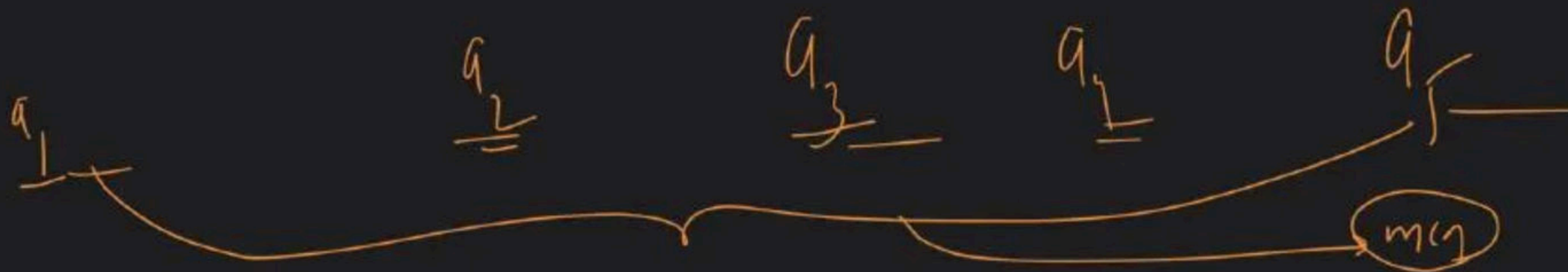
1-10

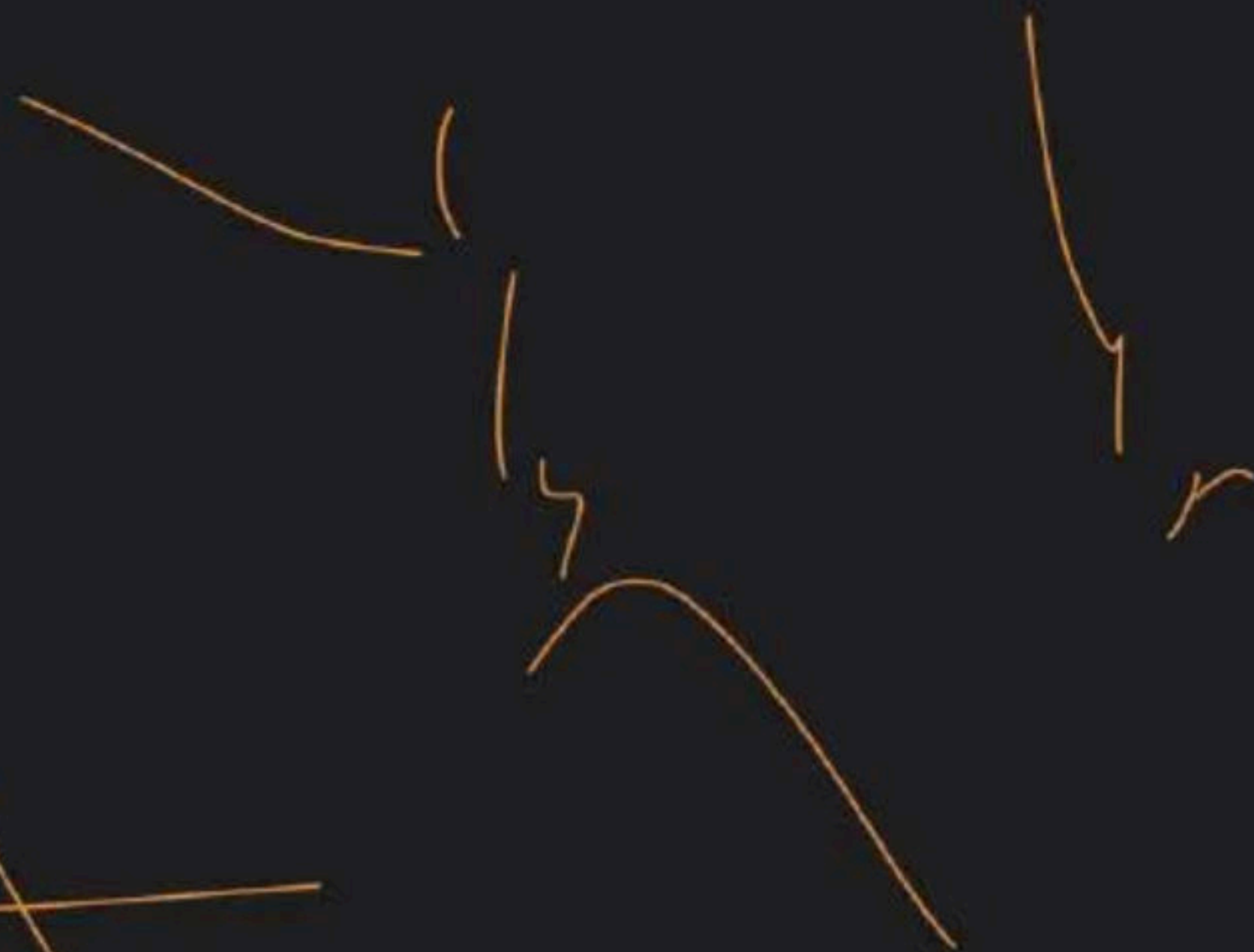




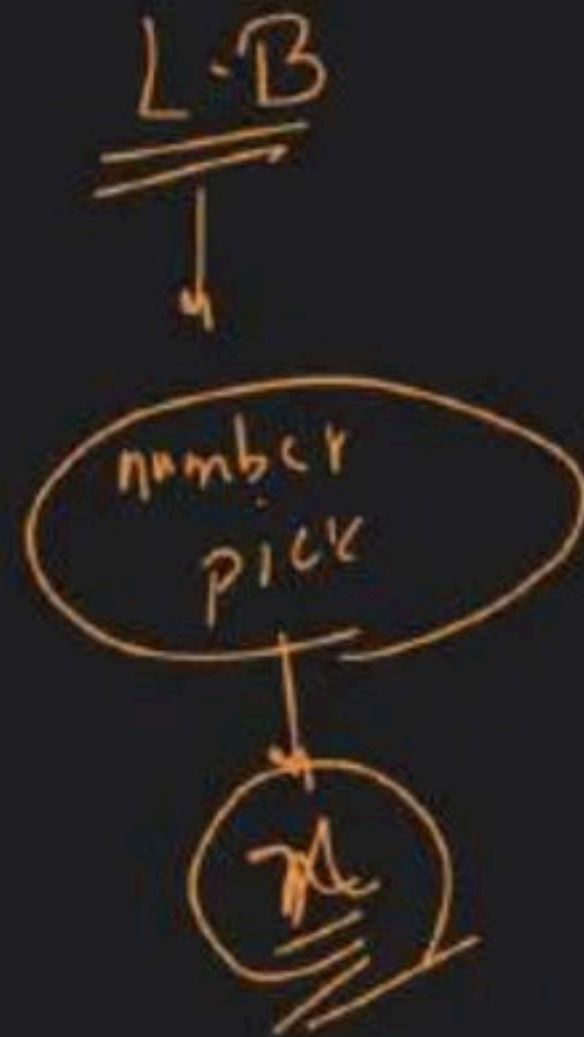




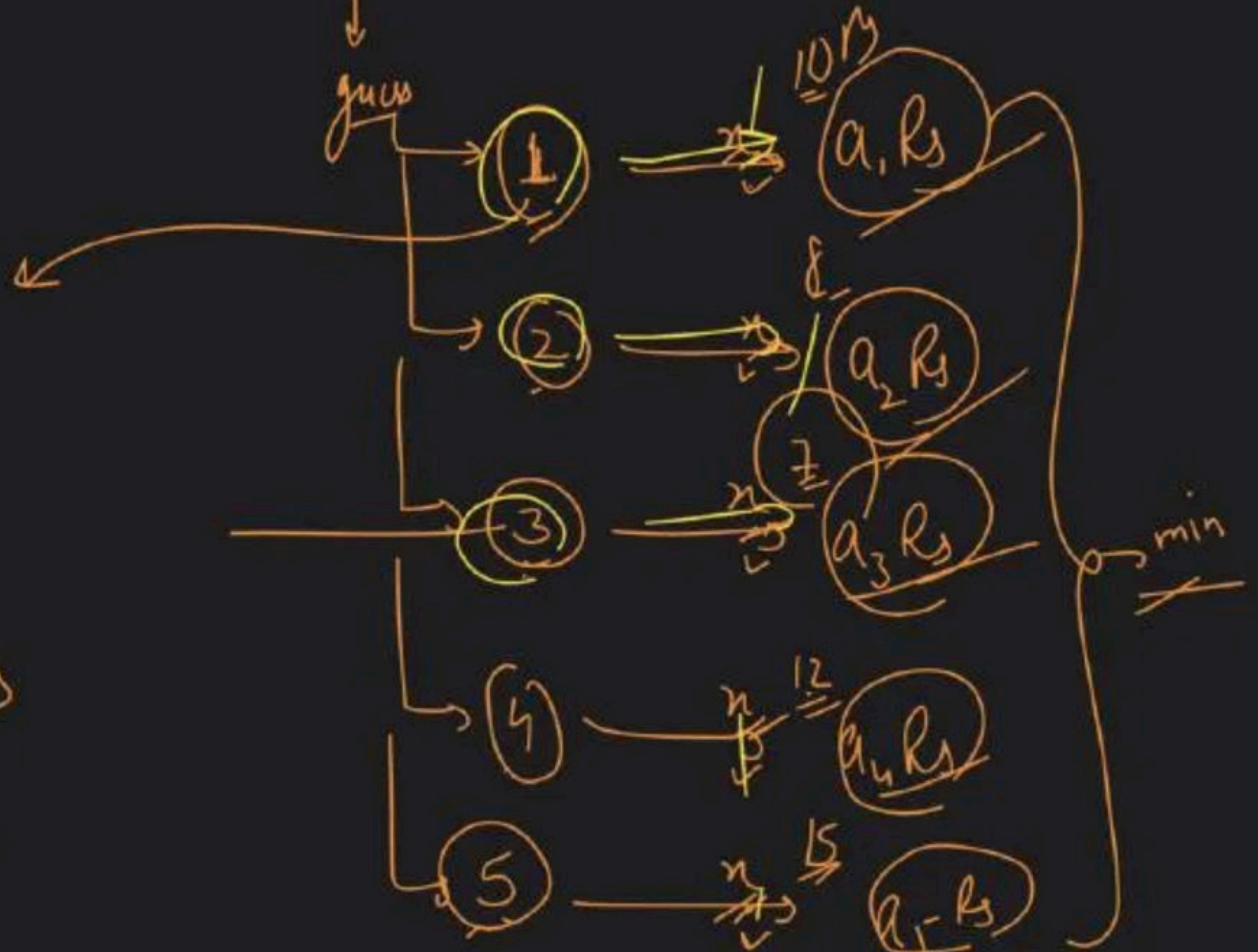


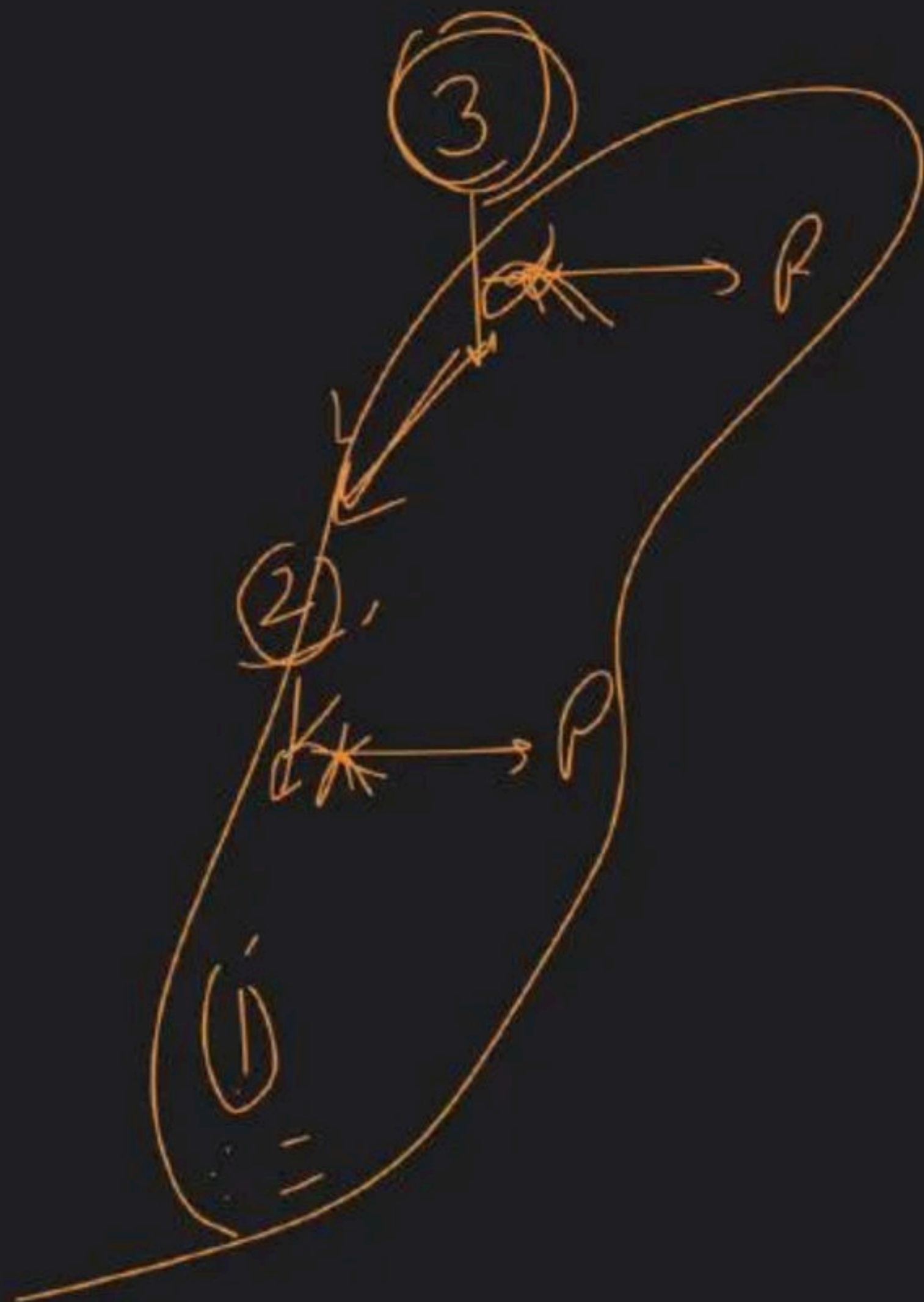


[1 → n]



Student -1 guess



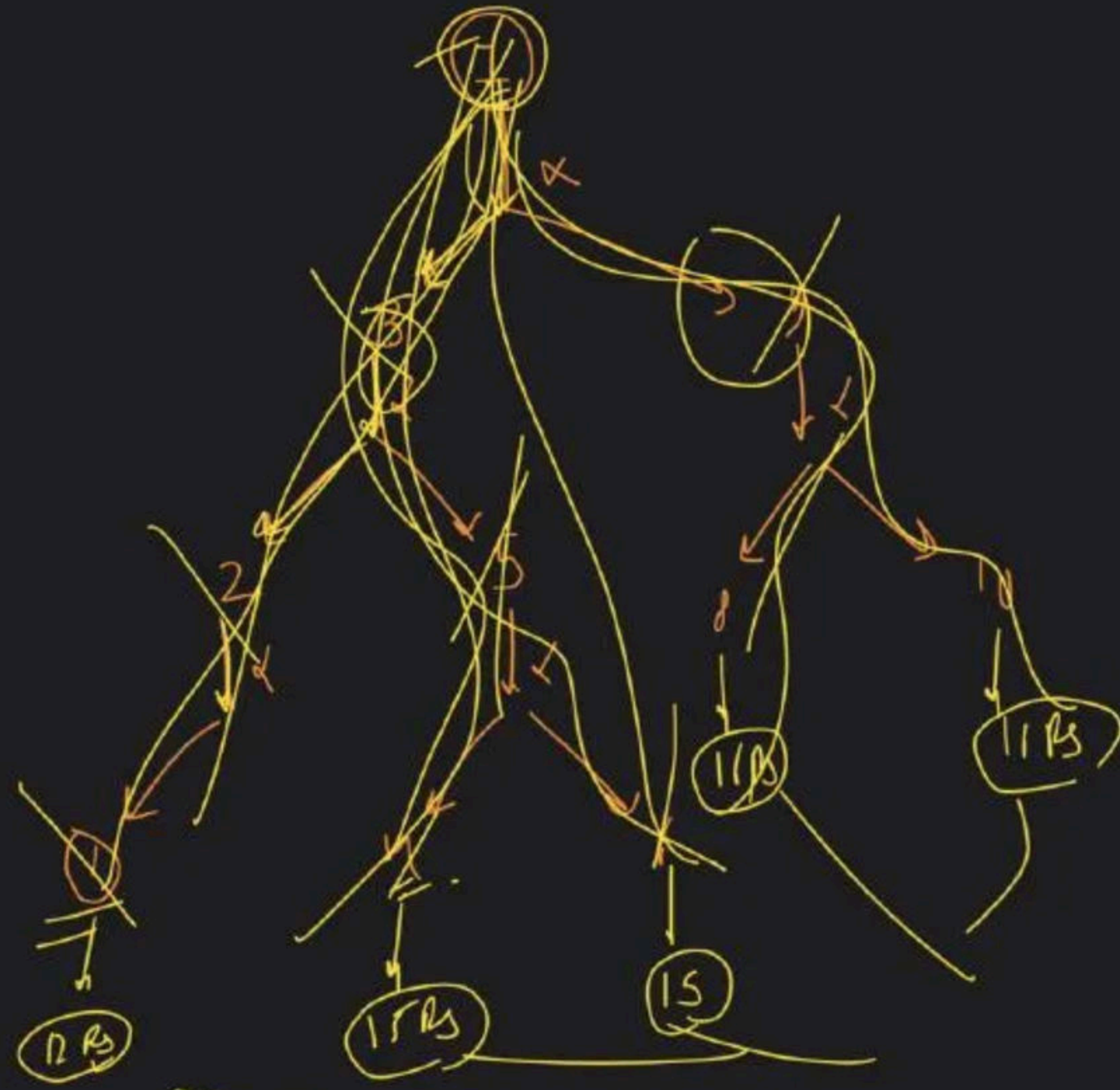


Gamble

- $I \propto \frac{1600}{\alpha}$
 - $II \propto 1600 \alpha$
 - $III \propto 1600 \alpha$
 - $IV \propto 1000 \alpha$
 - $V \propto 1000$
- (11) 11

L.B

1
2
3
4
5
6
7
8
9
10



↓
16 Rs

$$N = [1, N]$$

10

$[1, 10]$



S

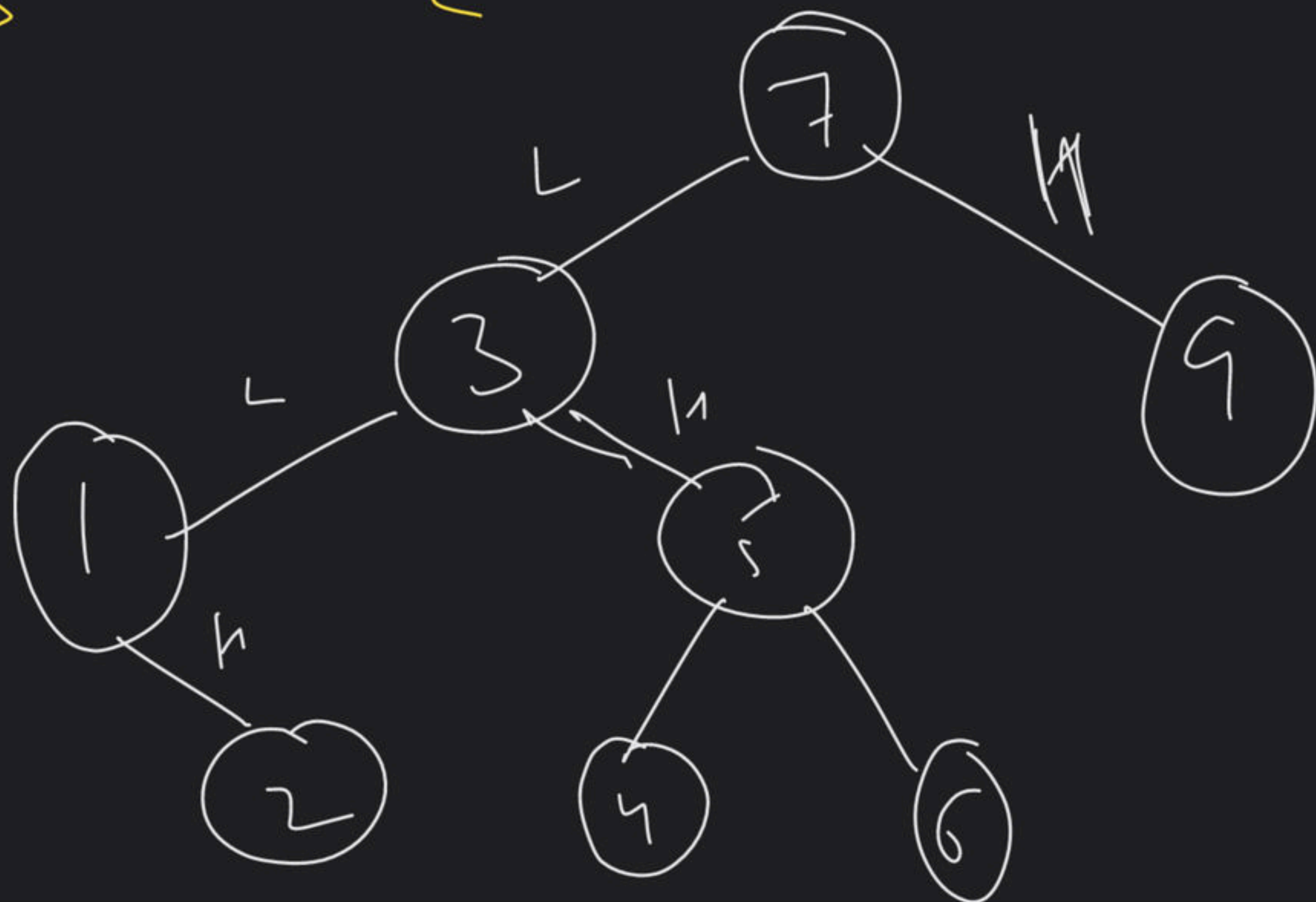
=)

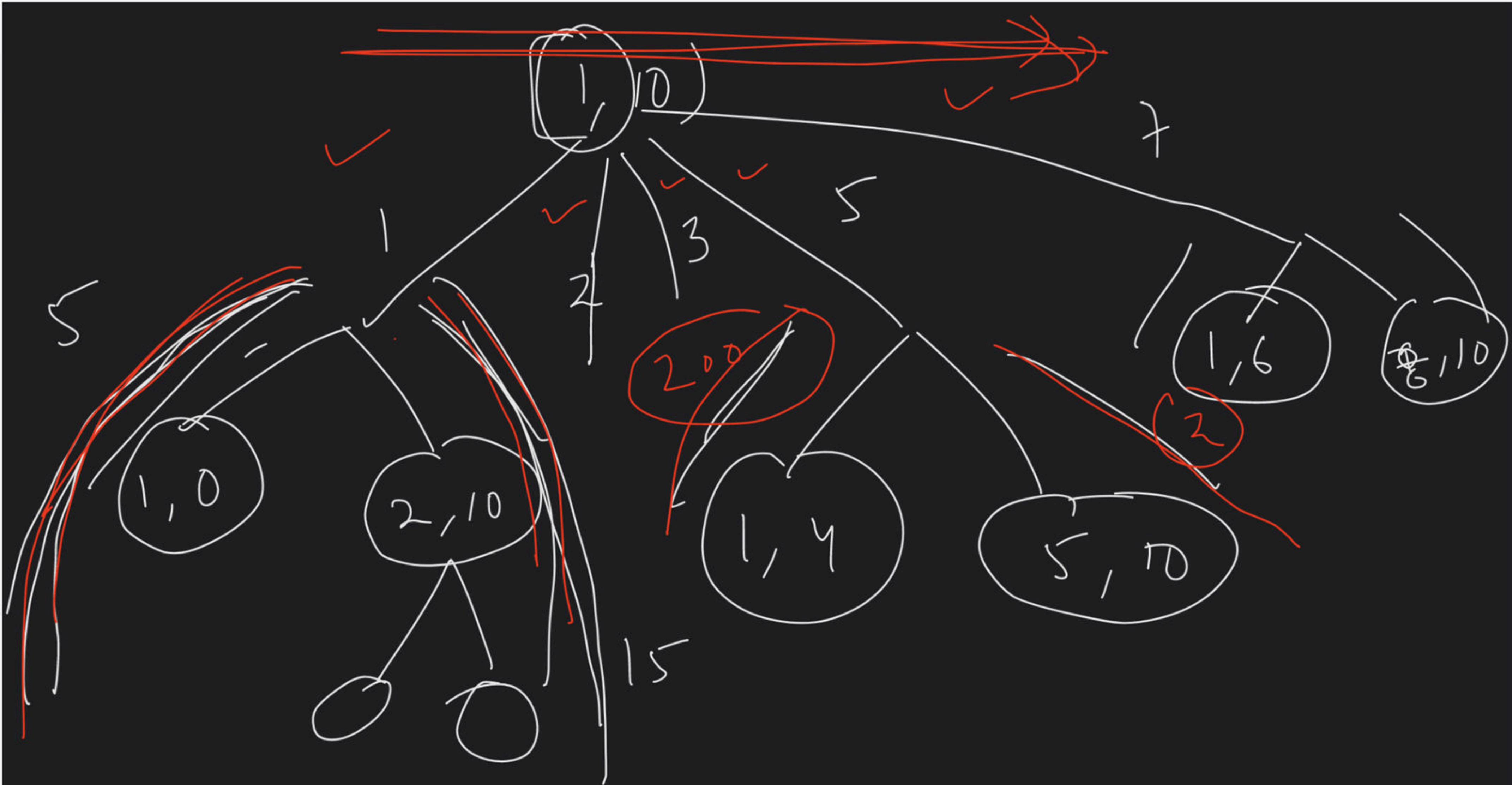
1

2

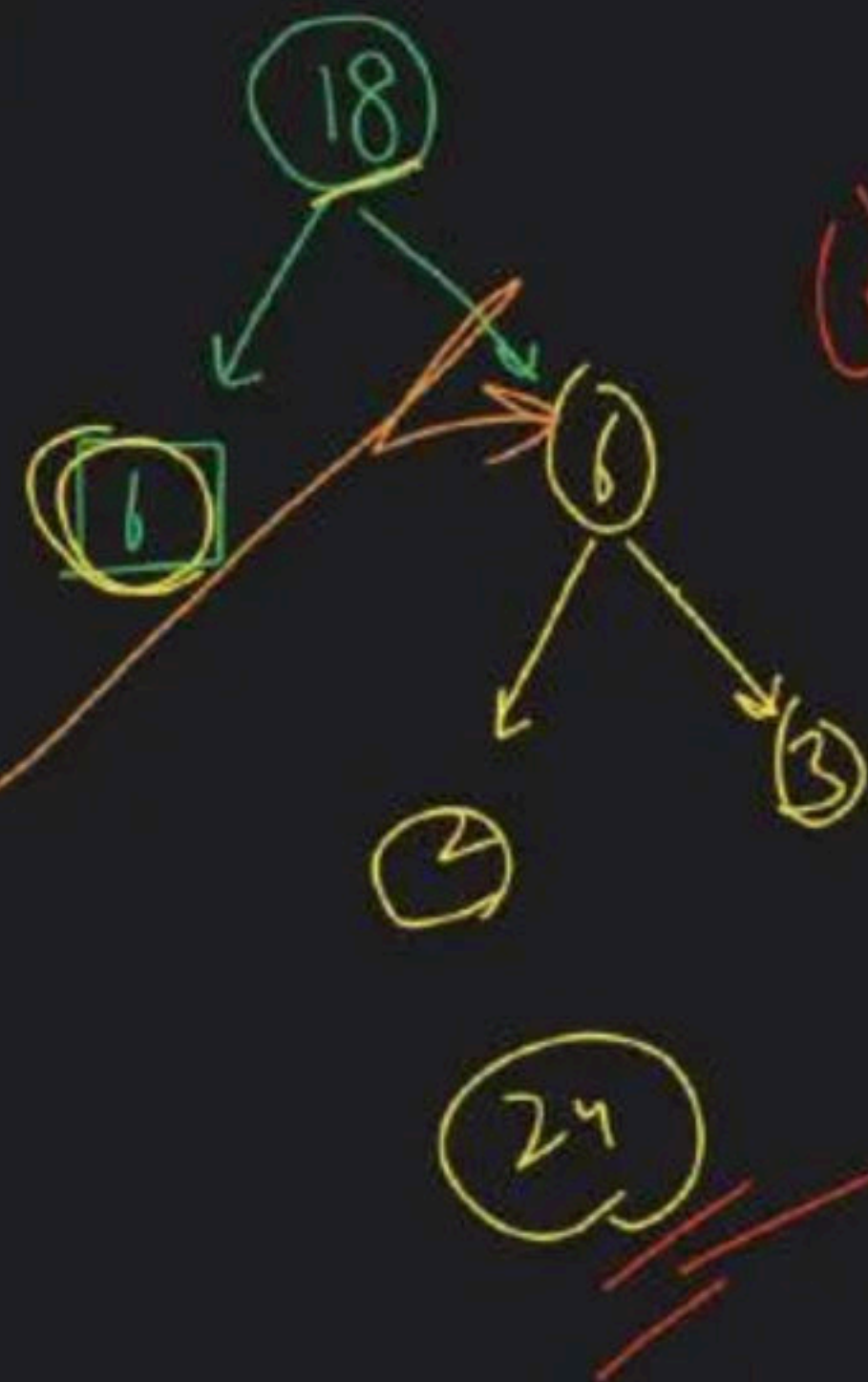
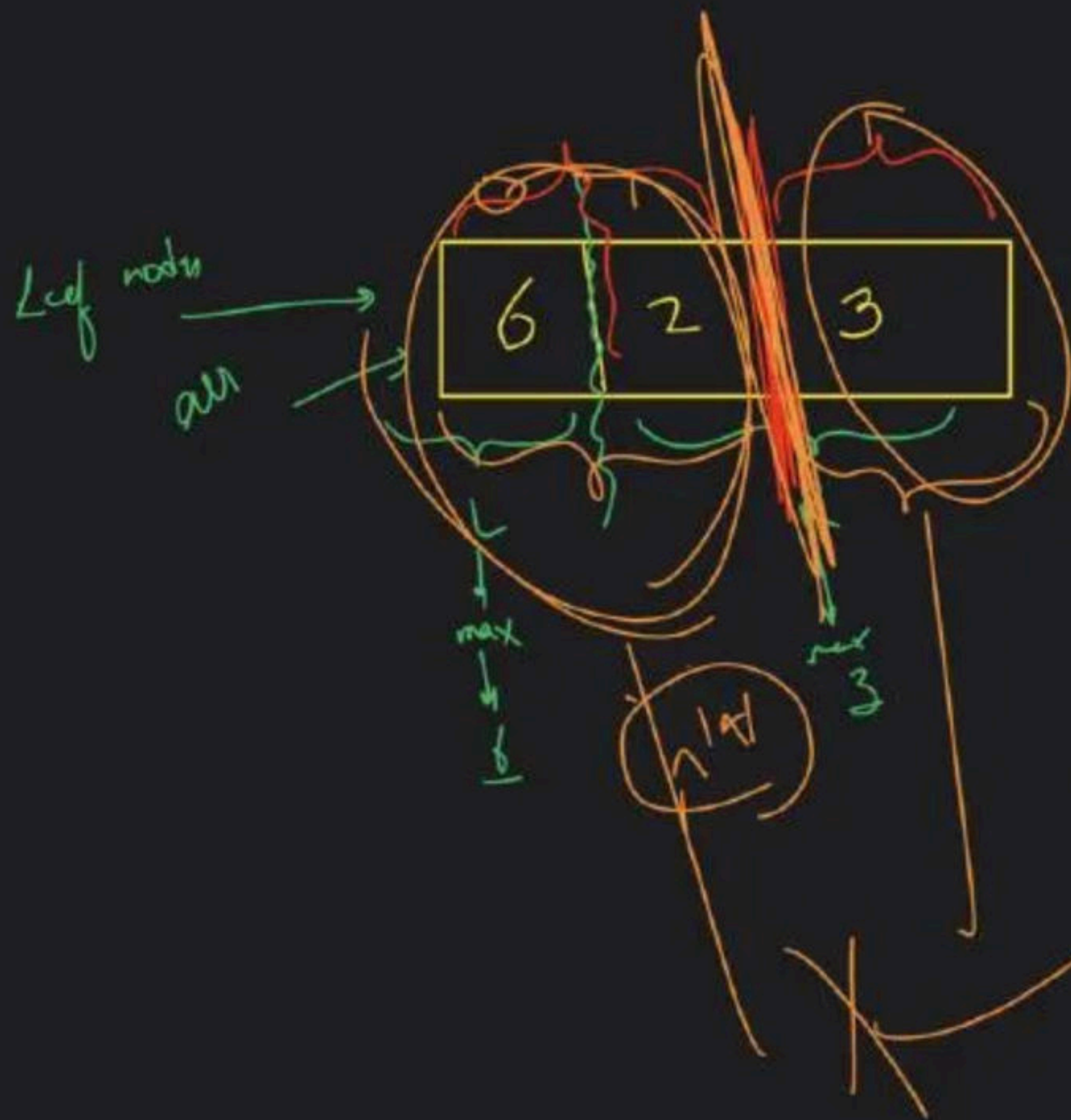
3

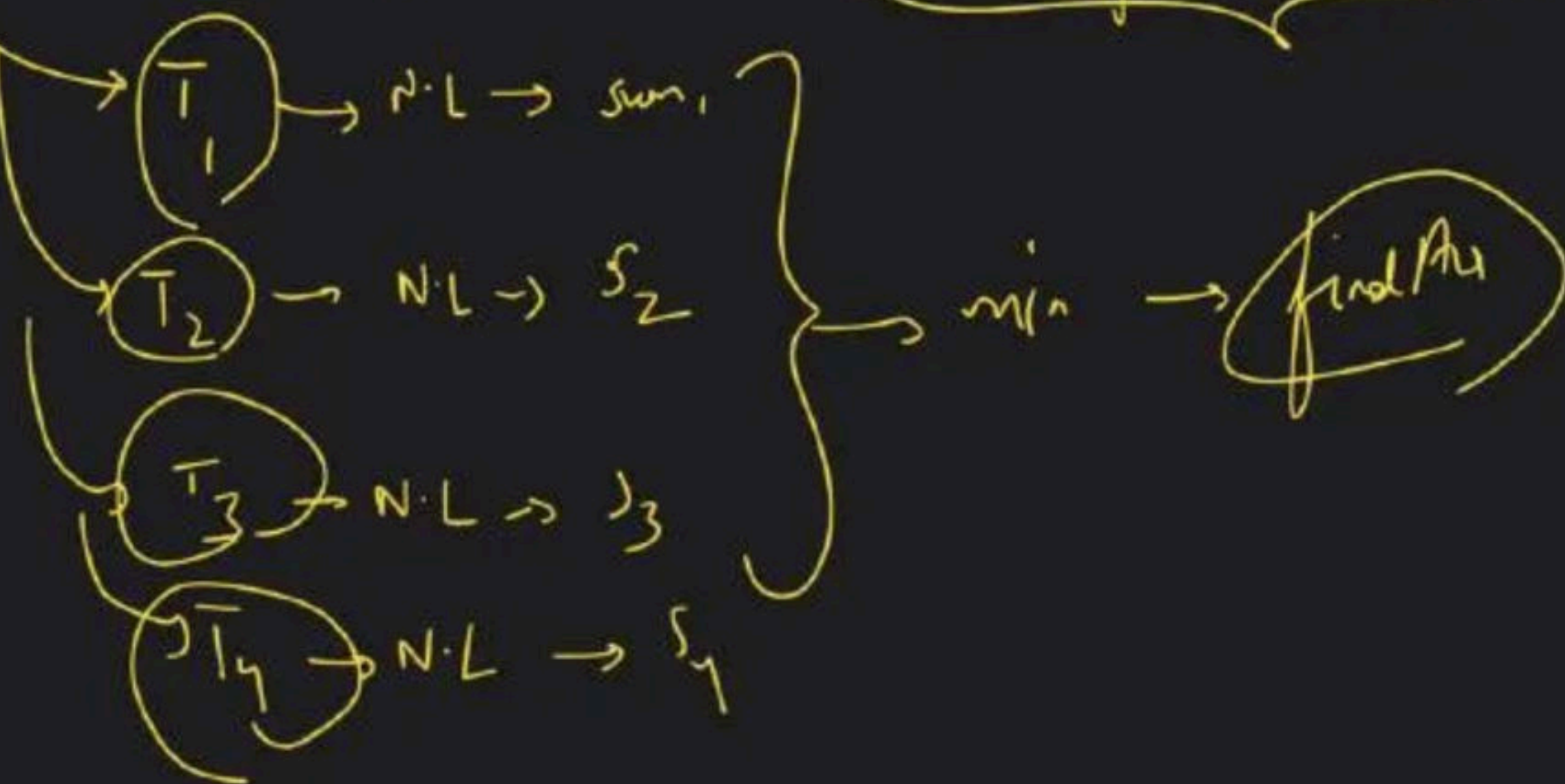
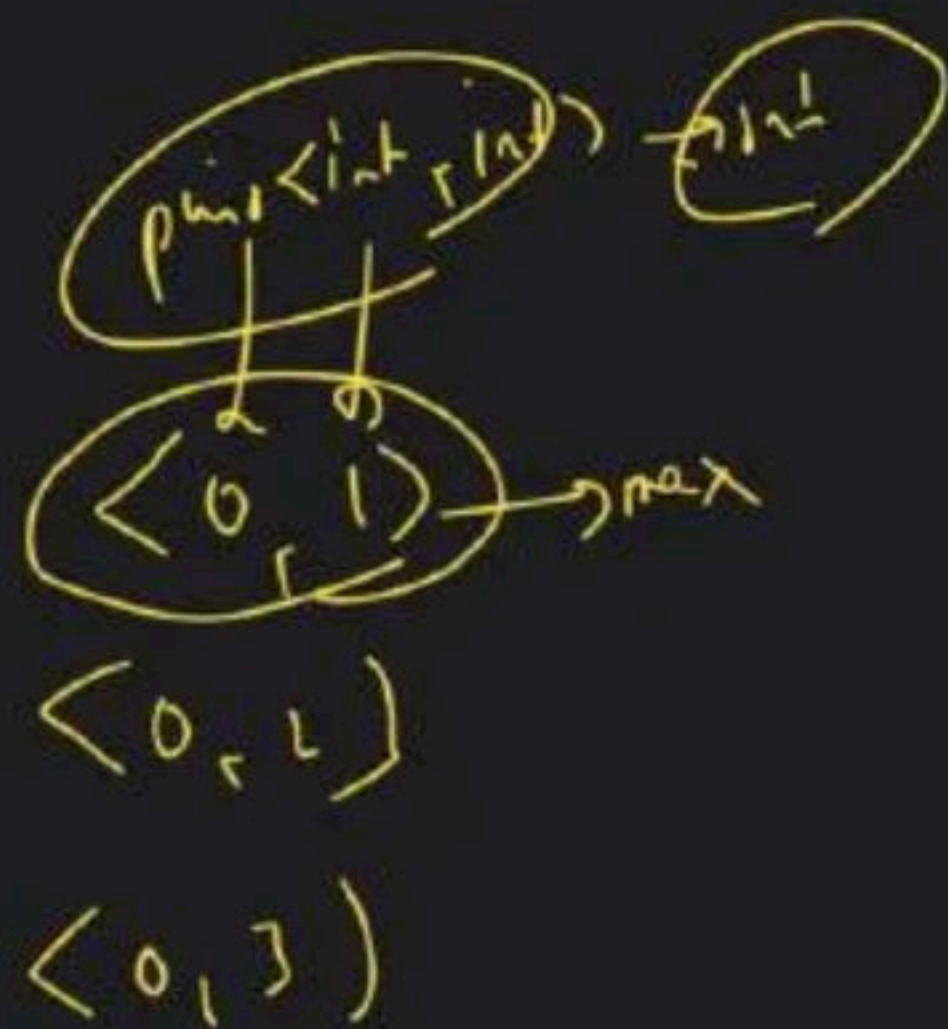
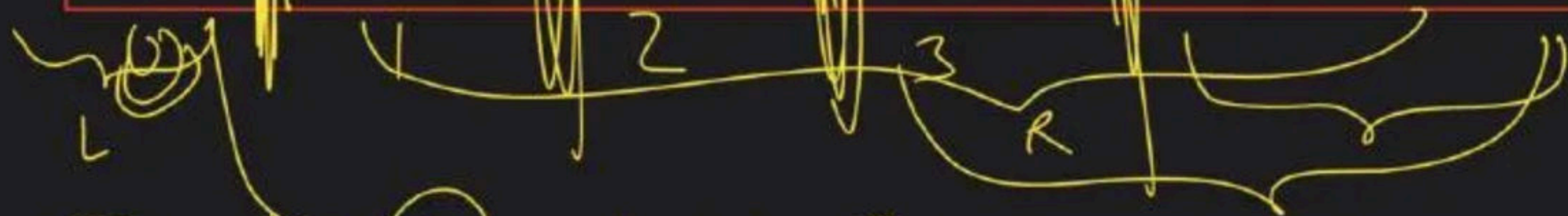
4

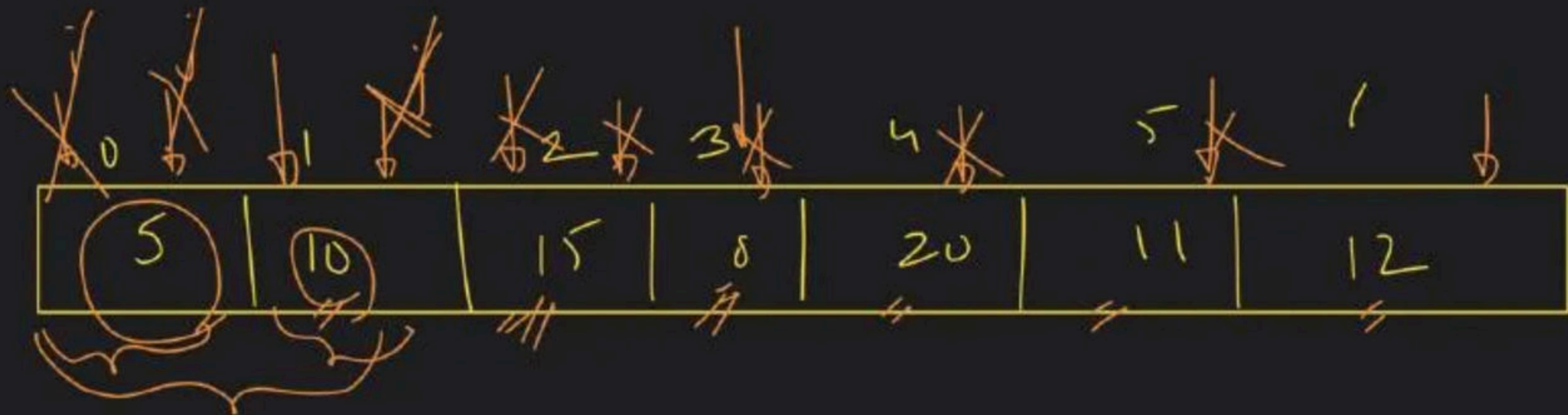




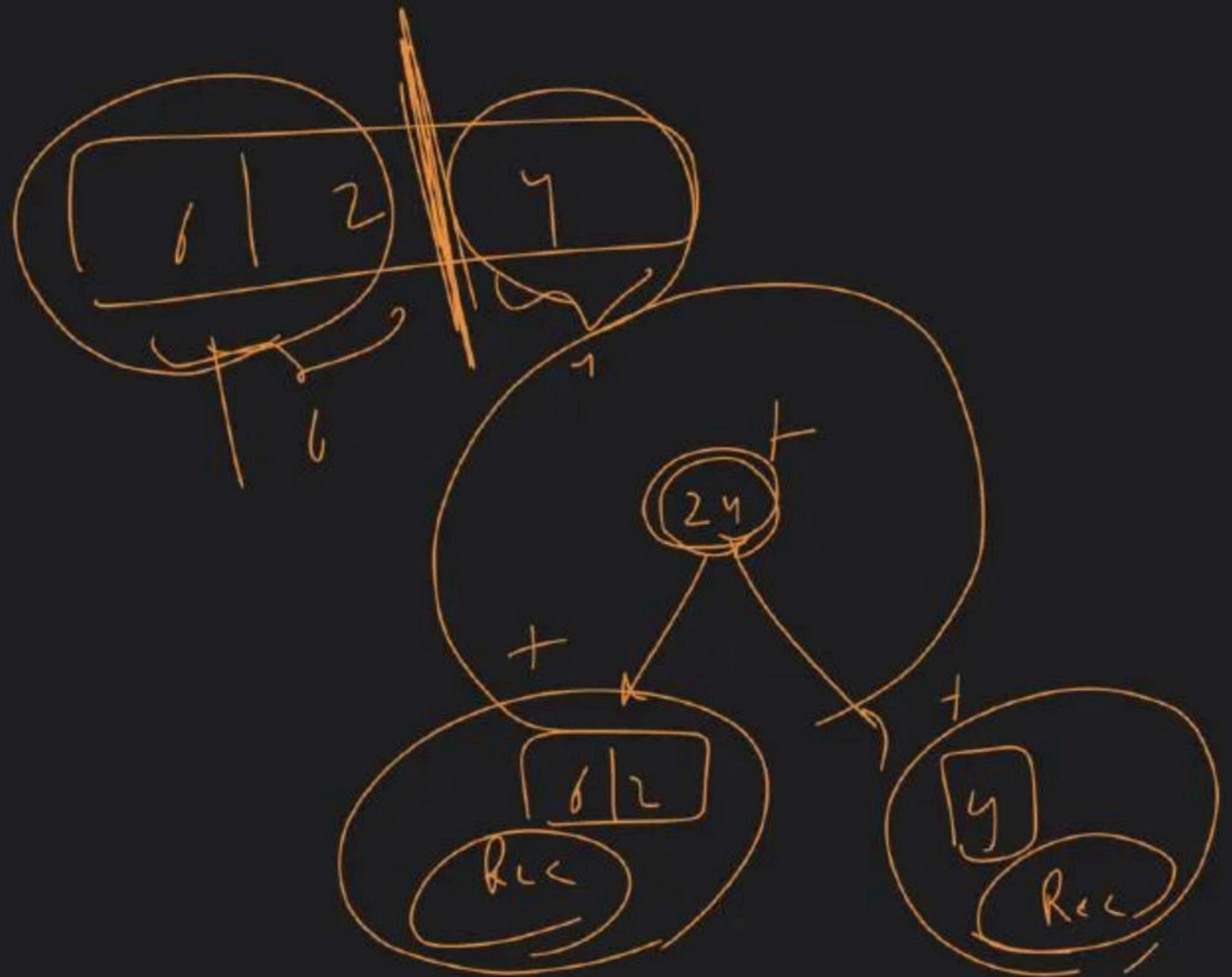
[6, 2, 4]







$\{0, 0\} \rightarrow 5$ $\{0, 1\} \rightarrow 10$ $\{0, 2\} \rightarrow 15$ $\{0, 3\} \rightarrow 15$ $\{0, 4\} \rightarrow 20$	$\{0, 1\} \rightarrow 20$ $\{0, 1\} \rightarrow 20$ $\{1, 1\} \rightarrow 10$ $\{1, 2\} \rightarrow 15$ $\{1, 3\} \rightarrow 15$		
--	---	--	--



Last DP

Ans

Kal → Tuesday

DP Class - 6

Special class

Dp $\rightarrow$ inc/exc ✓

$\rightarrow$ with strings

$\rightarrow$ with BS

$\rightarrow$ all possibility

$\rightarrow$ tree

$\rightarrow$ merge interval

$\rightarrow$ pattern matching

~~DP~~ LIS

→ wildcard pattern matching

str →

"babbar"

pattern →

"b?r*"

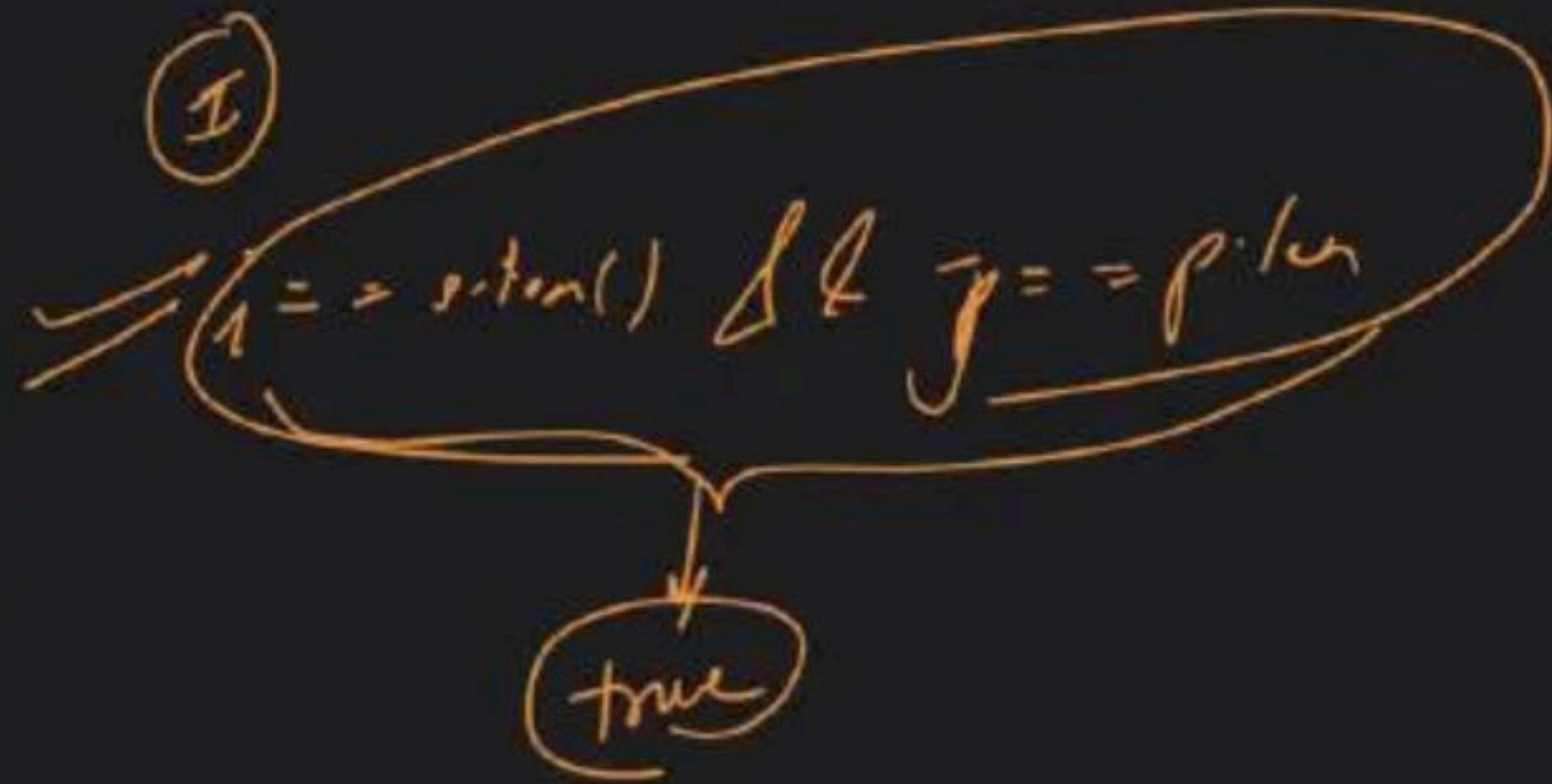
} True

? → any single character

{ a, b, c, ... }

* → matches any character
any no of character

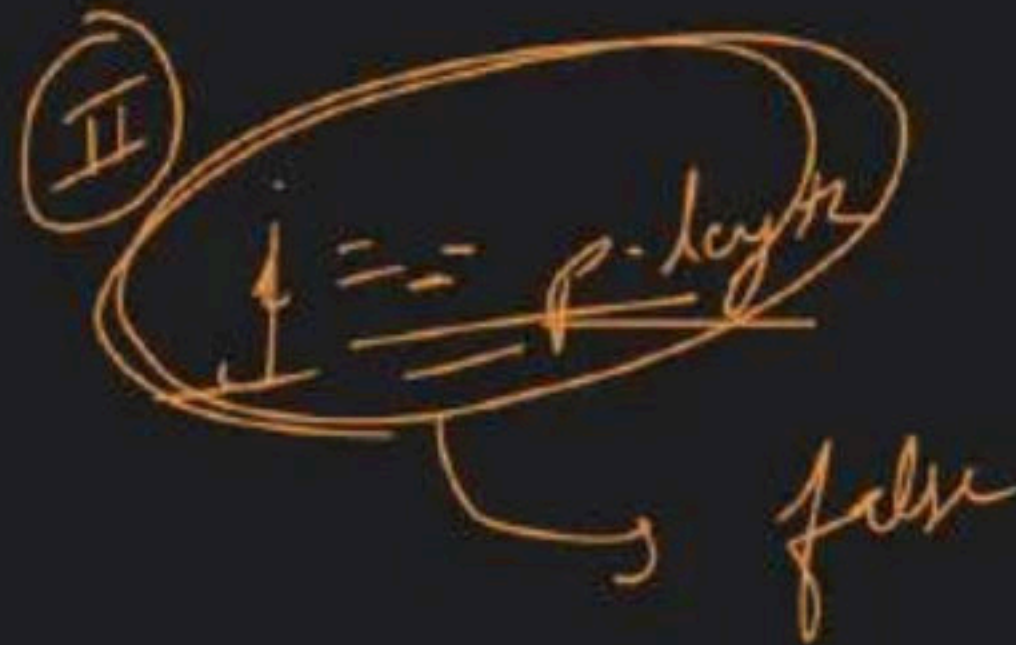
$s \rightarrow$ "babbar"
 $p \rightarrow$ "b**a**bbar"



?

★

$s =$ "babbar"
 $p =$ "ba**rr**bar"



s $\Rightarrow$

~~babbar~~



p $\rightarrow$

~~babbaring~~

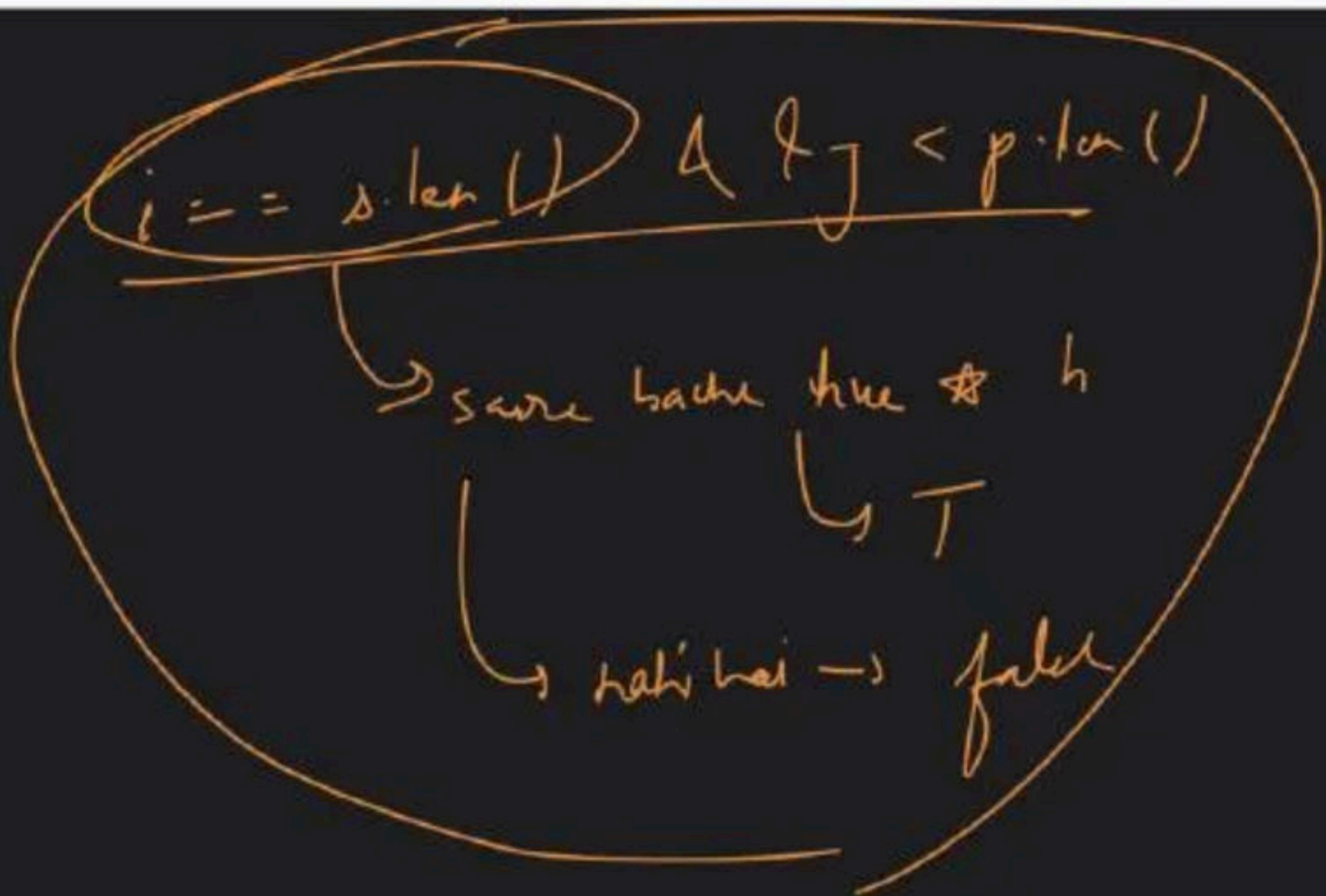
s $\rightarrow$

~~babbar~~

p $\rightarrow$

~~babbar~~

~~babbar~~



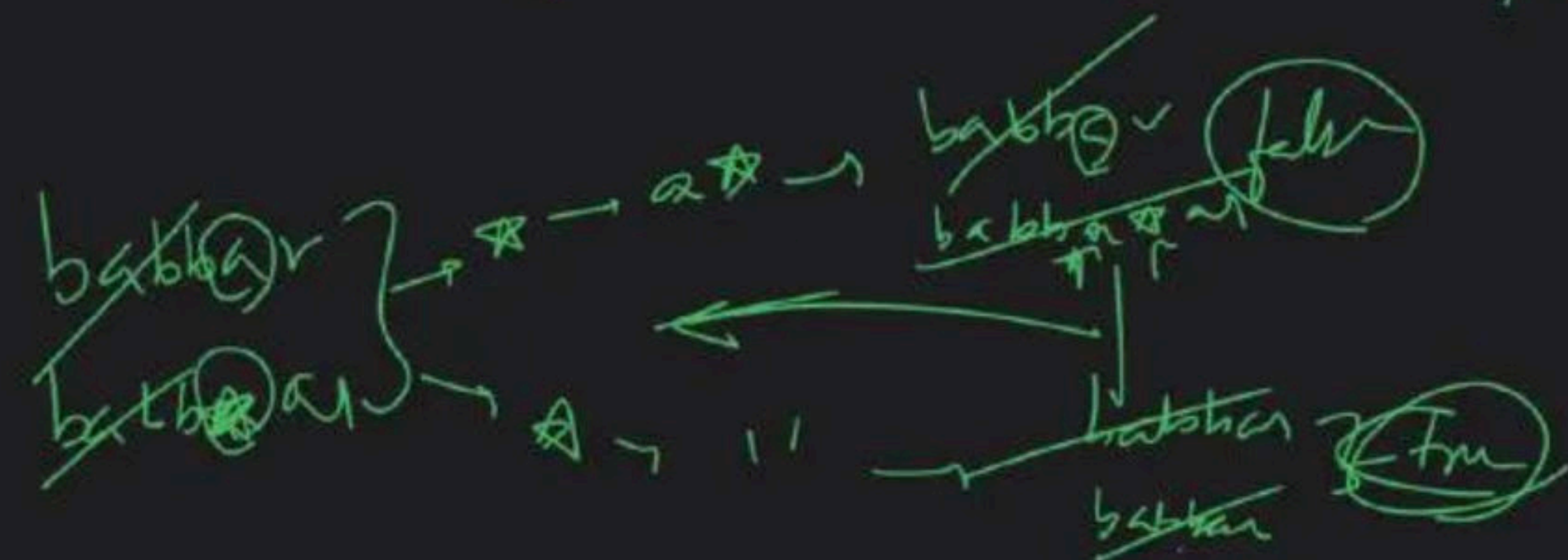
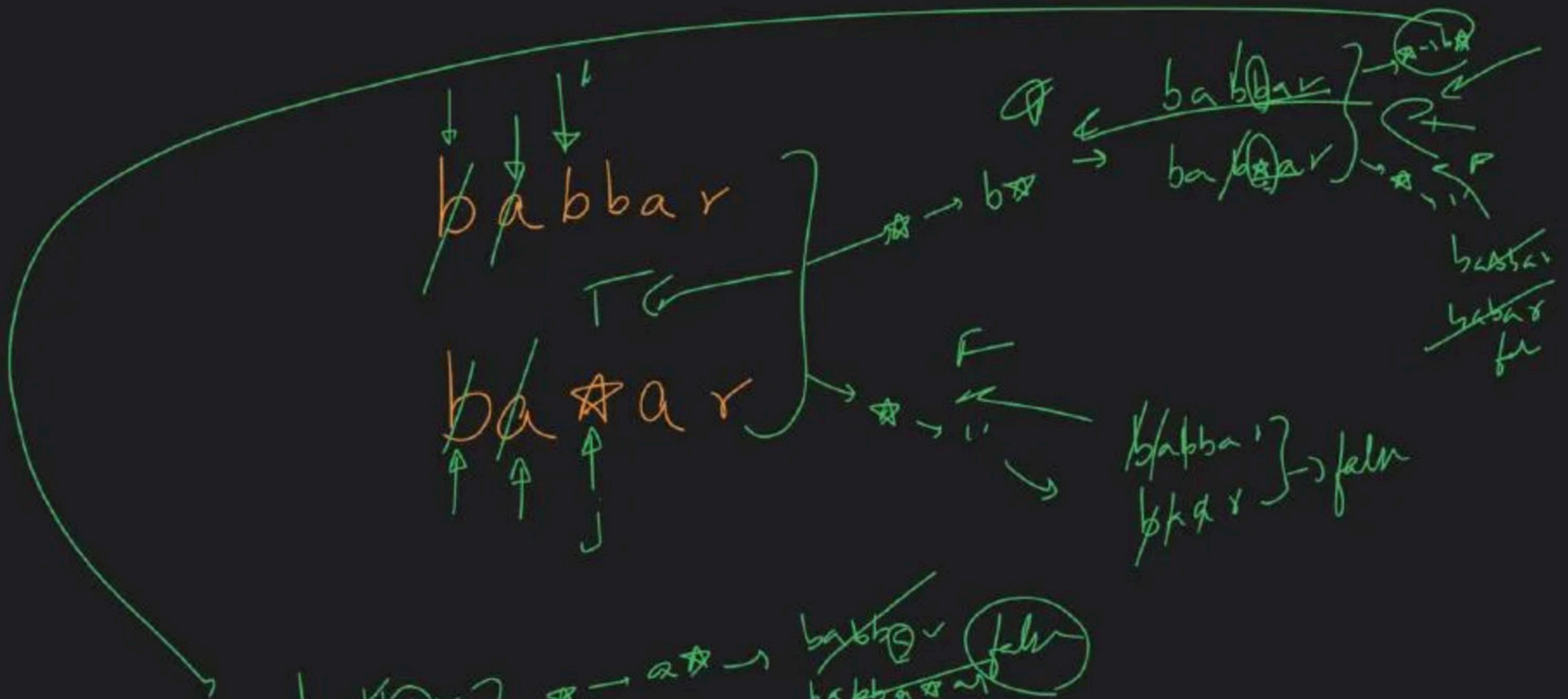
IV

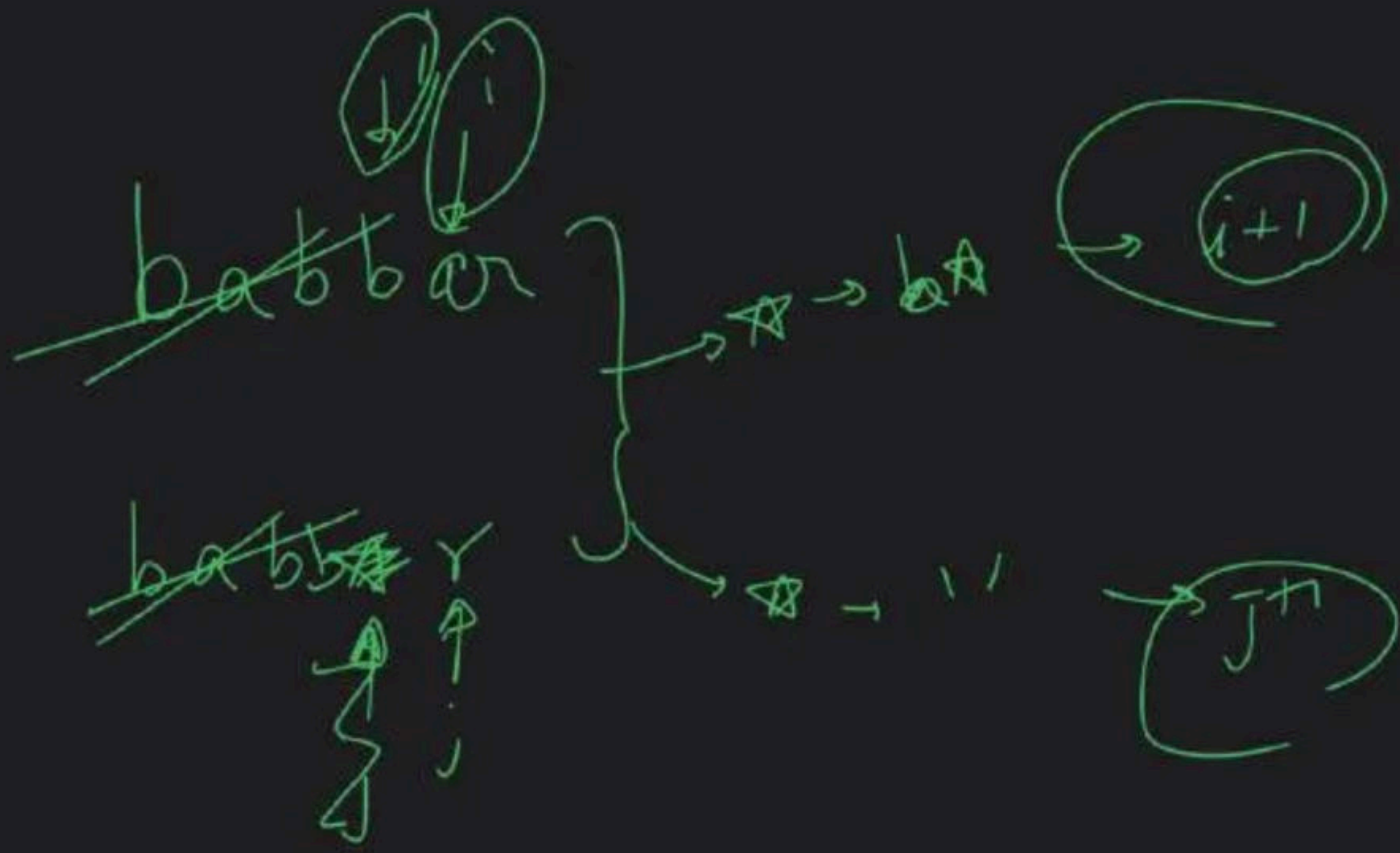
logic

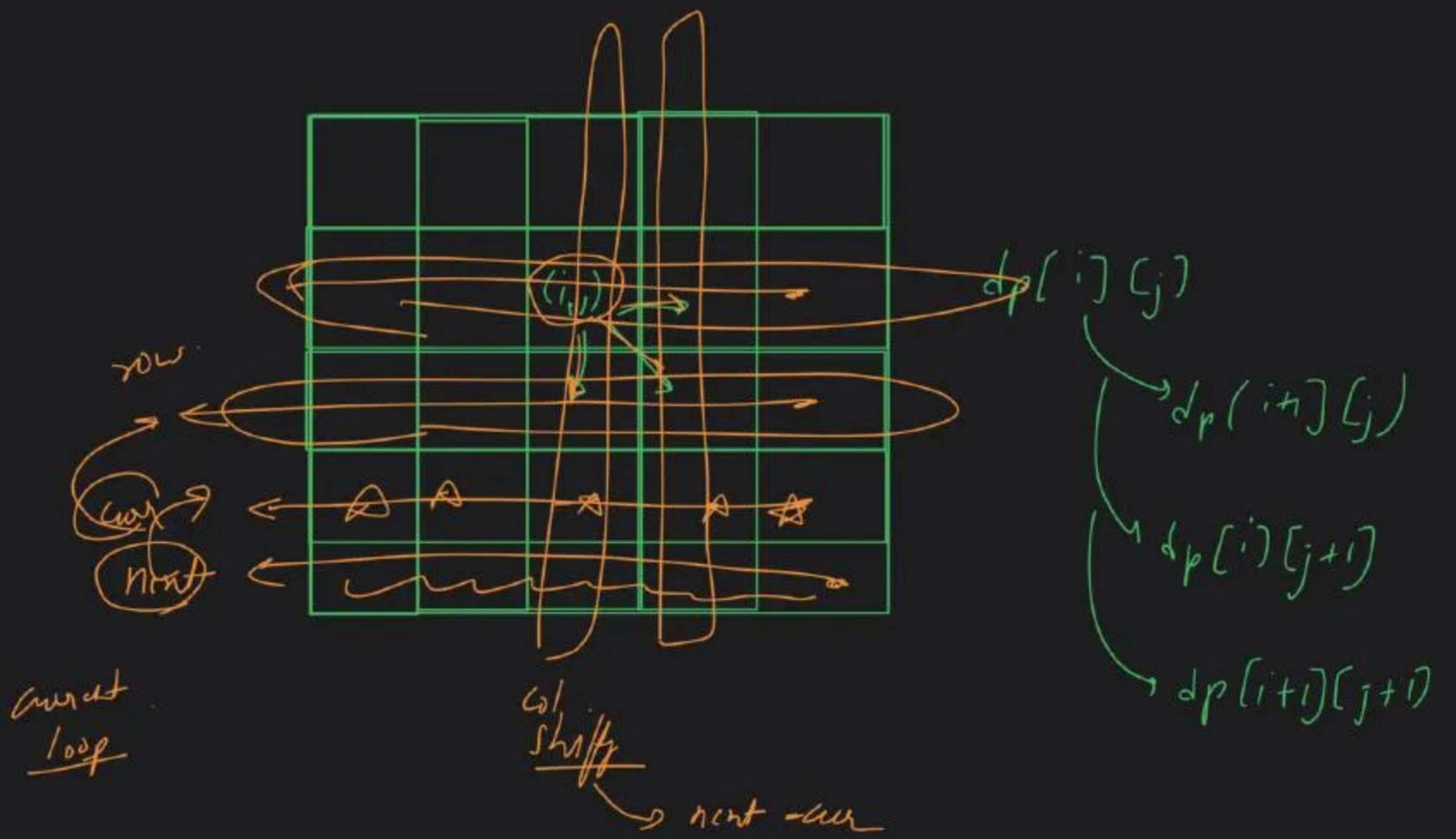
Match

no match

~~s~~ $\rightarrow$ false







$S \rightarrow$

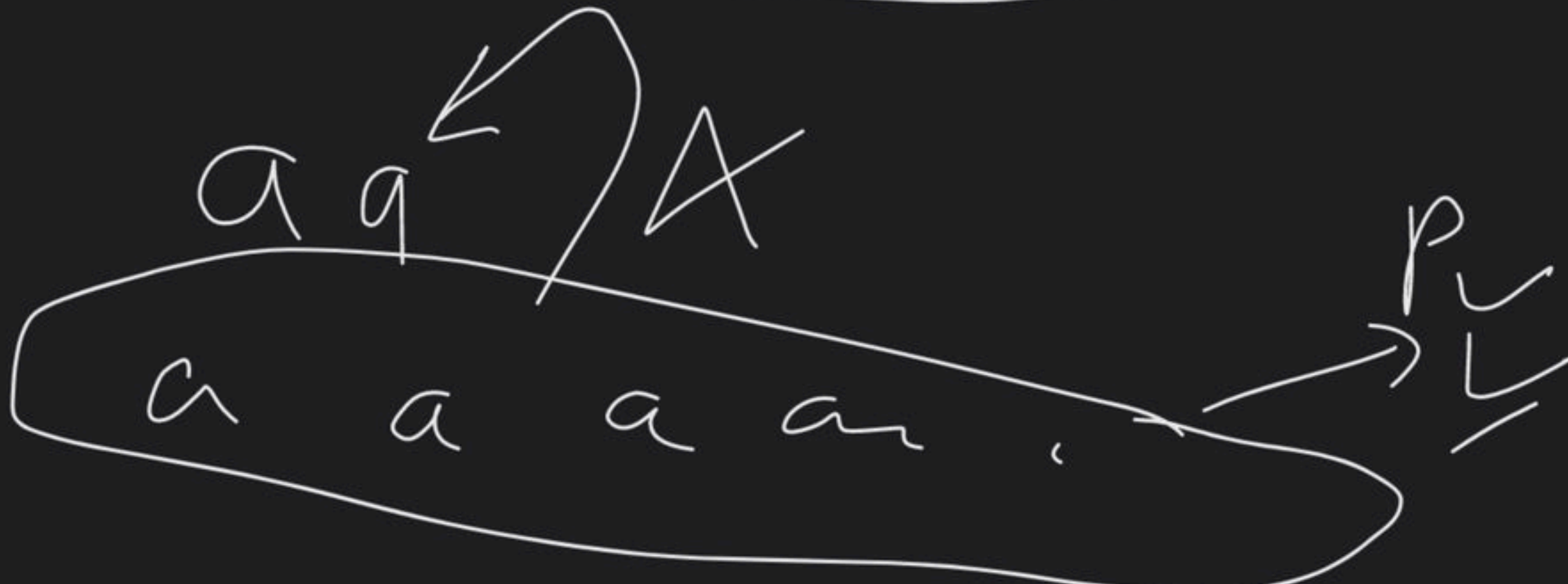
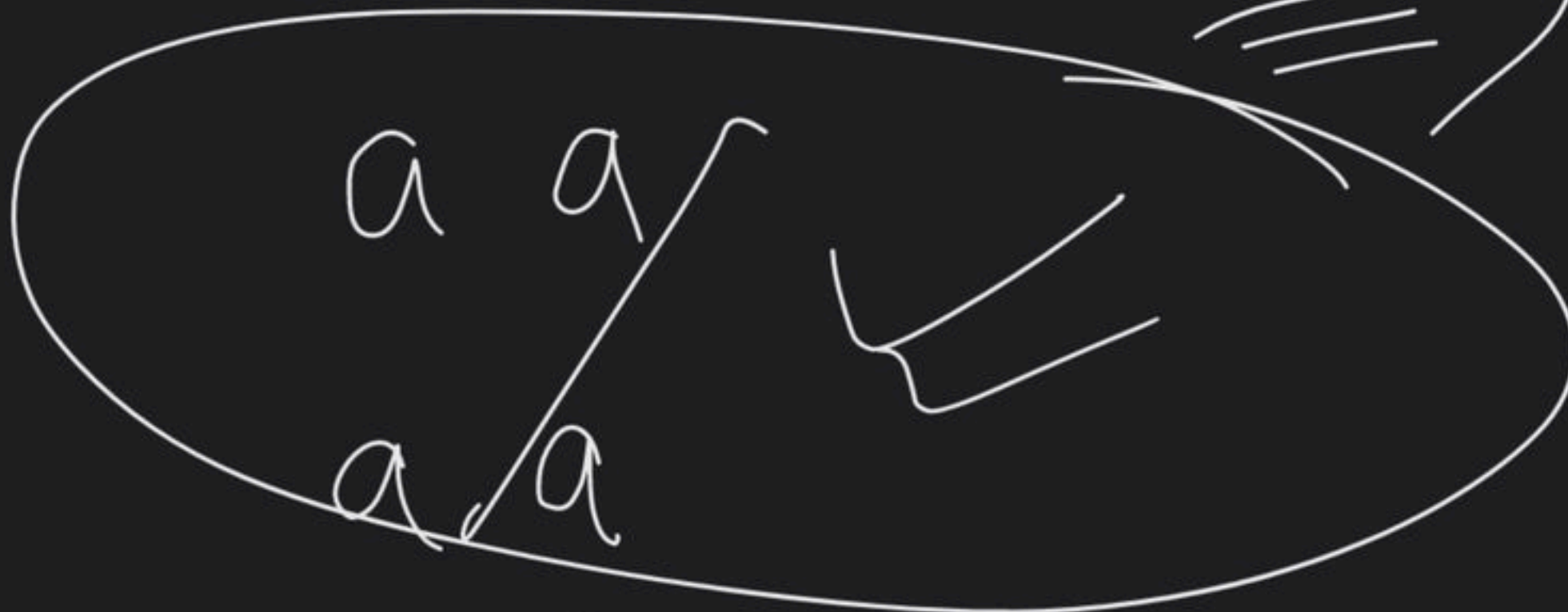
P_{-1}



$a a _$

$a q$

a

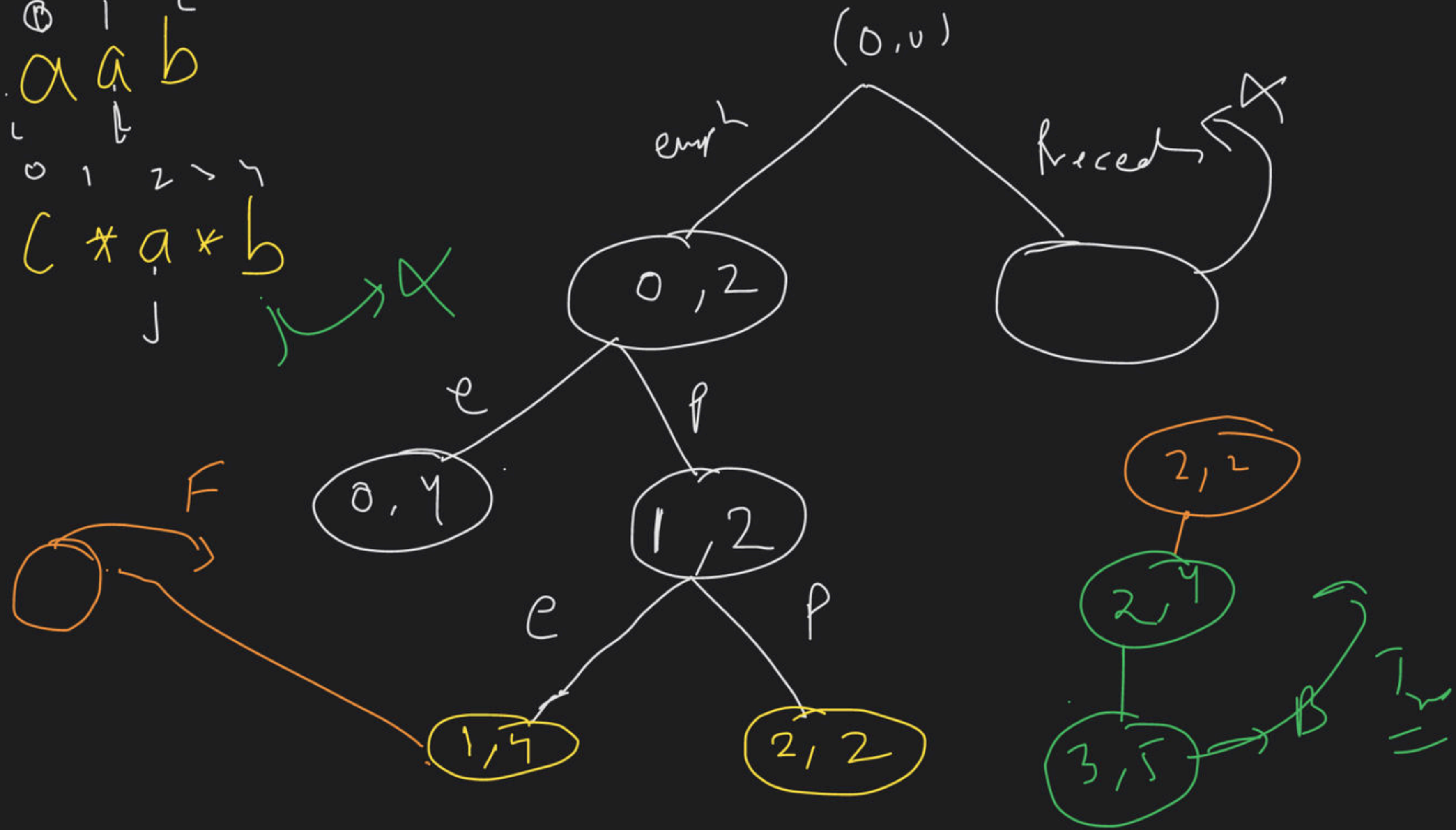


$S \Rightarrow$

	0	1	2
	a	a	b
	0	1	2

$P \Rightarrow$

	0	1	2	3
	c	*	a	*
			b	



0 1 2 3

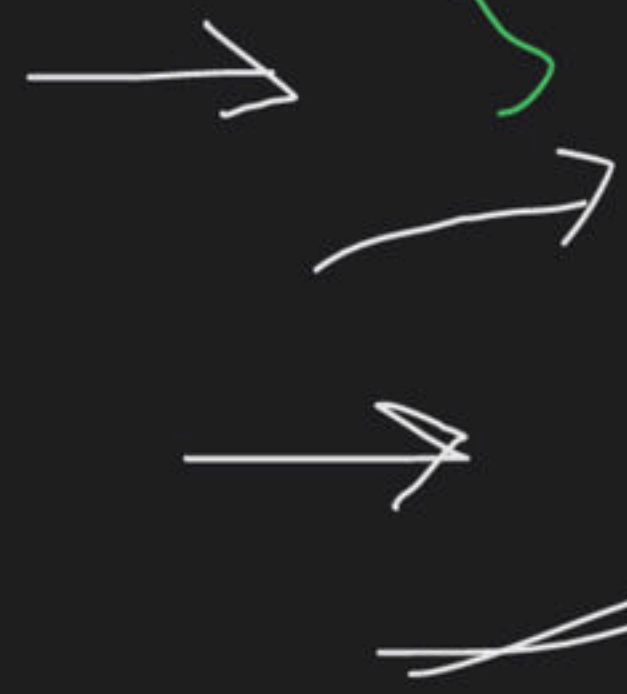
0

1

2

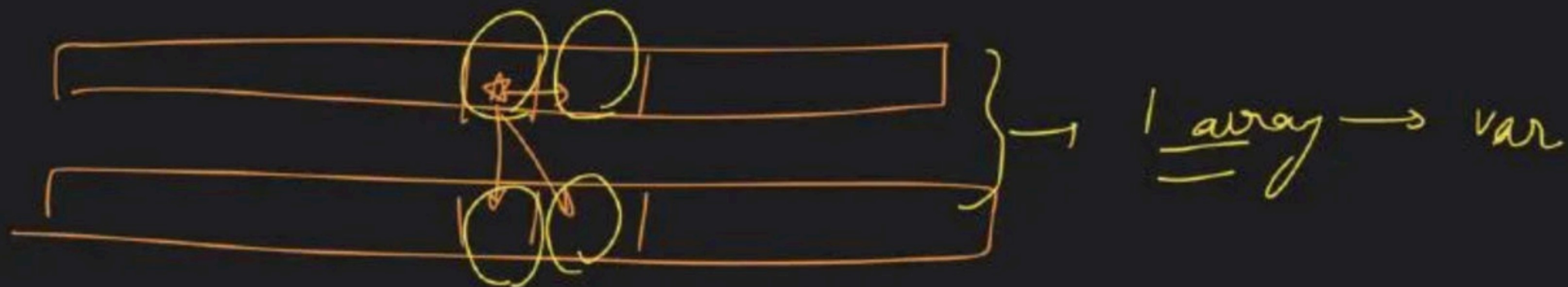
3

0	0	0	0	X
1	0	0	0	X
2	0	0	0	X
3	0	0	1	X

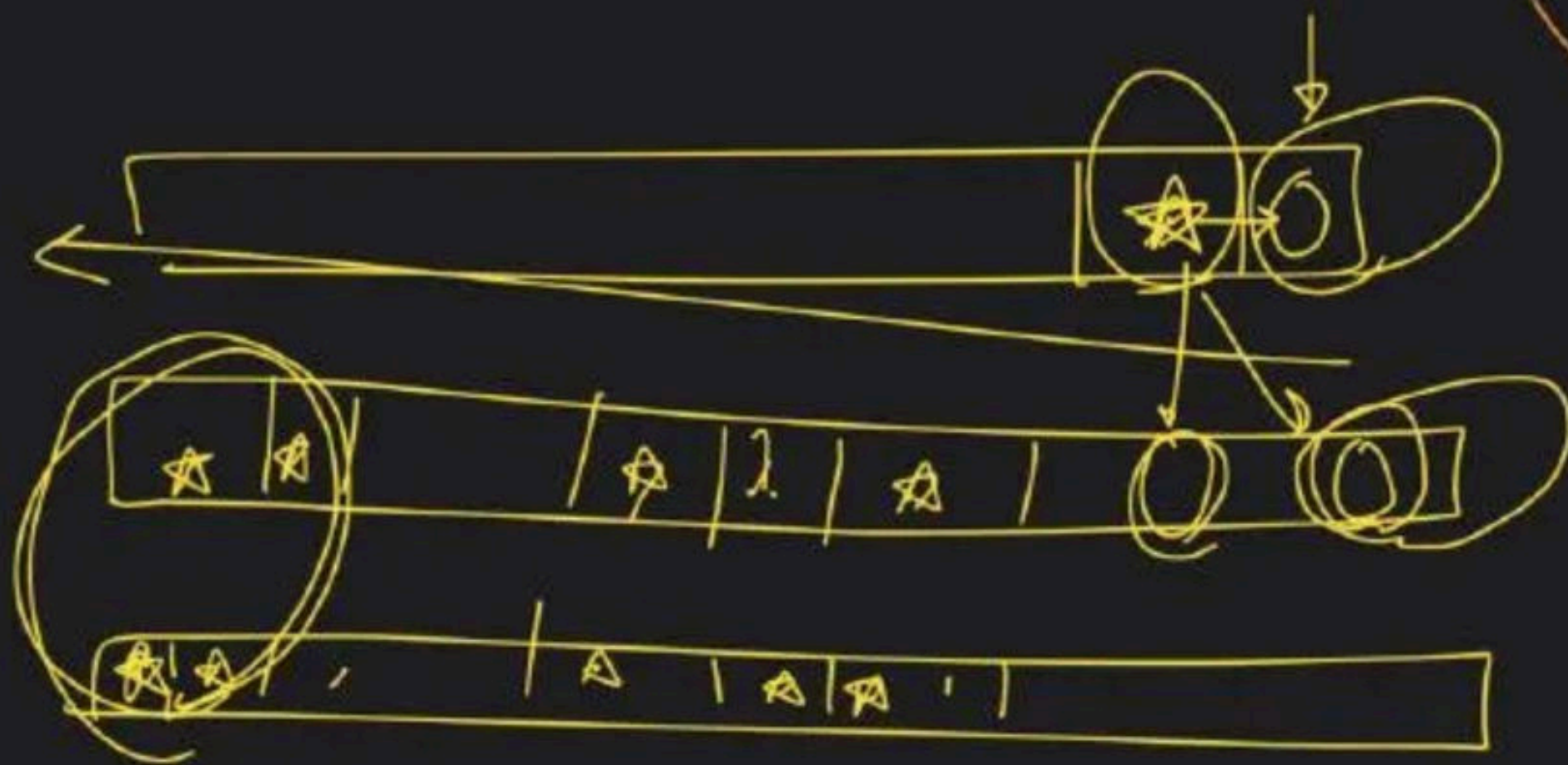


lost
=

m = 3
n = 3



$curr[j]$ $\rightarrow$ $win[j+1]$
 $\rightarrow$ $hint[j]$
 $\rightarrow$ $next[j+1]$

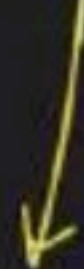




"b a b ~~b~~ a f"

(true)

"b a b ~~a~~ ~~a~~"



. b a

a ~~a~~

a ~~a~~

| < ~~a~~

wildcard

→ b a b ~~b~~ a r } false

→ b a ~~b~~ ~~a~~ ~~a~~ r

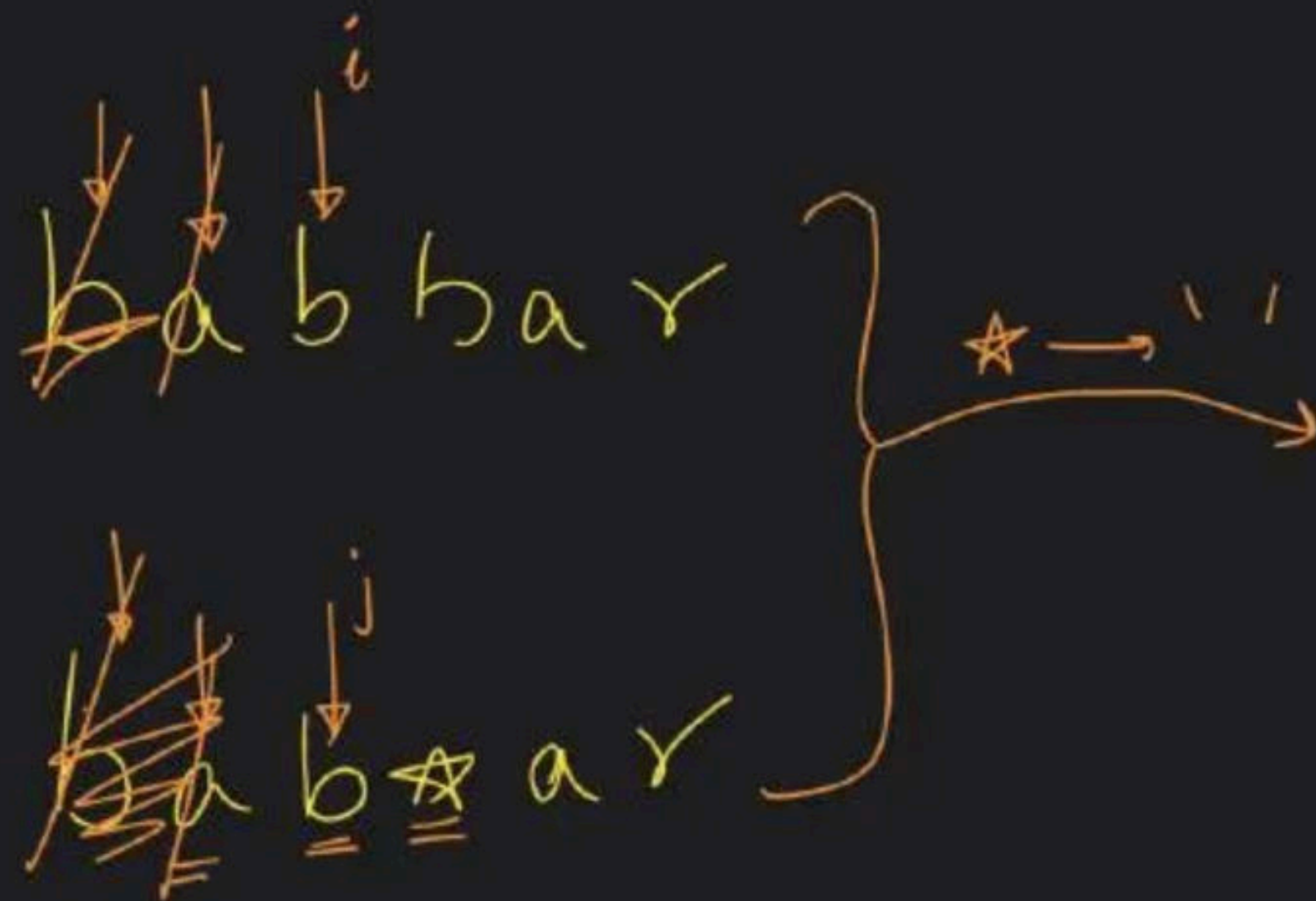
... b b, b b b, b b b b ...

~~b~~ ~~a~~ ~~b~~ ~~d~~ ~~r~~ } d
~~b~~ ~~a~~ ~~b~~ ~~a~~ ~~r~~ } d

I
 $i := s.\text{len}() \ \& \ j := r.\text{len}()$
 true

(II)
 $i == s.\text{len}()$
 true
 false

II
 $j^2 = r.\text{len}()$
 false



bbar

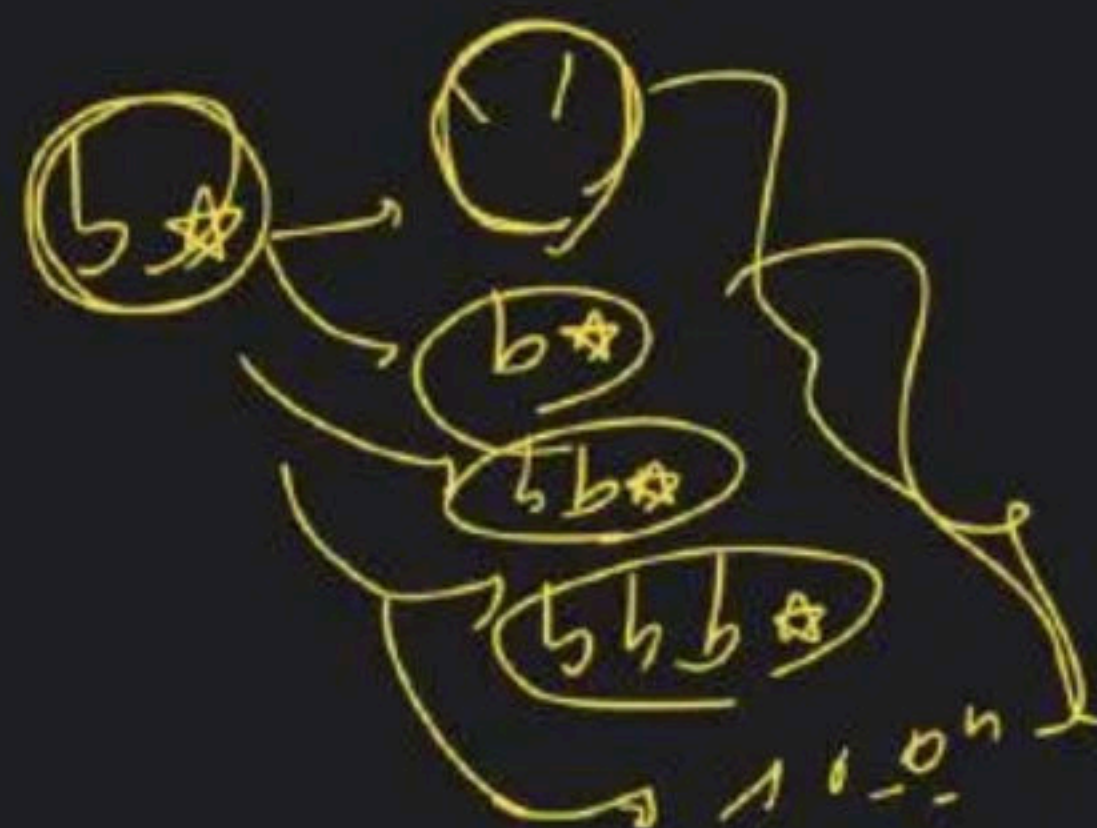
baar

$\downarrow \quad \downarrow \quad \downarrow$
~~b~~ ~~a~~ b b a r

$\downarrow \quad \downarrow$
~~b~~ ~~a~~ (b r) a r

empty $\rightarrow$ $f(i, j+2)$

$\downarrow$
 b a (b b a r)
 b a ~~x~~ ~~x~~ (a r)
 $\uparrow \quad \uparrow$
 $j \quad j+2$



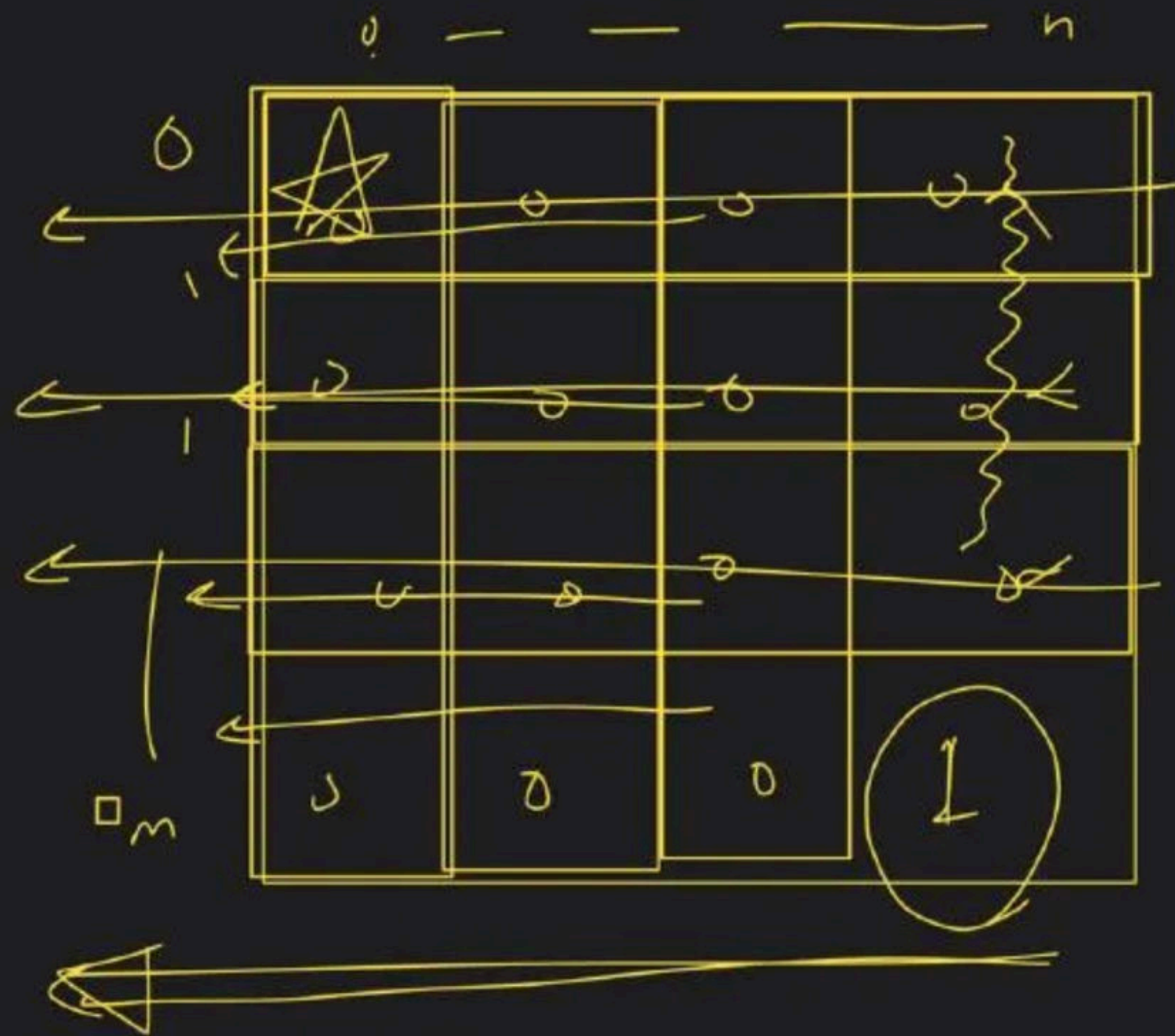
$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
~~b~~ ~~a~~ b b a r $\rightarrow$ "x
 $\downarrow$ $\downarrow$ $\downarrow$
~~b~~ ~~a~~ (b $\star$) a r

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 b a ~~b~~ b a r (it)

Cur Matrix & f(it, i)

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 b a (b) b $\star$ a r
 (j)





dp[m][n]

n-1

• → single ch



$\frac{C_{\text{max}} - C_{\text{min}}}{\Delta t} \quad (17)$

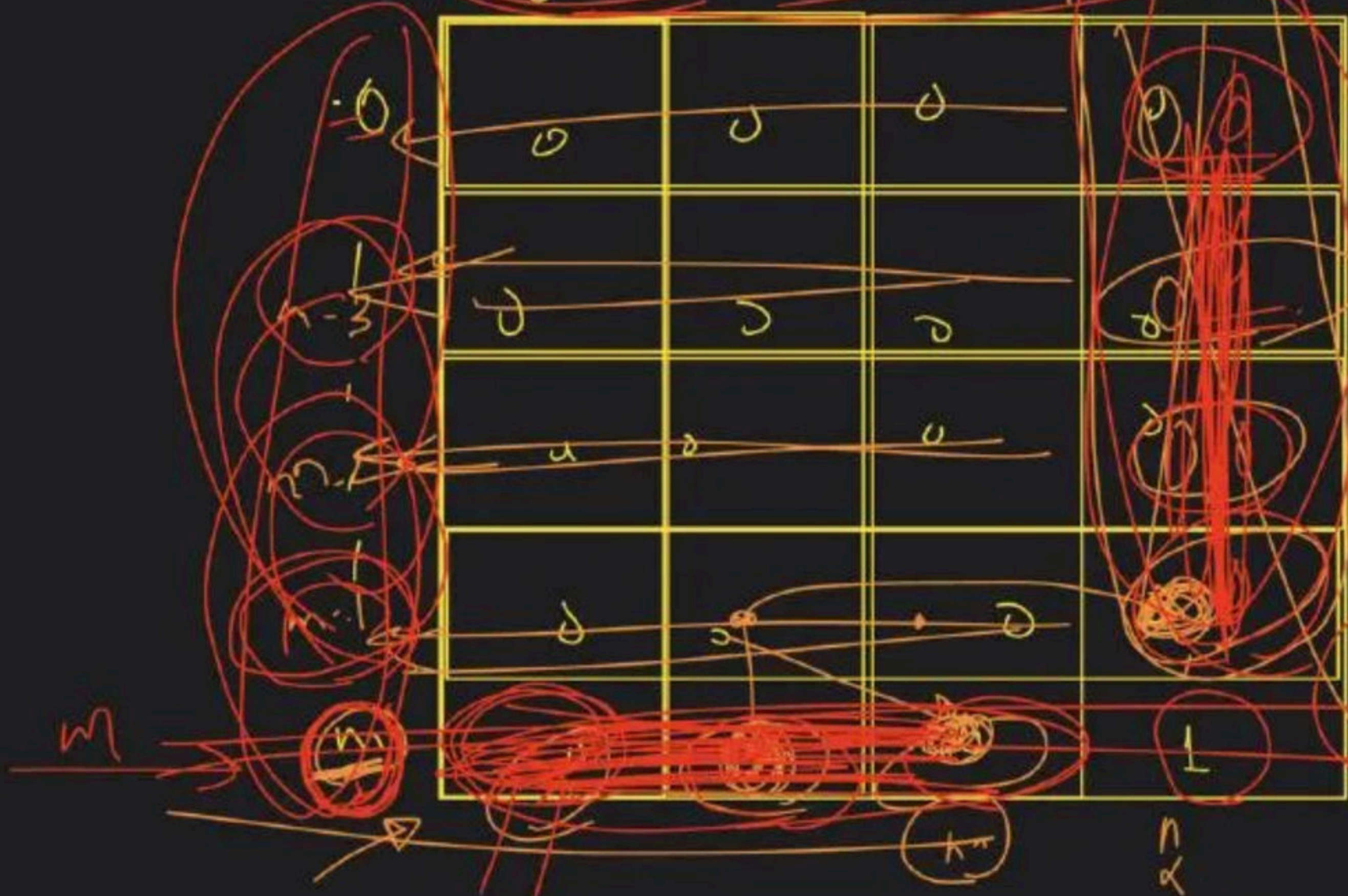
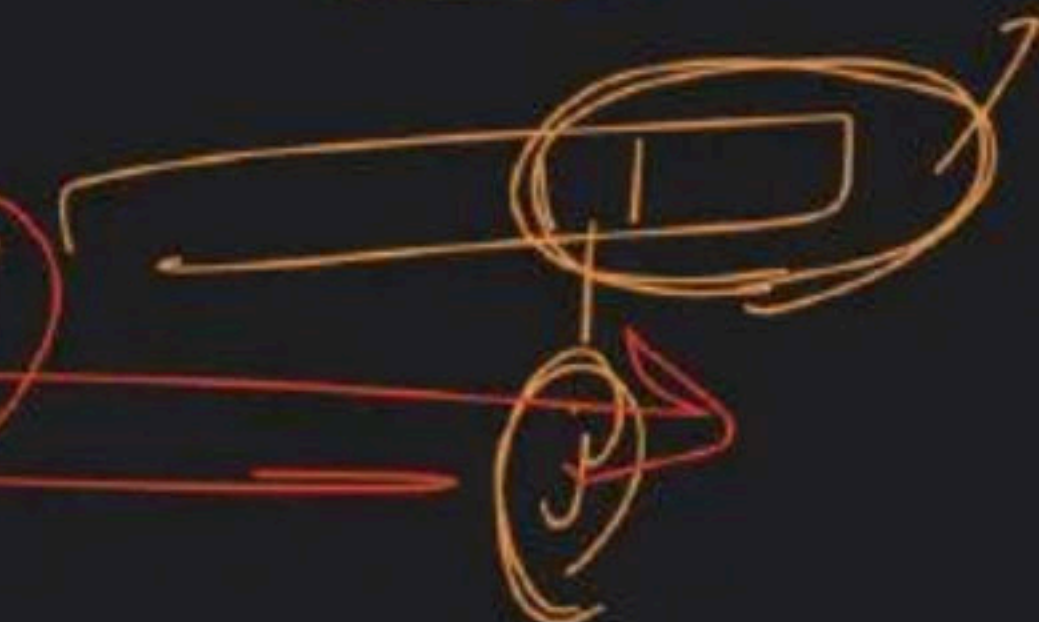
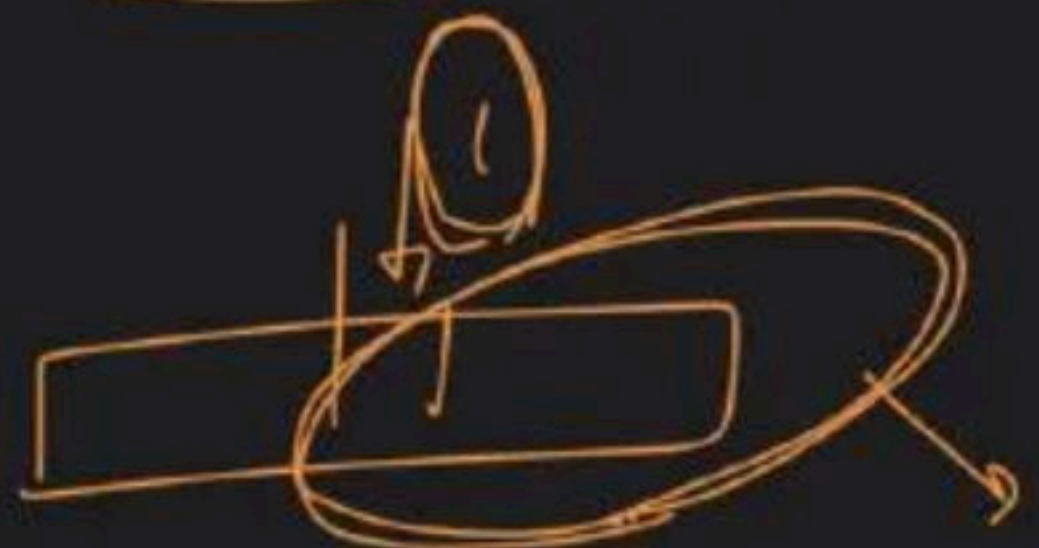
b a ~~b~~ b⁽¹⁴⁾ a r
 b a ~~b~~ ~~b~~ a r

$\Delta \rightarrow$
 $p-1 \quad a^*$ } $\rightarrow \text{True}$

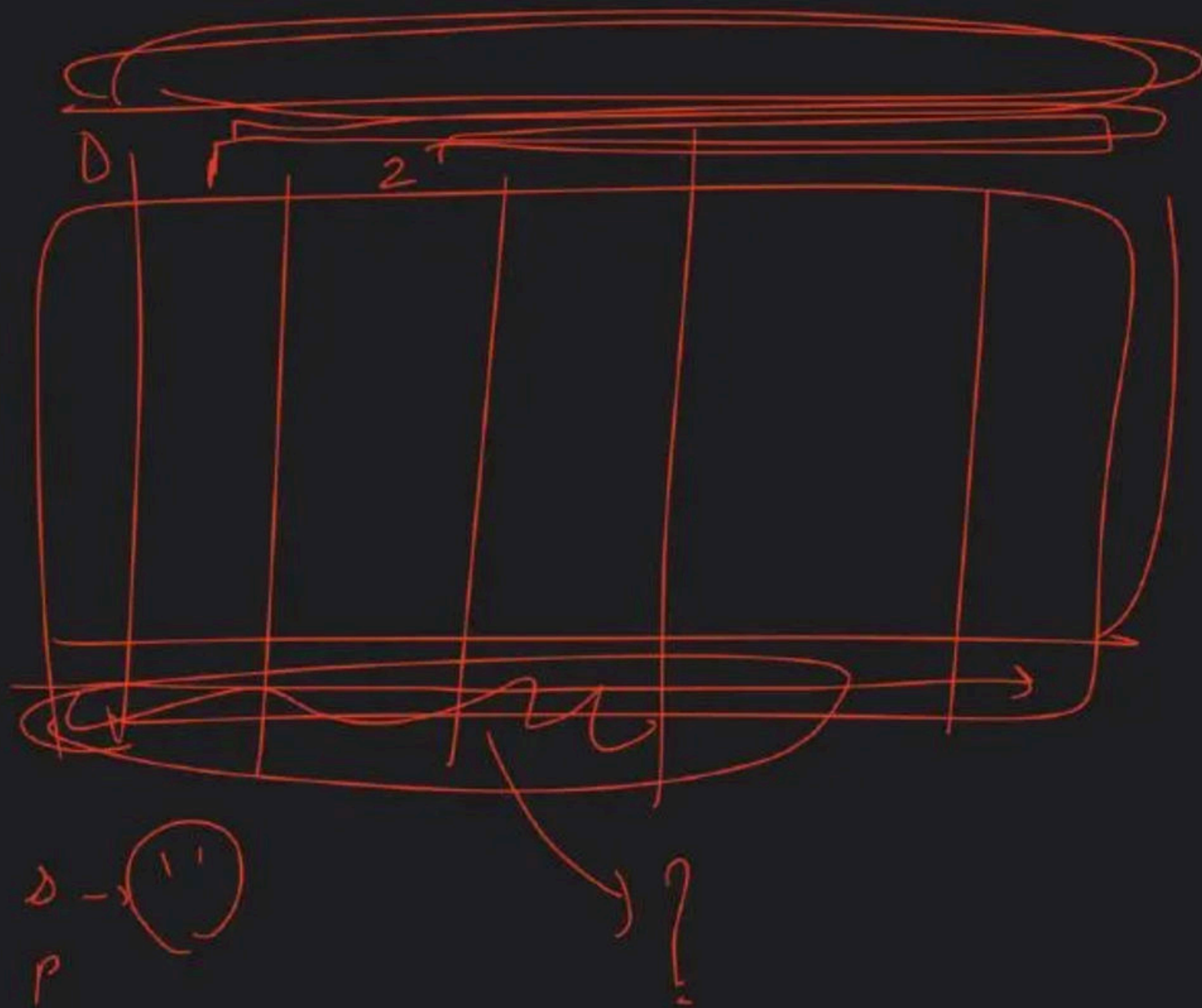
0 1 - - - n

$dp[m][n]$

$dp[i][j]$

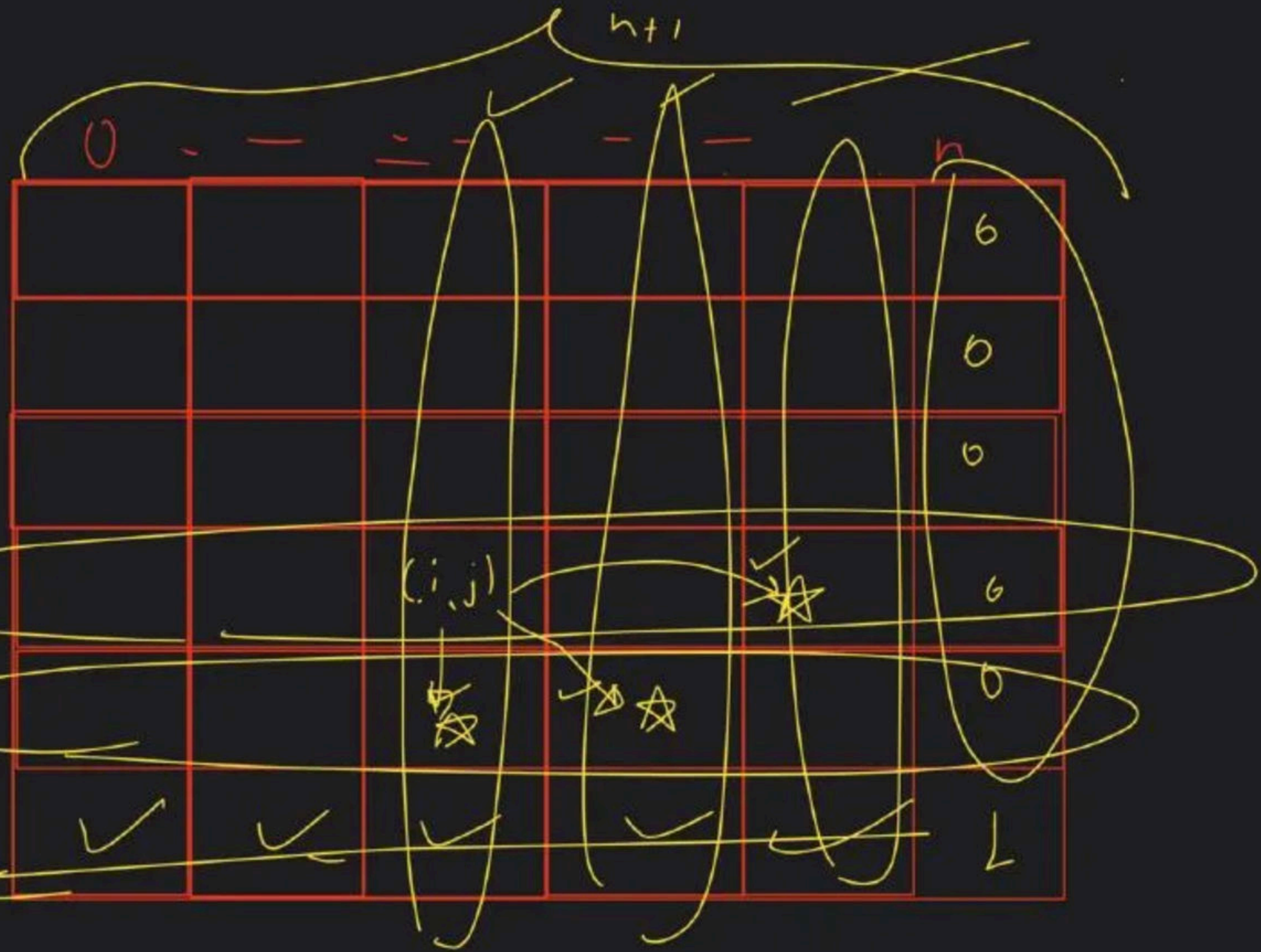


m



you

m



CW

0
0
0
0
0

curr

0	0	0	*	0	0
--------------	--------------	--------------	--------------	--------------	--------------

next

0	0	0	0	0	1
---	---	---	---	---	---

last row

$l == m$

curr

0	0	0	0	0
---	---	---	---	---

next

0	0	0	0	1
---	---	---	---	---

112m

Tabulation

$$1.2 = 4m$$


Tab

$$f_{\alpha}(m \rightarrow 0)$$

for $(n-1 \rightarrow 0)$

~~curr~~ $\rightarrow$ dp(i)

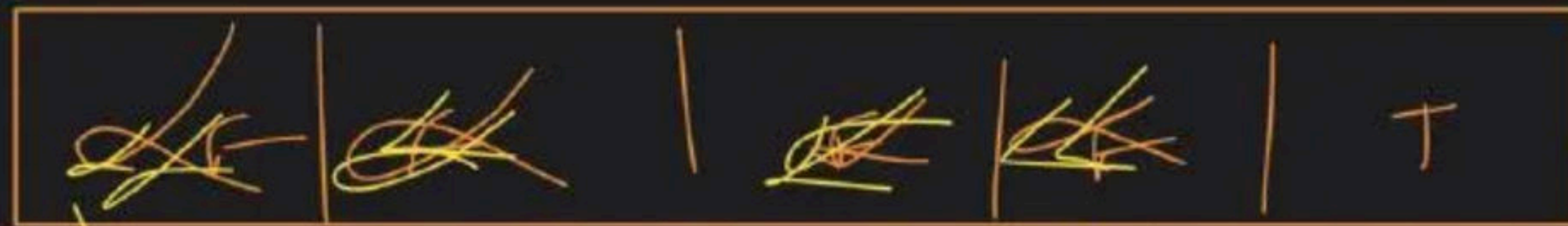
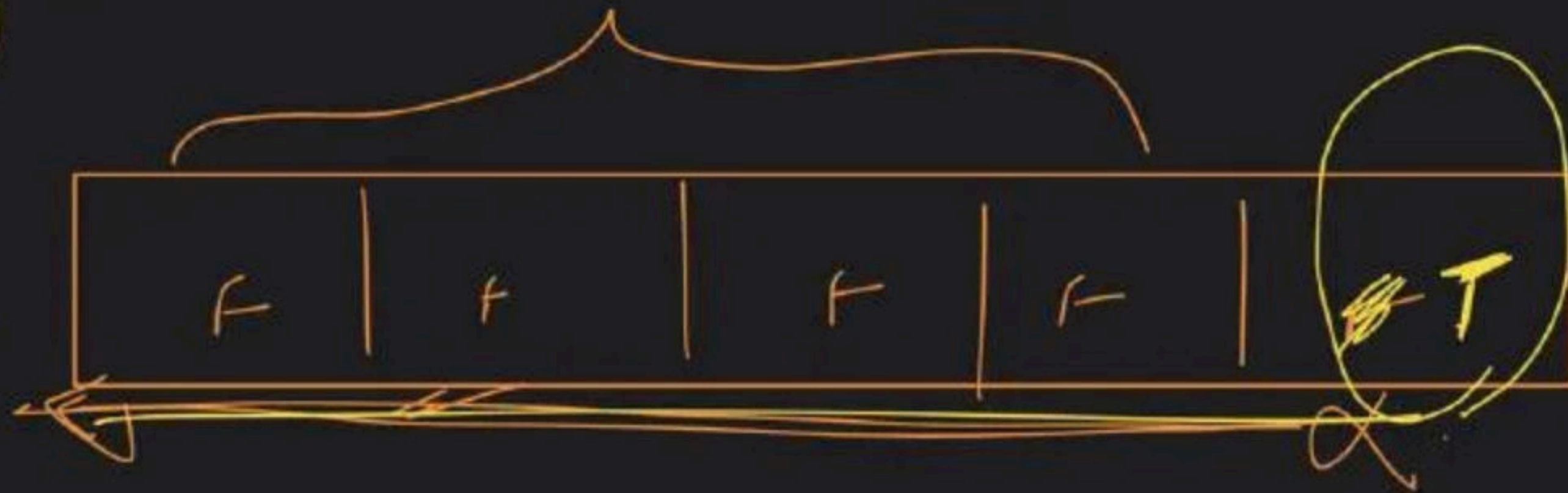
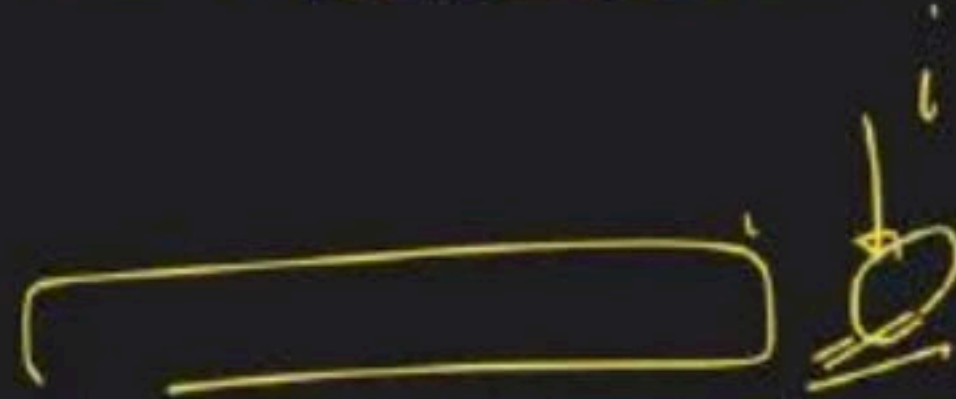
next $\Rightarrow$ dp(i+1)

curr $\rightarrow$ dp(i)

curr(n)
dp(i)(n)
(curr) $\rightarrow$

dp(m)(n)
 $\rightarrow$ T

next



f(i $\rightarrow$ m $\rightarrow$ 0)

~~f(j $\rightarrow$ n-1 $\rightarrow$ 0)~~



i = m
j = 4

curr(i) = T

$$if(i == n)$$



$$(cur \rightarrow dp[i])$$

$$if(i == m)$$
$$cur[n] = T$$

$$dp[i][n]$$

$$dp[m][n] = T$$